

Supplementary Material

1 Theoretical rigour assessment framework

Table 1: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[1]	3 -The paper shows high ecological relevance by mapping sediment transport pathways in coastal systems using GT. The nodes (geomorphic cells) and links (sediment fluxes) are well-defined and justified with empirical data.	3 -The paper systematically applies GT, clearly defining directed, weighted networks and justifying choices (e.g., adjacency matrices, modularity, centrality).	2 -The paper acknowledges uncertainties (e.g., grain size variability), but does not systematically quantify them. Sensitivity analyses are implied (e.g., different grain size classes), but not clearly stated.	2 -Method scales well for the case study (25 nodes) but lacks demonstration for larger systems. The paper notes challenges in defining geomorphic cells at broader scales but does not test computational limits or propose optimisations.	2 -The authors link their framework to coastal management questions, indicating stakeholder relevance, but provide no applied examples or direct stakeholder use.	3 -The paper uses clear visualisations (e.g., adjacency matrices, pathway diagrams) and connects metrics to ecological concepts. The discussion contextualizes results in terms of system-wide behaviour and management implications.	15
[2]	3 - The paper excels in ecological relevance. It maps graph structures to real-world marine ecosystems, where nodes represent habitat patches or subpopulations, and edges represent dispersal or connectivity pathways. The authors justify their graph construction with empirical data and ecological theory.	3 -The paper is rigorous in its theoretical foundations. It clearly defines the types of graphs used and justifies these choices based on ecological logic.	2 - The paper acknowledges uncertainty (e.g., variability in dispersal data, model assumptions) but does not extensively quantify it.	2 -The paper shows scalability to some extent, as graph-theoretic methods are inherently efficient for large networks. However, challenges with dynamic or multi-layered networks (e.g., temporal changes in connectivity) are noted but not deeply addressed.	3 -The paper strongly supports stakeholder utility. It translates network metrics (e.g., betweenness centrality) into actionable conservation recommendations (e.g., prioritising key stepping-stone habitats). The visualisations and results are intuitive and directly applicable to MPA design and management.	3 -The paper provides a highly interpretable framework and results. Network metrics are explained in ecological terms, such as using betweenness to identify key dispersal pathways.	16
[3]	3 -The study thoroughly applies GT to real marine ecological systems like the GBR, using empirically-justified nodes and edges to represent actual ecological units and processes.	3 -The methods build on established ecological and GT with clearly justified metrics and derivations.	2 -The paper includes some sensitivity analyses but generally does not rigorously quantify uncertainty.	2 -Shows spatial scalability by analysing the 321-reef GBR and showing logarithmic MLT increase with network size, but temporal scalability is limited as dynamic or multilayer networks for future work.	2 -It targets conservation applications, indicating potential stakeholder relevance. For instance, it "provides a theoretical framework to integrate complex patterns of dispersal connectivity marine conservation planning".	3 -The results are interpretable through ecological reasoning, though the methods are technical meaning they are not easily accessible to non-experts.	15

Table 2: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[4]	3 -Shows high ecological relevance by analysing co-occurrence patterns of white sharks using acoustic telemetry data, mapping social interactions and spatial associations in a coastal ecosystem.	3 -Applies social network analysis rigorously, defining nodes, edges, and metrics like SRI, assortativity, and modularity based on established ethological and graph theory principles.	2 -Acknowledges uncertainties such as tag performance variability and ecological drivers, but lacks systematic quantification or explicit sensitivity analyses.	2 -Scales to 185 individuals and 365880 detections in a case study, but does not demonstrate applicability to larger systems or address computational challenges.	3 -Links findings to conservation implications, such as avoidance behaviours and habitat preferences, providing actionable insights for management of aggregation sites.	3 -Uses clear visualizations (e.g., network graphs) and interprets metrics in ecological terms, discussing implications for behaviour and population dynamics.	16
[5]	3 -High ecological relevance through connectivity models for 16 fish species, integrating habitat distribution and anthropogenic disturbances to inform MPA expansion in coastal ecosystems.	3 -Builds on systematic conservation planning and connectivity theory, defining matrices and metrics for pairwise habitat connectivity justified by dispersal ecology.	2 -Acknowledges uncertainties in dispersal data and model assumptions but does not extensively quantify them; no explicit sensitivity analyses mentioned.	2 -Applied to Swedish Baltic Sea with multiple species, but scalability to larger systems or dynamic networks is noted as challenging without deeper addressing.	3 -Directly supports stakeholder utility by prioritizing areas for MPA expansion under 30x30 goals, translating connectivity and vulnerability into conservation recommendations.	3 -Provides interpretable results through maps and protection levels, explaining metrics in terms of biodiversity persistence and disturbance mitigation.	16
[6]	3 -High ecological relevance for marine mammal conservation, using networks to model MPA interactions for sea lions and whales across transboundary regions.	3 -Employs graph theory and complex network analysis, clearly defining nodes (MPAs), edges (interactions), and metrics like density and centrality based on social-ecological systems.	2 -Acknowledges limitations in self-reported data and geospatial validation but does not systematically quantify uncertainties.	2 -Applied to 27 MPAs across three countries, suitable for case study but lacks testing for larger or more complex networks.	3 -Provides recommendations for transboundary collaboration and governance, linking network metrics to conservation strategies.	3 -Interprets low-density networks and modularity in terms of geographic and hierarchical influences, using graphs for clear visualization.	16
[7]	3 -Ecologically relevant for giant kelp connectivity, mapping dispersal pathways and habitat patches in the Humboldt Current System amid climate and harvesting threats.	3 -Uses network analysis with connectivity matrices and centrality measures, justified by biophysical modeling and graph theory.	2 -Addresses temporal variability (seasons, ENSO) but implies rather than explicitly quantifies uncertainties through sensitivity analyses.	3 -Scales to 125 nodes over multiple years and scenarios, demonstrating robustness for regional metapopulations.	3 -Informs adaptive management, recommending temporal closures and protection of source-sink areas for resilience.	3 -Interprets metrics like strength and modularity in ecological contexts, using maps to visualize connectivity corridors.	17

Table 3: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[8]	3 -The paper excels in ecological relevance by linking GT to marine connectivity studies. It justifies the use of nodes and edges with empirical data from Lagrangian simulations and discusses their ecological implications (e.g., larval dispersal, nutrient transfer).	3 -The paper provides a strong theoretical basis for its methodology. It rigorously defines graph-theoretic concepts and the betweenness measure. It critiques the misuse of transfer probabilities as edge weights in prior studies and proposes a mathematically sound alternative to ensure consistency with shortest-path algorithms.	2 -The paper acknowledges variability in connectivity matrices due to ocean circulation patterns but does not quantify uncertainty (e.g., confidence intervals or sensitivity analyses for edge weights).	2 -The method is scalable to moderately large networks (e.g., 32 nodes in the Gulf of Lion case study) and uses efficient algorithms (Dijkstra, Brandes). However, the paper does not discuss computational limits for larger or dynamic network.	2 -The paper's results could inform marine conservation decisions (e.g., MPA design). However, it does not translate findings into actionable recommendations for stakeholders.	3 -The paper achieves high interpretability by clearly defining nodes, edges, and metrics (e.g., betweenness) in ecological terms. The results are presented with intuitive visualisations and are explained in detail.	15
[9]	3 -Applies GT to relevant ecological problems (HABs, food-webs). - it excels by bridging network theory and marine ecosystem dynamics, making it valuable for both theoretical and applied ecology.	3 -The paper clearly defines GT concepts and applies appropriate network metrics with consistent definitions for ecological questions.	0 -The paper does not quantify uncertainty in data or parameters.	0 -The analysis uses a small hypothetical network (34 species) and does not show or discuss its scalability to larger, complex real-world ecosystems.	0 -The paper does not address practical use by decision-makers or provide management to relevant outputs, offering only a general suggestion that network analysis could be "fruitful" for future applications.	3 -The model's logic is transparent, with nodes and edges having direct ecological meaning. Results and indices are explained with clear visuals and descriptions, they are clearly linked to ecological concepts.	9
[10]	3 -The paper achieves excellent ecological relevance through its biologically grounded network of habitat sites (nodes) and dispersal routes (edges), with direct applications for MPA conservation.	2 -The paper shows strong theoretical foundations but has minor gaps. The IBM and graph-theoretic approach are well-justified, and the choice of metrics (e.g., betweenness centrality, edge density) aligns with ecological questions.	1 -Uncertainty is minimally addressed. Although the paper acknowledges model limitations (e.g., lack of fine-scale density data, exclusion of ENSO events) and some sensitivity is qualitatively discussed, but does not rigorously quantify uncertainty.	2 -The connectivity framework is described as applicable at multiple scales, but only island-scale results are shown. The methods note that "graph-theoretic analysis is highly scalable", but it does not test dynamic or multi-layered networks or parallelization strategies.	3 -The paper supports management decisions by identifying priority regions and coast-specific connectivity patterns, translating results into actionable insights for MPA design and habitat protection, with community input enhancing practical relevance.	3 -The results are presented in a clear, visual manner. Connectivity is shown with network graphs and maps, and the authors explain all metrics in text and captions for understanding by non-experts.	14

Table 4: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[11]	3 -The paper thoroughly addresses ecological relevance by mapping genetic data to real aquatic systems (rivers, lakes, seas). The study links graph-theoretic approaches to biological processes like dispersal and recruitment, ensuring high ecological validity.	2 -The paper applies strong theoretical frameworks to aquatic systems rather than developing new theoretical foundations. It effectively synthesises and adapts existing concepts but does not introduce novel mathematical or computational innovations.	2 -The paper acknowledges uncertainty and uses sensitivity analyses (e.g., sliding-window IBD tests) and alternative model testing. However, it lacks explicit quantification (e.g., confidence intervals, bootstrapping) for some metrics, and uncertainty in graph construction is not fully explored.	2 -The methods (GT, individual-based simulations) scale to large, hierarchical systems, but challenges with dynamic networks or high-dimensional genomic data are noted without solutions, and computational efficiency at extreme scales is not tested.	3 -The study links genetic knowledge to real-world management, highlighting its importance for policy and conservation. Examples, like redefining fishing limits, illustrate clear benefits for stakeholders.	3 -The paper addresses interpretability of genetic patterns and methods, though not as an in-depth standalone topic. The authors highlight that certain analytical approaches enhance understanding by making complex patterns more observable.	15
[12]	3 -This paper shows high ecological relevance by addressing critical issues in marine conservation and resource management. The nodes (populations) and edges (genetic connectivity) are biologically meaningful, and the study justifies its approach.	2 -The paper uses proven genetic and phylogeographic methods. However, it lacks formal theoretical justifications for some assumptions, such as the mutation rates used for dating demographic expansion. The choices of metrics are appropriate but not deeply explored from a mathematical rigour perspective.	3 -The paper quantifies uncertainty by using published mutation rates and giving confidence intervals (e.g., 7,000–45,000 years for the Chilean expansion). It acknowledges the limitations of the data, the authors also admit that it is uncertain whether the genetic bottleneck can be attributed to overharvesting or domestication.	1 -The methods are robust for the studied populations but are not designed for large-scale or dynamic systems. The paper does not address computational efficiency or scalability to larger datasets or more complex ecological networks.	3 -The study provides direct information for conservation and aquaculture by highlighting genetic bottlenecks and risks of clonal propagation. The conclusions of the study, such as the need to preserve the genetic diversity of Chilean populations are actionable for policy makers and aquaculture managers.	3 -Results are clearly presented and tied to ecological concepts (bottlenecks, domestication). Genetic metrics are well-explained, and the narrative connects data to conservation implications. The study balances technical rigour with accessibility.	15
[13]	3 -The paper is highly ecologically relevant, reconstructing the food web from empirical data with clearly defined nodes and edges, and linking its structure to conservation actions like prioritising species with high centrality or unique topology for fisheries management.	3 -The paper shows strong theoretical foundations by rigourously applying GT to the food web. It tests three conceptual models (random, small-world, and scale-free) and justifies the choice of metrics (e.g., connectance, clustering coefficient, centrality indices) based on ecological theory.	2 -The paper acknowledges uncertainty by comparing the empirical food web to random networks and conducting sensitivity analyses. However, it does not fully quantify uncertainty in edge weights or node definitions (e.g., bootstrapping).	2 -The study handles a moderately sized network (256 nodes, 647 edges) and shows scalability by analysing subsets. However, the paper does not extensively discuss the computational performance (e.g., time complexity, memory usage) or the scalability.	3 -The paper directly addresses stakeholder utility by translating network metrics into actionable recommendations for fisheries management. The focus is on the conservation of high centrality species (e.g., shrimp) and top predators, which is consistent with the goals of ecosystem-based management.	3 -The paper is highly interpretable and clearly explains complex concepts of GT and their ecological implications. It contains a concise summary of the methodology and key results.	16

Table 5: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[14]	3 -The paper comprehensively addresses ecological relevance by systematically reviewing studies that apply GT to aquatic ecosystems. Graph structures (nodes, edges) are linked to ecological units (e.g., habitats, species) and processes (e.g., dispersal, larval connectivity).	2 -The paper provides a clear overview of GT applications but lacks a detailed discussion of the mathematical rigour or justification for chosen metrics. While it aggregates theoretically sound studies, it does not critically assess assumptions or internal consistency, such as sensitivity to edge-weight thresholds.	1 -The paper minimally addresses uncertainty. Although some cited studies mention sensitivity analyses or probabilistic edges, the review itself does not systematically evaluate how uncertainty is quantified across studies.	2 -The paper implicitly addresses scalability by including studies of varying spatial scales (e.g., local ponds to ocean basins) and network complexities. However, it does not discuss computational challenges or solutions (e.g., parallelization).	3 - The paper is highly relevant for stakeholders as it directly addresses "Human impacts on connectivity" and its implications for management. It references studies focusing on the role of MPAs in fisheries management and conservation.	3 -The review is very easy to interpret due to its clear structure and the systematic organisation of the information. The characteristics and results of the individual studies examined are presented in detailed tables, categorised according to water area, ecosystem type, taxa, life stage of the organism, region and type of human impact.	14
[15]	3 -The paper addresses a critical ecological challenge, enhancing GBR resilience and recovery from disturbances like cyclones, bleaching, and COTS outbreaks, by directly modeling larval dispersal and connectivity, key processes for ecosystem function.	3 -The paper has strong theoretical foundations. It uses graph-theoretical measures (e.g., outer degree, strength, outer component) to quantify the connectivity of reefs and clearly justifies these choices.	2 -The paper acknowledges significant uncertainties inherent in marine systems, such as the variability and transient nature of demographic connections, and empirical uncertainty surrounding dispersal parameters. They do not fully quantify uncertainty ranges.	3 -The study scales well to the 3800 reefs of the GBR using efficient graph algorithms and high-resolution oceanographic modelling. It handles spatial and temporal complexity (e.g., seasonal spawning events) and shows computational feasibility for large ecosystems.	3 -The paper provides actionable insights for marine management by identifying 112 "robust source reefs" that are critical to GBR resilience. The results directly inform conservation prioritisation, such as protecting these reefs to promote recovery of the entire system.	3 -The paper effectively communicates its complex findings through clear explanations and comprehensive visualisations.	17
[16]	3 -The paper tackles the ecological challenge of designing marine reserve networks under global warming, addressing larval connectivity to support ecosystem services, resilience, and recovery after disturbances.	3 -The paper justifies its use of GT, using centrality metrics (eigenvector centrality, distance-weighted fragmentation) to identify hubs and stepping stones for connectivity. The choice of directed, weighted graphs aligns with asymmetric larval dispersal patterns.	2 -The paper acknowledges uncertainties like limited empirical data for dispersal patterns and the high uncertainty in future ocean warming impacts on larval duration. While addressing some by exploring extreme scenarios, it lacks comprehensive quantification for all identified uncertainties.	2 -The method scales well spatially and handles multi-species connectivity, but dynamic or multi-layer networks are not addressed. Computational constraints are noted (e.g., downscaling coarse connectivity matrices), but are not discussed.	3 -The paper directly supports MPA planning by translating graphical metrics into actionable recommendations (e.g., prioritising upstream nodes). It considers socio-economic costs (opportunity costs for fisheries) and compares trade-offs between different protected area configurations.	3 -The paper is highly interpretable, effectively communicating complex concepts and results.	16

Table 6: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[17]	3 -The paper is highly ecologically relevant, capturing COTS outbreak dynamics and linking nutrient enrichment and hydrodynamic conditions to larval survival and connectivity.	3 -The paper shows strong theoretical foundations. It uses GT to analyse larval connectivity, with clear justifications for the choice of metrics (e.g., cluster coefficient, CI) and their relevance to COTS outbreaks.	2 -The paper acknowledges some uncertainties (e.g., in identifying which flood events trigger plankton blooms, and it references confidence levels in model calibration), but it stops short of a full uncertainty analysis.	1 -The methods (GT and hydrodynamic modelling) are scalable to some extent, as showed by their application across a 30-year period and multiple reefs. However, the paper does not address computational challenges or scalability to larger spatial or temporal scales.	3 -The paper provides clear, actionable insights for marine management. It identifies the reduction of terrestrial nutrient loads (~20–40%) as a feasible measure to mitigate COTS outbreaks, consistent with existing water quality objectives.	3 -The results are very easy to interpret. GT metrics are linked to ecological concepts such as larval retention and local reef connectivity. The paper uses visualisations to illustrate relationships between environmental triggers and outbreaks.	15
[18]	3 -The paper addresses a highly relevant ecological problem on how spatial connectivity influences vital ecosystem functions, specifically herbivory, in the context of disturbances.	3 -The paper has a solid theoretical basis. It uses network theory (closeness centrality) to quantify connectivity, with clear justifications for the construction of graphs (static, undirected patches).	2 -The study addresses uncertainties moderately, considering variability in grazing rates and behavioural interactions, but does not provide confidence intervals or conduct further sensitivity analyses.	2 -The microcosm approach is in principle scalable, but limited by its small, controlled system. The methods could theoretically be scaled to larger systems, but the study does not show this empirically, or discuss the computational challenges for complex, dynamic ecosystems.	2 -The paper contains clear and actionable recommendations for marine managers. It shows that "greater connectivity significantly improves herbivory", which is a crucial ecological function for maintaining ecosystem resilience in disturbed environments.	3 -The paper is very interpretable. Network metrics are clearly linked to ecological concepts (patch accessibility), and the results are presented intuitively (e.g., herbivory rates vs. connectivity). The discussion contextualises the results for ecosystem management.	15
[19]	3 -The paper emphasises the ecological importance of <i>Posidonia oceanica</i> , highlights its role in important coastal services and the urgency of taking action to address its decline. It models realistic ecological processes such as seasonal dispersal using a high-resolution oceanographic framework.	3 -The paper shows strong theoretical foundations. It justifies the use of GT metrics (e.g., In-strength, Betweenness, Eigenvector centrality) to analyse connectivity, which are well-suited for ecological questions about dispersal and metapopulation dynamics.	2 -The paper addresses uncertainty, but could be more comprehensive. Sensitivity analyses are conducted to test the impact of key parameters (e.g., drift duration, coastal interaction types) and interannual variability in dispersal probability is recognised. However, uncertainty is not fully quantified.	2 -The methods are scalable, but have their limits. The graph-theoretic approach efficiently handles the spatial complexity of the Balearic archipelago, and the IBM simulates a large number of particles. However, the study does not address the computational challenges or scalability to larger regions or more complex ecosystems.	3 -The paper provides clear utility for stakeholders. It translates network analysis results into actionable conservation recommendations, such as prioritising Alcudia Bay for restoration due to its high In-strength centrality.	3 -The paper is easy to interpret, the GT metrics (e.g., centrality measures) are ecologically meaningful and clearly linked to concepts such as "stepping stone" patches or settlement hotspots. The methods are transparent and the results are presented in a way that combines technical rigour with practical insights.	16

Table 7: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[20]	3 -The paper shows a high ecological relevance by linking graph-theoretical methods to real-world marine conservation scenarios.	3 -The paper provides a rigorous theoretical foundation for using GT in marine reserve design. It clearly justifies the choice of graph metrics (e.g., Google PageRank, eigenvector centrality) and their ecological interpretations.	2 -The authors address the most important sources of uncertainty and variability in their analysis, but do not analyse the computational scaling in detail.	2 -The methods were applied to moderately large networks and multiple species, suggesting some scalability, but the paper does not analyse the computational limits in depth.	1 -The study makes only minimal connection to stakeholder or management perspectives.	2 -The methods and results are explained with moderate clarity, supported by descriptive summaries and examples, but are still technically complex.	
[21]	3 -The paper excels in ecological relevance by thoroughly mapping network concepts to marine ecosystems. It justifies graph structures (e.g., food webs) with empirical examples and aligns network dynamics with ecological theory.	3 -The paper shows strong theoretical rigour by clearly defining the network types and justifying the metrics with ecological logic. It integrates established theories (e.g., Lotka-Volterra models) and discusses limitations.	1 -The paper acknowledges data limitations and computational challenges but lacks explicit techniques to quantify uncertainty.	2 -The paper emphasises the computational requirements for large networks, but it does not offer any concrete scaling strategies.	3 -The network methods are linked to practical management. The paper provides examples where network analysis informs decisions, for example identifying keystone species in fisheries.	2 -The methods provide interpretable metrics, but also require expertise to understand them. The quantitative results have clear ecological meaning to researchers, but are somewhat technical.	14
[22]	3 -The paper effectively connects ecological theory to real-world dynamics in MPAs by mapping nodes and edges to specific species and interactions.	2 -The paper uses GT with clear formulations and reasoned decisions based on ecological logic. However, it lacks mathematical depth as it lacks discussions on limitations and sensitivity analyses.	1 -The study acknowledges data limitations (e.g., reliance on expert validation) but does not systematically quantify uncertainty in edge weights or network metrics.	2 -The paper effectively show scalability by applying its methodology to four distinct MPAs and then integrating them into a "global network", but computational efficiency is not discussed.	3 -The paper excels in translating network results into actionable insights. visualisations and centrality metrics directly inform management priorities.	3 -The results are presented in a clear, visual, and qualitative way.	14
[23]	3 -The paper shows high ecological relevance by linking GT to real-world marine conservation.	3 -The paper provides a rigorous theoretical foundation for its methods. It clearly defines the graph type and justifies the choice of centrality metrics based on their ecological significance.	2 -The paper acknowledges uncertainty via interannual variability and sensitivity analyses, but key parameter uncertainties are not fully quantified, and confidence intervals for centrality metrics are not provided.	2 -The methods are scalable to the Mediterranean basin but face limitations for larger or more complex systems. The paper does not discuss computational efficiency or parallelization.	3 -The results are linked to conservation planning. The visualisations and tables translate complex metrics into interpretable insights and help prioritise conservation.	3 -The model and results are explained in clear ecological terms. The paper avoids black-box methods, ensuring transparency.	16

Table 8: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[24]	3 -The paper excels in ecological relevance. It models the dispersal of Sabellaria alveolata larvae, a reef-building polychaete, using a biophysical transport model.	3 -The paper shows strong theoretical foundations. It uses GT and hydrodynamic modelling (MARS3D) with clear justifications for methodological choices.	1 -The study acknowledges several sources of uncertainty and data limitations. However, it does not provide explicit quantitative uncertainty measures for the network metrics derived.	1 -There is no discussion of scaling methodology beyond this case, suggesting that scalability was only considered to a limited extent.	3 -The results directly inform conservation strategies, such as prioritising high-centrality reefs and stepping-stone habitats.	3 - Model outputs are presented and interpreted in clear ecological terms. The results are straightforward to interpret in ecological and conservation contexts.	14
[25]	3 -The paper shows strong ecological relevance by directly addressing a challenge in marine conservation, which is the need to incorporate ecological connectivity, particularly through adult movement, into the design of MPAs between climate change and data limitations.	3 -The paper shows strong theoretical foundations for GT, clearly justifying the use of multiplex networks to represent multiple habitat types and centrality metrics (e.g., betweenness, eigenvector centrality) to assess connectivity.	2 -The paper shows a moderate level of uncertainty quantification. It acknowledges various sources of uncertainty and data limitations inherent in marine ecological modelling but does not carefully quantify uncertainty.	2 -The methods scale moderately well to the regional level, with 100 km planning units and multiplex networks handling 16 habitat categories. The paper does not explore dynamic networks or parallelization, which would be needed for larger/more complex systems.	3 -The paper directly informs the ongoing MPA planning process by translating results into actionable insights (e.g., identifying connectivity hotspots for candidate MPAs).	3 -The paper excels in interpretability, making considerable efforts to clearly explain the internal logic and outcomes of the models in accessible, ecologically meaningful terms.	16
[26]	3 -The paper demonstrates high ecological relevance by directly addressing the critical need for understanding ecosystem connectivity in the design and adaptive management of MPA networks.	3 -The paper shows strong theoretical foundations by carefully applying GT (EDENetworks) and population genetics (FST metrics) with clear justifications for methodological choices. It clearly defines the graph components, with sampling localities as nodes and pairwise genetic differentiation (FST) quantifying connections.	2 -The paper demonstrates a moderate level of uncertainty quantification. It acknowledges limitations (e.g., species-specific idiosyncrasies, autocorrelation between variables) but does not carefully quantify uncertainty.	2 -The paper shows moderate consideration for scalability. The methods are scalable to larger datasets, as evidenced by the integration of multiple species and localities. However, the study focuses on 16 localities and 5 species, leaving open questions about performance with finer-scale or more taxa.	3 -The study provides actionable insights for marine park management, such as identifying key hubs and recommending monitoring priorities. Results are translated into clear conservation implications (e.g., protecting connectivity pathways for climate adaptation).	3 -The paper excels in interpretability, clearly explaining its methodology and findings in terms that are accessible and meaningful within an ecological context.	16

Table 9: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[27]	3 -This paper shows high ecological relevance by directly addressing the critical need for understanding ecosystem connectivity in the context of marine conservation and management, particularly for coral reefs in the GBR.	3 -The paper shows strong theoretical foundations by rigourously applying a biophysical modelling approach coupled with GT. The choice of GT is well-justified for identifying spatially isolated reef clusters.	2 -The paper shows a moderate level of uncertainty quantification. It acknowledges various sources of uncertainty and limitations inherent in its modelling approach but does not rigourously quantify them.	2 -The unstructured-grid model efficiently handles the GBR's spatial complexity, but scalability to larger or more dynamic systems is not fully tested. The Lagrangian model's computational demands are noted, but limitations for very large-scale applications remain.	3 -The study directly supports marine conservation by identifying reef communities and dispersal barriers, which can inform MPA design. The results are presented as spatially explicit maps and connectivity metrics, making them actionable for managers.	3 -The paper excels in interpretability, clearly explaining its complex methodology and findings in accessible and ecologically meaningful terms. It provides clear definitions of key ecological and modelling concepts such as "connectivity network", "larval dispersal", "genetic connectivity", and "demographic connectivity".	16
[28]	3 -The paper is ecologically relevant, addressing how temperature and dispersal shape population connectivity and species range boundaries, key for predicting distributions and assessing ecosystem resilience.	2 -The paper shows strong theoretical foundations by carefully applying biophysical dispersal modelling and GT, but the paper lacks formal justification for choosing specific metrics (e.g., betweenness centrality) over alternatives.	2 -The paper shows a moderate level of uncertainty consideration. It acknowledges inherent variability and influencing factors but does not rigourously quantify uncertainty.	2 -The study shows scalability within the tested system but does not discuss computational limits for larger networks or higher-resolution models.	3 -The paper clearly shows its potential for stakeholder utility by informing conservation and management of species living in variable environments. Results directly inform conservation by identifying critical sites for prioritising monitoring or protection.	3 -The paper excels in interpretability through its clear explanation of methods and direct linkage of results to ecological concepts.	15
[29]	3 -This paper shows high ecological relevance by directly addressing how climate-modulated ocean circulation influences larval transport and population connectivity, which are fundamental processes impacting species distribution and survival.	2 -The paper shows strong theoretical foundations by using Regional Ocean modelling System. It clearly defines the types of graphs used and justifies these choices based on ecological logic.	2 -The study shows a moderate level of uncertainty consideration. It acknowledges that connectivity is influenced by a multitude of factors, but it does not detail the use of formal uncertainty quantification techniques.	2 -The methods show scalability for regional systems. However, the study doesn't discuss computational limits or generalisability to larger/sparser networks.	2 -The paper directly addresses conservation applications and provides actionable insights. Results are translated into management recommendations though stakeholder engagement processes are not detailed.	3 -The paper excels in interpretability through its clear explanation of methods and direct linkage of results to ecological concepts. It defines its core concepts in easily understandable terms.	14

Table 10: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[30]	3 -The paper excels in ecological relevance by linking graph-theoretical concepts with ecological dynamics.	3 -The paper has a solid mathematical foundation and uses GT and linear algebra to develop its ideas. It builds on solid, established theories and concepts.	1 -The paper acknowledges variability in food-web structure but does not quantify uncertainty in its network metrics.	2 -The methods scale well to the 17 food webs analysed, but the paper does not address computational limits for larger or dynamic networks.	1 -The paper identifies functionally integrated species subgroups and discusses conservation implications but does not provide actionable management recommendations.	3 -The graph-theoretic framework is highly interpretable, with clear links between network metrics and ecological concepts, supported by clear figures and tables.	13
[31]	3 -The paper excels in ecological relevance. The paper justifies its graph construction with empirical data (carbon flow networks) and ecological theory.	3 -The paper is rigorous in its theoretical foundations. It clearly defines the use of directed graphs for energy flow networks and justifies the choice of SCCs to identify cycling behaviour.	2 -The paper moderately addresses uncertainty. It tests robustness by pruning weak edges and observing SCC fragmentation, but this is limited to a single ecosystem.	2 -Scalability is moderately addressed. The linear-time algorithms are efficient for the studied networks, but the paper does not test scalability to larger or more dynamic systems.	1 -The study offers moderate utility for stakeholders, but the paper does not translate results into explicit management recommendations.	3 -The paper is highly interpretable. and GT concepts are clearly linked to ecological processes. visualisations and table summaries enhance transparency, making the results accessible to ecologists.	14
[32]	3 -The paper shows high ecological relevance by applying the dominator tree method to the Barents Sea food web, which was constructed from empirical data on species and their interactions.	3 -The paper rigorously justifies its use of GT (dominator trees) with clear references to foundational algorithms. The choice of a rooted network and immediate dominators is well explained.	1 -The paper minimally addresses uncertainty. Although prediction observation discrepancies are noted, uncertainty in the food web structure and dominator tree results is not quantified.	2 -The method is applied to a moderately complex food web, showing scalability to real-world systems. However, the paper does not discuss computational efficiency for larger/dynamic networks.	2 -The paper provides usable findings, but lacks explicit recommendations for conservation. The results can be interpreted by ecologists, but the implementation of management actions is implicit.	3 -The paper excels in interpretability. It clearly defines the abstract concept of a dominator into simple understandable ecological terms.	14
[33]	3 -The paper shows high ecological relevance by constructing food webs from empirical diet data of top pelagic predators in the Pacific Ocean.	3 -The paper uses well-established graph-theoretical methods and justifies the selection. Overall, the work integrates recognised theoretical foundations from ecology and network science throughout.	1 -The study acknowledges uncertainty by testing different link-weight thresholds and removing isolated taxa, but provides very limited formal quantification of uncertainty.	2 -The methods handle moderate-sized networks using aggregation. While the REGE algorithm scales to the data, the paper does not address computational limits for larger or dynamic ecosystems.	3 -The results directly inform ecosystem-based fisheries management, predicting climate change impacts on tuna and billfish.	3 -The paper shows high interpretability. The results are presented with clear tables and explanations, which makes them easier to interpret.	15
[34]	3 -The paper excels in ecological relevance by linking GT to metapopulation dynamics in marine ecosystems, particularly coral reefs.	3 -The paper shows strong theoretical foundations by adapting Drechsler's (2009) metapopulation model to a graph-theoretical framework.	1 -The paper acknowledges variability in edge weights and patch areas but does not fully quantify uncertainty.	3 -The study shows scalability by testing networks of varying sizes (10–160 vertices) and discusses computational trade-offs	2 -The method has promise for practitioners but its practical utility is only moderately showed.	3 -The results on paper can be interpreted and the visualisation of the GBR network increases clarity.	15

Table 11: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[35]	3 -The paper is highly ecologically relevant. The paper deals with a real ecological system and establishes a direct link between connectivity, population resilience and fisheries.	3 -The paper has a strong theoretical foundation, it uses biophysical modelling and connectivity matrices with clear justifications for methodological choices.	2 -The paper acknowledges uncertainties and conducts sensitivity analyses for PLD, spawning timing, and dispersion coefficients but does not quantify it.	2 -The paper shows strong scalability. However, the study does not test or discuss scaling up (e.g., multi-species or larger domains) and is computed for a single 7-year period of one fjord.	3 -The results are directly framed in terms of fisheries management and stock protection.	3 -The paper explains complex graph-theory results in understandable ecological terms. GT results are clearly mapped to ecological concepts.	16
[36]	3 -The work shows high ecological relevance by linking its biophysical and bioenergetic models to the real ecological dynamics of <i>Pinctada margaritifera</i> in French Polynesia.	3 -The work is methodologically rigorous and combines Lagrangian particle tracking with dynamic energy budget (DEB) theory. The choice of graph metrics is well justified, and the DEB model is based on established principles of ecology.	1 -The paper analyses the variability between scenarios (e.g., seasonal, interannual effects), but does not quantify uncertainty (no confidence intervals or error bounds are given).	2 -The model efficiently handles a large spatial scale and 10-year simulations by using ocean parcels for Lagrangian tracking. However, the scalability to more complex systems has not yet been tested.	3 -The study provides direct information for pearl oyster management by identifying connectivity clusters and important atolls for spat collection. The results are translated into actionable insights and the discussion emphasises policy relevance.	3 -The graph-theoretical approach and the metrics (e.g., local retention, self-recruitment) are ecologically interpretable. The results of the DEB model are clearly linked to environmental factors.	
[37]	3 -The paper excels in ecological relevance. The structure of the graph is well reasoned, with the nodes representing coral reef habitats and the edges representing larval dispersal links across ocean currents.	3 -The paper shows strong theoretical foundations, it clearly defining graph concepts and connectivity metrics.	2 -The paper acknowledges uncertainty by testing different PLDs and inter-annual variability, but it lacks explicit quantification of uncertainty	2 -The model handles a large spatial domain and 457 reef sites, demonstrating moderate scalability. The graph-theoretic approach is computationally efficient, but the paper does not discuss limitations for larger datasets.	3 - The results are directly applicable to marine conservation, identifying key stepping-stone reefs and dispersal barriers, and offering actionable insights for prioritising reef protection based on connectivity metrics.	3 -The results are interpretable and graph models (nodes=reefs, edges=larval flows) are explained in plain terms and network metrics (degree, betweenness) are mapped to ecological meanings.	
[38]	3 -The paper excels in ecological relevance. The graph structure is well-justified, with nodes representing hydrothermal vent sites and edges based on species assemblage similarity.	2 -The paper shows strong theoretical foundations by using established graph-theoretic methods and justifying their application to metacommunity dynamics, but could benefit from deeper theoretical justification	1 -The study recognises data limitations (e.g., presence-only species records), but does not consistently quantify uncertainty.	2 -The methods handle the studied region (36 vent sites, 117 species), but the scalability to larger systems is not tested. The paper does not address computational efficiency or potential challenges.	3 -The paper directly supports conservation decisions by identifying key vent sites critical for regional connectivity. Results are translated into actionable insights and align with management needs.	3 -The network representations are highly interpretable, with clear mappings of nodes and edges . Metrics like centrality and modularity are well-explained in ecological terms.	

Table 12: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[39]	3 -The paper excels in ecological relevance by linking GT to metapopulation dynamics in marine systems.	3 -The work builds on established metapopulation and network theory and uses Leslie matrices and spectral graph analyses to derive clear persistence criteria that indicate a strong theoretical foundation.	1 -The paper focuses primarily on deterministic models and does not address parameter uncertainty (e.g., dispersal rates, survival probabilities) or network structure.	2 -The method scales well for theoretical analysis of metapopulation networks using efficient graph-theoretic metrics but does not address computational challenges for very large or dynamic networks.	3 -The results directly inform conservation priorities, such as identifying "critical links" in cyclic components for MPA design.	3 -The results are very easy to interpret and the work avoids black box methods. The relationship between the topology of the network and the viability of the population is transparent.	
[40]	3 -The study excels in ecological relevance. The graph structure (nodes as estuaries, edges as larval dispersal pathways) directly maps to the blue crab metapopulation dynamics.	3 -It justifies the use of GT as a mathematical field that "analyses the encoding and properties of the connections among a set of discrete objects" and offers "many usable metrics for analysis of Lagrangian model output".	2 -The authors acknowledge and partially address sources of uncertainty, but do not fully quantify it.	2 -The model handles a spatially explicit network of 14 estuaries and scales for Lagrangian tracking of millions of particles, but does not address dynamic graphs or multi-layer networks.	3 -The results are clearly linked to stakeholder needs and decision-making. The graph metrics (e.g., betweenness centrality) identify critical estuaries for maintaining connectivity, which can guide MPA design.	2 -The authors present many of their results in simple terms (percentage of retained, settled or oiled larvae), which makes interpretation easier, but full understanding requires some technical background knowledge.	15
[41]	1 -The paper focuses on mathematics, particularly network science and linear algebra. While it notes that large networks appear in biology, it does not explore real ecological systems or issues in detail.	3 - The work is built on solid graph-theoretic and linear-algebraic results. Definitions and measures are connected to classic centrality concepts and spectral GT.	0 -As a purely theoretical and mathematical paper, it does not address empirical data limitations or quantify uncertainty in the way conservation ecology models typically would.	2 - The authors acknowledge large, sparse networks, but the paper lacks complexity analysis or large-scale algorithms, leaving practical scalability to bigger ecological networks uncertain.	0 -The paper's explicit aim is to unify and extend mathematical concepts for network analysis, not to provide direct, actionable recommendations for conservation or management.	2 -The metrics make sense when looking at graphs, but they still feel a bit abstract or hard to relate to for most users.	8
[42]	3 -The paper thoroughly addresses ecological relevance by defining connectivity in terms of biological and ecological dynamics. It discusses direct and indirect methods for quantifying connectivity.	3 -The paper provides a strong theoretical foundation for connectivity research, distinguishing structural, functional, and genetic connectivity, and justifies various modelling approaches based on ecological questions.	2 -The paper acknowledges uncertainties in connectivity research, however while it mentions challenges like spatial autocorrelation and imperfect detection, it does not extensively quantify uncertainty.	2 -The paper discusses scalability in terms of handling large spatial and temporal extents. However, it notes computational challenges for high-dimensional landscapes and does not deeply address solutions.	3 -The paper strongly links connectivity research to conservation management. It contains clear recommendations for the integration of connectivity into operational strategies, such as the prioritisation of habitat patches or corridors.	3 -The paper excels in interpretability by clearly defining connectivity concepts and the use of intuitive metrics. It builds a bridge between theoretical models and ecological applications and makes the logic accessible.	16

Table 13: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[43]	3 -The paper excels in ecological relevance, it directly addresses a crucial aspect of marine conservation: integrating larval dispersal connectivity into MPA network design.	3 -The work is based on GT and metapopulation modelling with a clear mathematical formulation and justifying their use for marine connectivity.	2 -The study acknowledges stochasticity in larval dispersal and uses sensitivity analyses for parameters. However, the uncertainty in the topology of the graph is not fully quantified.	2 -The model handles a 50-site network efficiently and discusses scaling to larger systems. However, dynamic or multi-species networks (common in marine ecosystems) are not addressed.	2 -The paper is designed to provide practical guidance for marine conservation planning. But there is no indication of direct stakeholder engagement in the model's development or validation process.	3 -The study excels at providing clear, accessible interpretations of complex results, translating findings into practical guidance for protected area placement for conservation planners.	15
[44]	3 -The paper excels in ecological relevance, it directly addresses marine ecology by focusing on larval dispersal and MPA network connectivity, key processes for species persistence.	2 -The paper uses GT well and explains its choices clearly, but it does not fully test or validate its graph methods with detailed math.	1 -The study acknowledges uncertainty in dispersal distance estimates (e.g., PLD-to-distance conversion) but does not rigorously quantify it.	2 -The graph-theoretic approach scales well to the Gulf of California's 54 habitat patches, and the methods are computationally feasible for this case. However, the paper does not discuss scalability to larger networks.	3 -The paper excels in stakeholder utility by providing actionable insights for the design of MPAs. It evaluates existing spacing rules and proposes habitat-specific adaptations.	3 -The graph-based approach is highly interpretable as nodes and edges are mapped directly to habitats and connectivity. The figures visually illustrate the connectivity patterns, making it easier for stakeholders to understand.	
[45]	3 -This study is highly ecologically relevant as it addresses the critical issue of MPA network effectiveness, particularly for an overexploited, conservation dependent species.	3 -The paper uses GT rigorously, with clear definitions of nodes MPAs and edges (larval dispersal probabilities). Metrics such as neighbourhood size, cluster identification and betweenness centrality are well chosen for the ecological question.	2 -The paper acknowledges uncertainty through sensitivity analyses and reports variability in connectivity metrics. However, uncertainty is handled in a piecewise way and the study assumes an optimistic scenario.	3 -The study shows scalability by simulating larval dispersal across 115 MPAs over a decade, using high-performance computing. The paper discusses computational constraints but proves the approach is feasible for large-scale systems.	3 -The results directly inform MPA network design, identifying critical nodes and highlighting gaps in larval supply to fished areas.	3 -Model outputs and metrics are clearly defined and interpreted. The visualisations (connectivity matrices, larval export maps) enhance clarity.	17
[46]	3 -This paper is highly ecologically relevant as it directly addresses a key challenge in marine conservation.	3 -The paper shows strong theoretical foundations. It clearly defines the graph type and justifies its construction using established genetic distance metrics. GT concepts are applied appropriately to address ecological questions about connectivity.	1 - The paper acknowledges general challenges in marine connectivity research and limitations of hydrodynamic models, but it does not quantify uncertainty in its core findings or data.	2 -The study handles a moderate-sized network (47 nodes, 1081 edges) efficiently using GT tools. While the methods are scalable to similar or slightly larger systems, the paper does not address computational limits.	2 -The results provide actionable insights for conservation (e.g., identifying resilient clusters and long-distance dispersal pathways). However, the paper does not translate findings into explicit management recommendations.	3 -The paper excels in interpretability. Graph metrics are clearly linked to ecological concepts. The network visualisation and comparison to null models make the results accessible.	

Table 14: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[47]	3 -This paper is highly ecologically relevant as it directly addresses the complexity of ecosystems by focusing on food webs, which describes predator-prey relationships and flow of energy between species.	3 - The paper shows strong mathematical rigour and internal consistency by rigourously applying GT concepts to ecological networks. The authors justify their choices and reference prior work to validate their approach.	1 -The study acknowledges variability in food web construction (e.g., species aggregation biases) but does not quantify uncertainty in graph metrics or edge definitions.	2 -The methods scale moderately well to the 21 food webs analysed, but the study does not address computational challenges for larger or more dynamic networks.	2 -The study identifies structural patterns (e.g., common sub-graphs) that could inform conservation, but it does not translate findings into explicit management recommendations.	3 -The results and the paper are highly interpretable, metrics like clique number and diameter are clearly linked to ecological concepts	14
[48]	3 -The study is highly ecologically relevant as it addresses the critical problem of estimating animal abundance from passive acoustic data, which is useful for conservation in the presence of environmental stress and studying behaviour.	3 -The paper rigourously applies GT to solve the problem of bounding animal counts. The paper cites foundational works and justifies choices.	2 -The paper addresses uncertainty by using confidence intervals for call locations. However, it does not quantify uncertainty in other parameters.	2 - The method scales well for moderate call numbers (e.g., 12 calls in simulations) but does not show performance for very large datasets.	3 -The study has clear and direct utility for conservation and management. It translates graph-theoretic outputs into clear lower-bound estimates, which could inform policy.	3 -The results (minimum number of animals and possible call groupings) are intuitively clear and are illustrated with straightforward examples and figures	16
[49]	3 -The paper directly addresses the critical ecological problem of understanding the relationships between habitat heterogeneity and benthic species diversity, which is linked to the diversity of habitats.	3 -The paper shows strong theoretical foundations by clearly formulating the type of graph and justifying its use for studying species-habitat interactions.	1 -The paper provides only limited quantification of uncertainty. It acknowledges variability in ecological data and uses standardized centrality measures to account for uneven sampling.	2 -The method handles a moderately complex system (442 species, 6 habitats, 146 sites) using graph-theoretic algorithms. However, it does not address scalability challenges for larger or more dynamic systems.	3 -The paper connects its findings to conservation and management, showing clear stakeholder relevance.	3 -The graph-theoretic results are highly interpretable in ecological terms. The paper defines all network metrics in understandable ways.	15

Table 15: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[50]	3 -The paper excels in ecological relevance, this directly addresses a critical ecological and conservation challenge on how overfishing and localized depletion impact the population connectivity of queen conch.	3 -The graph-theoretic approach is well-formulated (directed, weighted edges for larval connectivity) and justified by oceanographic and biological data.	2 -The paper acknowledges uncertainty through bootstrapping and compares two hydrodynamic models. However, it does not provide formal confidence intervals for specific derived network metrics.	2 -The CMS model handles 1359 habitat polygons and multi-year simulations, demonstrating moderate scalability. However, dynamic changes are not tested, and computational limits are unclear.	3 - The results directly inform regional management. The paper translates network metrics into actionable recommendations.	3 -The authors provide detailed explanations of their methods and results, aiding interpretability. The paper clearly links results to genetic studies and management needs.	16
[51]	3 -The paper excels in ecological relevance as it directly addresses a significant real-world ecological problem.	3 -The study builds on well-established theory, using it to assess habitat connectivity. It applies established metrics, such as the proportion of patches in the largest component and the IIC.	1 - The study acknowledges data limitations (e.g., patch detection biases, salinity interpolation) but does not rigorously quantify uncertainty.	2 -The paper shows scalability within a single estuary but does not discuss broader applicability (e.g., to other coastal systems). Dynamic connectivity is not addressed computationally.	3 - The results directly inform conservation (e.g., prioritising areas for water quality improvements to enhance connectivity).	3 -The methods and findings are clearly explained. Key metrics are defined in simple terms and results are interpreted in ecological terms.	15
[52]	3 -The paper shows high ecological relevance by constructing a food web from empirical data. The study justifies its approach with ecological theory and aligns with real-world conservation concerns (e.g., overfishing impacts).	3 -The paper is rigorous in its theoretical foundations, it uses metrics like clustering coefficient, average path length, and topological redundancy.	1 -The paper acknowledges data limitations (e.g., using binary interactions due to inconsistent prey quantification) but does not deeply explore and quantify uncertainty.	2 -The method handles a moderately sized network (139 nodes, 837 edges) using standard GT tools. It is scalable to similar ecosystems but does not discuss computational limits or adaptations for larger, more complex networks.	2 -The results are relevant to fisheries management (e.g., highlighting sharks as low-redundancy species and squids as highly connected). However, does not translate findings into explicit conservation actions or policy recommendations.	3 -The results are well-interpreted in ecological terms, making them accessible. visualisations and straightforward simulations (species removal) enhance interpretability.	14
[53]	3 -The paper shows strong ecological relevance by mapping nodes (habitat mosaics) and edges (fish movement pathways) to real-world ecological units and processes.	3 -The paper has a solid mathematical foundation and uses GT (network analysis) with clear definitions of nodes (habitat mosaics) and edges (movement probabilities).	2 -The paper addresses uncertainty but could be more comprehensive, does not quantify uncertainty in edge weights or node attributes.	2 -The method shows moderate scalability, efficiently handling a regional seascape with static network analysis. Dynamic changes and multi-layer networks are not addressed, limiting scalability for larger or more dynamic systems.	3 -The study excels in stakeholder utility, it compares results directly with existing marine reserves, highlighting gaps while proposing actionable recommendations.	3 -The results and metrics are explained in a clear way, enhancing its interpretability. It thoroughly defines key concepts and terms from both ecology and GT. visualisations make results accessible to non-experts.	16

Table 16: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[54]	3 -The paper shows high ecological relevance by mapping GT to real-world marine conservation issues.	3 -The paper is theoretically well grounded, using established graph-theoretic methods and conservation optimisation. It clearly justifies the use of GT (connectivity via Probability of Connection) and MARXAN for systematic conservation planning.	1 -The study acknowledges uncertainty, but it does not systematically quantify uncertainty in its results	2 -The method scales well to the Gulf of Naples (10454 planning units), using GT algorithms and MARXAN's heuristic approach. However, scalability to larger or more dynamic systems is not tested.	3 -The paper excels in translating results into actionable insights for marine spatial planning. MARXAN outputs identify priority areas, and the inclusion of costs ensures relevance to local stakeholders.	3 -The paper is well-structured and highly interpretable. It clearly defines key ecological and graph-theoretic terms, the results are presented with clarity and visuals increasing interpretability.	
[55]	3 -The paper shows high ecological relevance by modelling seagrass dispersal and connectivity in Torres Strait, a real-world marine ecosystem.	3 -The paper is methodologically rigorous, combining a hydrodynamic model (SLIM) with Lagrangian particle tracking and network analysis. IBMs and connectivity metrics are well justified and grounded in ecological theory.	2 -Key uncertainties are acknowledged and explored but also there are limitations that introduce unquantified error. Thus uncertainty is partially quantified but not exhaustively.	3 -The model handles spatial and temporal complexity, the study discusses extending the approach to other systems.	3 -The study provides actionable insights for marine conservation, such as the optimal number of particles for dispersion modelling and connectivity metrics for the design of protected areas.	3 -The paper excels in interpretability. Network metrics are clearly linked to ecological concepts, visualisations and regression analyses improve understanding.	
[56]	3 -The paper excels in ecological relevance as it directly addresses a critical aspect of marine conservation which is understanding connectivity between MPAs for effective regional management.	3 -The paper builds on established frameworks in biophysical modelling and network analysis. The GT approach is well formulated, with clear justification for using connectivity matrices, self-recruitment rates, and betweenness centrality.	1 -The study minimally addresses uncertainty. While it acknowledges stochasticity in larval dispersal, it does not quantify uncertainty in key parameters or model outputs.	2 -The method shows moderate scalability. The particle-tracking model efficiently simulates millions of larvae over large spatial scales and multi-year periods, using HYCOM's global output. However, the study does not clearly test scalability for more complex scenarios.	3 -The study provides high stakeholder utility. Results are directly applicable to MPA network design, highlighting key "source" and "sink" regions.	3 -The paper is highly interpretable, presenting its methods and results in a clear and logical manner.	15
[57]	3 -The paper excels in ecological relevance by using habitat types and depth ranges as proxies for species connectivity, which aligns well with real-world ecological dynamics. The study links GT to adult movement in marine ecosystems, justifying the choice of nodes and edges.	3 -The paper shows strong theoretical foundations by clearly defining the graph structure and justifying the use of centrality metrics for connectivity analysis. The methodology is internally consistent and aligns with established GT principles.	2 -The paper acknowledges uncertainty, particularly in the limitations section, where it discusses data constraints and assumptions. However, it lacks formal quantification of uncertainty, such as sensitivity analyses.	2 -The method scales moderately, handling 1597 planning units and 16 habitat categories. GT and multiplex networks are computationally feasible, though the paper does not address efficiency or challenges with larger datasets.	3 -The paper strongly supports stakeholder utility by providing actionable insights for MPA network design. The results are translated into clear recommendations (e.g., prioritising connectivity hotspots, upgrading protection for existing MPAs).	3 -The paper achieves high interpretability by using intuitive GT concepts and straightforward metrics. The ecological meaning of the results is transparent, making it accessible to both scientists and managers.	16

Table 17: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[58]	2 -The study shows high ecological relevance, however the method itself relies on social-media hashtags as proxies for human values rather than modelling ecological processes or conditions. Thus it connects to real ecosystems but does not directly incorporate ecological variables.	3 -The methodology is grounded in established theory as it clearly formulates its use of GT and justifies choices (e.g., hashtag networks for CES assessment). The paper provides a clear and justified theoretical basis for its application of GT to social media data for CES assessment.	1 -While data biases are acknowledged and a data volume sensitivity analysis is performed, the paper lacks explicit quantification of uncertainty in the graph-theoretic outputs.	3 -The paper highlights scalability as a key advantage and shows it through its chosen methodology and application across multiple large datasets.	3 -The study is strongly oriented towards practical application and stakeholder relevance in conservation and management. The paper directly links results to stakeholder needs.	3 -The methods and findings are clearly explained, comparisons with photo content analysis enhance transparency. The results are presented with clear network visualisations that facilitate understanding.	15
[59]	3 - The study shows high ecological relevance by accurately representing the biological and ecological dynamics of the Stony Coral Tissue Loss Disease (SCTLD) in the Florida Reef Tract (FRT).	3 -The paper shows strong mathematical rigour. It clearly formulates the modified SIR model equations, including inter and intra-reef transmission rates and a novel smooth step function for the intra-reef "infection" threshold (I_0), which accounts for coral resistance.	1 -The paper acknowledges uncertainty (e.g., in coral resistance thresholds and current parameterizations) but does not rigorously quantify it.	2 -The model handles a large spatial domain (FRT) with high-resolution reef polygons, demonstrating scalability in terms of spatial complexity. However, the paper does not discuss computational efficiency or generalisability to other systems.	3 -The study provides actionable insights for marine conservation, such as identifying neutrally buoyant material as the likely disease vector and highlighting reefs at risk. The results are directly applicable to disease management strategies.	3 -The model is highly interpretable, with intuitive visualisations of connectivity networks and disease spread. The use of GT metrics aligns well with ecological concepts, and the SIR framework is transparent.	15

Table 18: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[60]	3 -The paper excels in ecological relevance as it directly addresses a significant real-world ecological problem by understanding the connectivity of seagrass populations and how multiple disturbances affect their resilience.	3 -The paper shows strong mathematical and conceptual rigour. It clearly formulates the biophysical model (SLIM) for hydrodynamics and particle tracking, which forms the basis for connectivity matrices.	2 -The paper acknowledges variability in propagule dispersal through 68 simulations capturing wind, tide, and current variability. However, it lacks explicit quantification of uncertainty.	2 -The method scales well to the spatial and temporal complexity of the GBR ecosystem. The study shows scalability by analysing large datasets and dynamic oceanographic processes, though it does not address computational efficiency.	3 -The results provide actionable insights for conservation, identifying critical meadows for protection. The study links findings to management strategies, such as prioritising environmental recovery over active restoration in well-connected areas.	3 -The GT approach is highly interpretable, with nodes and edges directly mapping to meadows and dispersal pathways. Metrics like "local retention" and "betweenness centrality" are clearly linked to ecological concepts.	16
[61]	3 -The paper excels in ecological relevance as it directly addresses a real-world problem, which is the decline of coral reef ecosystems in the Western Indian Ocean (WIO) and the role of connectivity in MPA design for resilience.	2 -The paper uses well-established graph-theoretic metrics and Lagrangian particle-tracking models, which are grounded in ecological and oceanographic theory. However, it lacks formal justification for certain assumptions.	1 -The paper notes some modelling assumptions that could bias results, but they do not perform formal uncertainty or sensitivity analyses.	3 -The model handles a large spatial scale (Western Indian Ocean) and multiple PLD scenarios, showing moderate scalability. The authors mention that their approach allows for evaluation of connectivity at a "reasonable computational cost and time", indicating practical scalability.	3 -The study provides clear conservation recommendations, identifying priority reefs for MPAs. The results are directly actionable for policymakers, and the discussion emphasises the practical implications for MPA network design.	3 -The paper excels in interpretability by using clear graph visualisations to explain technical outputs into clear and understandable ecological insights.	15
[62]	3 -The paper excels in ecological relevance as it directly addresses a critical and real-world problem in conservation by accurately inferring connectivity across landscapes to understand population persistence, particularly when ecological data are limited.	3 -The paper justifies the use of social network models (sender-receiver and latent space models) for ecological connectivity based on stochastic network theory. It contrasts these with traditional methods and clearly explains the assumptions and limitations.	2 -The paper addresses uncertainty through cross-validation (randomly removing data to test robustness) and model selection criteria. However, it lacks explicit quantification of uncertainty in network metrics.	2 -The methods are tested on two spatially distinct systems (a small cactus patch network and a larger wetland network) and show that they are applicable across scales. However, the paper does not discuss computational efficiency or challenges of very large networks.	3 -The paper provides direct and actionable insights for conservation managers. It warns against the widespread use of maximum distance construction, which it shows consistently overpredicted connectivity and led to misleadingly high estimates of metapopulation lifetime.	3 -The results are presented and explained in a transparent, understandable way. Key network metrics are clearly defined and related to ecological concepts.	16

Table 19: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[63]	3 -The paper excels in ecological relevance as it directly addresses a significant conservation problem: the effectiveness of MPA networks in stemming biodiversity decline and promoting resilience in European marine ecosystems.	3 -The paper is built on robust theoretical foundations, combining a well-established biophysical modelling approach with GT. The choice of network metrics is appropriate for assessing connectivity, and the model's assumptions are clearly stated.	2 -The paper recognizes sources of uncertainty, noting that assuming passive propagule drift may oversimplify small-scale movements and active larval behaviour, potentially biasing estimates. However, the paper does not quantify the uncertainty of its derived network metrics.	3 -The method scales well to the European marine reserve network, covering large spatial and temporal scales. The use of detailed ocean data and efficient particle-tracking algorithms ensures computational feasibility.	3 -The paper provides highly relevant and actionable recommendations for marine resource managers and conservation planners. It directly identifies large connectivity gaps and highlights regions that require additional marine reserves to function as stepping-stone connectivity corridors.	3 -The paper excels in interpretability, they effectively communicate their complex scientific findings to a broader audience.	17
[64]	3 -The paper shows high ecological relevance as it addresses a crucial issue in conservation ecology by understanding metapopulation persistence in fragmented marine landscapes to inform conservation planning.	3 -The paper is firmly based on established theoretical frameworks. It applies metapopulation theory and uses the dominant eigenvalue of the migration matrix as a quantitative measure of the persistence and growth rate of metapopulations, a concept widely used in ecological modelling.	1 -The study acknowledges several sources of uncertainty and simplification, however the paper does not explicitly present a formal quantification of uncertainty.	2 -The method is scalable but with limitations, it is successfully applied to a large and complex geographical region demonstrating computational feasibility, but dynamic or multi-layer networks are not addressed.	3 -The paper provides highly practical and actionable recommendations for marine resource managers and conservation planners. It directly identifies two important hotspots of local connectivity in southeast Australia that contribute strongly to the persistence of multiple species.	3 -The paper effectively communicates complex scientific concepts and findings. It also uses visual network maps to help interpret results.	15
[65]	3 -The paper excels in ecological relevance as it directly addresses a critical and real-world conservation problem by understanding the spatial ecology of highly mobile marine species, specifically sea turtles, to guide conservation prioritisation and management.	3 -The paper uses GT metrics (e.g., closeness, betweenness) appropriately and justifies their selection for analysing connectivity. It clearly defines the basic elements of GT, such as nodes and links.	1 -The paper acknowledges sampling biases (e.g., uneven species representation) and data limitations but does not rigorously quantify uncertainty in network metrics.	3 -The study handles a large dataset (1235 individuals across six species) and global-scale networks efficiently. GT methods are inherently scalable, and the paper shows this by analysing regional and species-specific subnetworks.	3 -The paper provides highly practical and actionable recommendations for marine conservation and management, such as prioritising areas with high closeness or protecting high-betweenness links.	3 -The results are presented in an interpretable way. Metrics like closeness and betweenness are well-defined and linked to conservation concepts	16

Table 20: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[66]	3 -The paper is ecologically relevant as it addresses a long-standing debate in ecology and biogeography on whether marine ciliate taxa exhibit global dispersal or are structured by habitat and geography.	3 -The paper rigourously justifies its use of GT, detailing the construction of SSNs and the choice of metrics (e.g., assortativity, closeness).	2 -The authors provide comprehensive qualitative discussions of limitations and potential biases, but does not fully quantify uncertainty.	3 -The method handles large datasets (85482 sequences) efficiently, using graph partitioning and scalable tools.	2 -While the study identifies biogeographical patterns and suggests primer design strategies for future surveys, it lacks direct translation into conservation actions.	3 -Network visualisations and intuitive metrics enhance interpretability. The paper clearly links network properties to ecological concepts, making the results accessible.	16
[67]	3 -The paper excels in ecological relevance by linking GT to marine metapopulation dynamics. The study justifies its focus on node self-connections with ecological theory and empirical evidence.	3 -The paper rigourously modifies and introduces network metrics (e.g., Total Flux, Self-influx) to account for node self-connections, with clear mathematical formulations.	2 -The authors assess robustness of their results through systematic sensitivity analyses, but the paper does not include formal statistical uncertainty estimates.	2 - The methods are applied to a 33-node network and tested on 10-node subnetworks, demonstrating feasibility for moderate-sized systems. While the paper does not address computational limits or dynamic networks, the metrics are computationally tractable and could scale to larger systems with similar structures.	3 -The paper directly informs marine reserve design by identifying key nodes for persistence. Results are presented as actionable insights and are linked to conservation goals.	3 -Network metrics are clearly defined and visualised with ecological interpretations. Correlations among metrics are clearly discussed, enhancing understanding of their ecological implications.	16
[68]	3 -The paper excels in ecological relevance as it addresses a crucial challenge in marine conservation by accurately predicting species distributions by incorporating ecological connectivity.	3 -The study is firmly grounded in established ecological theory, in particular GT applied to habitat connectivity. It describes habitat networks as graphs of nodes and edges and uses centrality metrics (betweenness, eigenvector, degree) based on classical definitions.	2 -The paper addresses uncertainty through cross-validation and sensitivity analyses, but no confidence intervals for the prediction are presented or the uncertainty of the model inputs formally applied to the results.	2 -The study applies methods to a moderate-sized network (236–264 habitat patches) and shows computational feasibility. However, scalability to larger or dynamic networks is not clearly tested.	3 -The study offers directly actionable insights for conservation and management. By integrating connectivity into SDMs and identifying hotspots of connectivity and critical habitats based on centrality measures.	3 -The BRT results (relative influence, interaction diagrams) are transparent, and the paper discusses how connectivity interacts with environmental variables (e.g., temperature, depth). The methods and results are also understandable for non-experts.	16

Table 21: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[69]	3 -The paper excels in ecological relevance. The study justifies its graph construction with empirical hydrodynamic data and larval dispersal simulations, aligning closely with ecological theory.	3 - The paper shows strong theoretical foundations. It clearly formulates the graph as directional and weighted, justifying these choices based on larval dispersal dynamics.	2 -The paper moderately addresses uncertainty. It tests robustness by varying larval dispersal parameters and examines yearly hydrodynamic variability. However, it lacks formal uncertainty quantification techniques for graph metrics.	3 -The paper shows its ability to handle large and complex ecological systems by modelling the connectivity of 321 reefs. The authors point out that large and complex network structures can now be processed and visualised in milliseconds, showing scalability and computational efficiency.	3 -The paper provides high stakeholder utility. It translates graph metrics into actionable insights, identifying key reefs for conservation. The results are directly applicable to MPA design, and the discussion emphasises practical implications for GBR management.	3 -The paper achieves a high level of interpretability. The GT concepts are clearly linked to ecological processes. Visualisations and intuitive metrics make the results accessible to ecologists and managers.	17
[70]	3 -The paper excels in ecological relevance as it directly addresses key ecological issues of larval dispersal and connectivity in a vulnerable reef fish, emphasizing implications for marine reserves and fisheries conservation.	3 -The study is firmly grounded in GT and uses directional weighted networks to model larval connectivity. It applies ecological interpretations of metrics such as betweenness centrality and clustering and validates the model with a population graph from genetic data.	3 -The paper shows comprehensive uncertainty quantification by acknowledging assumptions like passive particles and the lack of explicit description of the habitat for larval recruitment.	2 - The method handles the spatial complexity of the Gulf of California well, but scalability to larger or more dynamic systems is not clearly tested. The approach is theoretically scalable, computational demands for larger networks or dynamic graphs are not discussed.	3 -The results are directly applicable to marine conservation, with clear recommendations for MPA design. The study highlights practical implications for fisheries management and provides actionable insights, such as identifying key larval source and sink areas.	3 -The graph-based approach is very easy to interpret, whereby nodes and edges can be clearly assigned to ecological units. Network metrics are intuitively linked to conservation concepts. Visualisations improve understanding.	17
[71]	3 -The paper demonstrates strong ecological relevance by linking graph-theoretic methods to tropical marine fish movement. Bipartite and spatial graphs are justified using empirical telemetry data, with results aligned to species-specific behaviours.	3 -The paper provides a clear rationale for using bipartite and spatial graphs, aligning with ecological questions about connectivity and movement.	1 -The study acknowledges technical challenges (e.g., detection probability variability) but does not rigorously quantify uncertainty in graph structures or metrics.	2 -The methods handle the dataset (31 fish, 39 receivers) efficiently, and the use of R's igraph package suggests scalability to larger arrays. However, the paper does not clearly discuss computational limits or test scalability.	1 -The results could inform marine reserve design (e.g., identifying key habitats via community detection), but the paper does not translate findings into explicit management recommendations.	3 -The paper excels in interpretability. Graph structures (bipartite/spatial) and metrics are clearly explained and mapped to ecological concepts. Figures make the results accessible.	13

Table 22: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[72]	3 -The paper shows high ecological relevance by linking GT to marine conservation. The study justifies its graph construction with empirical data on larval connectivity and reproductive output of <i>Serranus scriba</i> , ensuring alignment with ecological theory.	3 -The paper rigourously applies GT, clearly defining directed, weighted graphs and justifying centrality measures for ecological questions.	1 -The study qualitatively acknowledges several sources of uncertainty and limitations, but does not fully quantify probabilistic uncertainty.	2 -The method handles a regional MPA network (12 nodes) efficiently, but scalability to larger or dynamic systems is not clearly tested. While GT itself is scalable, the coupled biophysical models may face computational limits with broader spatial extents.	3 -The paper excels in translating results into actionable insights, such as identifying key nodes and proposing MPA reconfigurations. Visualisations and clear prioritisation criteria enhance utility for marine managers.	3 -Graph metrics are simply linked to ecological concepts. The paper provides clear explanations of centrality measures and their conservation implications, making the methodology accessible to non-experts.	15
[73]	3 -The paper shows high ecological relevance by defining its graph structures to represent real-world ecological units and processes.	3 -The paper shows strong theoretical foundations. It formulates the graph as a connectivity matrix derived from ocean circulation models and justifies the use of GT metrics to analyse dispersal barriers and recolonization pathways.	2 -The paper acknowledges variability in larval dispersal due to ocean currents and environmental conditions but does not fully quantify uncertainty. While it uses multiple years of data to capture temporal variability, it lacks explicit techniques to assess confidence in connectivity metrics.	2 -The paper shows scalability within the tested region but does not address how the method would perform in larger or more complex systems. The paper does not discuss computational limits or strategies for handling larger datasets.	2 -The paper provides actionable insights but does not fully translate results into explicit management recommendations. The findings could inform MPA design or recolonization strategies, but the link to stakeholder decision-making is implied rather than developed.	3 -The paper is highly interpretable. The GT metrics are easily linked to ecological concepts such as dispersal barriers and population connectivity. The results are presented with clear visualisations and straightforward explanations.	15
[74]	3 -The study excels in ecological relevance. It establishes an explicit link between oceanographic connectivity and genetic diversity in marine forests by using biophysical models to simulate dispersal and GT to visualise connectivity networks.	3 -The paper shows strong theoretical foundations. It employs a biophysical model coupled with GT, justified by prior work in marine connectivity.	2 -The study qualitatively acknowledges several sources of uncertainty and limitations. However, there is no discussion of confidence intervals for network metrics or propagation of uncertainty from input data to the final model results.	2 -The methodology is highly scalable, with a global biophysical model (16489 hexagons) and GT efficiently handling multi-generation connectivity. It is applied to 30 species across different marine regions, though computational costs for finer resolutions or dynamic networks are not discussed.	3 -The paper positions its findings as crucial for guiding conservation efforts and informing spatial conservation planning.	3 -The paper is very easy to interpret. The GT metrics are clearly linked to ecological concepts. Plots (e.g., connectivity networks, dispersal weighting) enhances understanding.	16

Table 23: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[75]	3 -The paper shows high ecological relevance by constructing a symbiotic graph model that accurately represents reef fish communities.	2 -The paper provides a clear formulation of the symbiotic graph model and justifies its choices. However, it lacks formal proofs for the graph kernel method (PMA) and does not discuss convergence properties for the Louvain algorithm.	1 -The study acknowledges data limitations but does not systematically quantify uncertainty in the graph model or PMA method. While it mentions repeated experiments (20 trials) for similarity calculations, it does not provide confidence intervals.	2 -The method handles a multi-year dataset with up to 1427 nodes and 122696 edges, demonstrating moderate scalability. However, the paper does not discuss computational efficiency or performance under larger datasets.	3 -The study directly supports conservation decision-making by identifying priority areas for protection. Results are linked to actionable insights in an easy way to understand.	3 -The graph-based approach is very easy to interpret. The results are visualised using multi-year symbiotic graphs and module subdivisions, which enhances interpretation.	14
[76]	3 -The excels in ecological relevance as it shows comprehensive ecological relevance by focusing on mesozooplankton, which are identified as critical components of marine ecosystems.	2 -The paper clearly formulates the GNN approach, including spectral-temporal GNNs (StemGNN) and transfer entropy for interaction analysis. However, although GNNs are justified for capturing multivariate dependencies, the paper lacks deeper theoretical discussion on why the chosen graph metrics are optimal for ecological forecasting.	2 -The study acknowledges variability in mesozooplankton dynamics but does not systematically quantify uncertainty in model predictions.	2 -The StemGNN model handles multivariate time-series data effectively, but scalability is only moderately addressed. The paper notes computational efficiency improvements over traditional models but does not test the method on significantly larger datasets.	2 -The results could inform marine management, but the translation to actionable conservation strategies is limited. While the GNN's interpretability is highlighted, there are no explicit recommendations for policymakers.	2 -The paper highlights the interpretative power of GNNs and visualises the learned graph to infer variable relationships. Interpretability is discussed qualitatively but not quantified or formally evaluated.	13
[77]	3 -The paper shows high ecological relevance by mapping hydrothermal vent systems and species distributions to real-world ecological dynamics. The study justifies its graph construction with empirical data and aligns with ecological theory.	3 - The paper builds upon strong theoretical foundations by clearly applying GT to examine connectivity among vent systems.	1 -The study acknowledges that sampling effort across vent systems is not consistent and may influence beta-diversity outcomes, but does not rigorously quantify uncertainty.	2 -The methods are scalable to the 11 vent systems studied, but the paper does not address computational efficiency or dynamic network extensions. The analysis is feasible for the dataset but lacks discussion of limitations for larger or more complex systems.	3 - The study excels in stakeholder utility by translating results into actionable conservation recommendations. The network "hub" and endemic-rich systems are clearly identified, providing direct guidance for seabed mining policies.	3 -The paper achieves high interpretability through intuitive network representations and clear ecological mappings. Metrics such as corrected weighted endemism and β -diversity components are well explained and linked to the conservation results.	15

Table 24: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[78]	3 -The paper shows high ecological relevance by linking its GT approach to real-world ecological dynamics. The study justifies its habitat grid and spawning locations using empirical data and aligns with ecological theory on metapopulation connectivity.	3 -The paper has a strong theoretical foundations. It clearly defines the structure of the graph and justifies its choice to answer ecological questions about connectivity and invasion dispersal.	2 -The paper acknowledges uncertainties through sensitivity analyses (e.g., testing scenarios for the duration of planktonic larvae, effects of particle retention) and discusses limitations. Uncertainty is addressed moderately, but not in detail.	2 -The study scales well spatially (180 grid polygons) and temporally (13 generations), but computational constraints are noted (e.g., 7 km hydrodynamic resolution). The paper does not clearly discuss parallelization or approximations for larger systems.	3 -The results directly inform management priorities by identifying highly connected nodes and stable dispersal pathways. The paper translates findings into actionable insights for invasive species control.	3 -The graph metrics are ecologically interpretable, and the paper clearly links them to concepts like source-sink dynamics and biogeographic barriers. The methods are transparent, with detailed descriptions of model inputs and outputs.	16
[79]	3 -The paper excels in ecological relevance as it addresses a relevant ecological problem such as the decline of marine and coastal ecosystems due to human pressures and the need for effective conservation strategies that transcend artificial maritime boundaries.	2 -The paper uses GT and spatial prioritisation with clear justifications for the methodological choices. While the graph-theoretic approach is well justified, the theoretical guarantees are not explored in depth.	1 The study acknowledges several limitations and uncertainties but does not quantitatively assess their impact on the main findings or include sensitivity analyses to evaluate how changes in assumptions might affect the identified connectivity patterns.	3 -The methods scale well to the Western Indian Ocean’s regional scope. The study shows scalability by integrating multi-country EEZs and high seas, though dynamic or multi-layer network complexities.	3 -The paper excels in stakeholder utility by translating results into actionable conservation priorities. The Marxan output directly informs policy negotiations, and the emphasis on minimising fishing conflicts enhances feasibility.	3 -The paper achieves a high level of interpretability. The graph-based metrics are ecologically interpretable, and visualisations clearly link results to conservation targets.	15
[80]	3 -The study excels in ecological relevance by accurately representing coral larval dispersal dynamics using a biophysical model.	3 -The paper shows strong theoretical rigour, it uses a Lagrangian particle tracking algorithm in conjunction with an ocean circulation model (ROMS). GT metrics (e.g., betweenness centrality) are used appropriately to analyse connectivity and their use in a conservation context is clearly justified.	2 -The study acknowledges several limitations and simplifications. While these limitations are clearly identified, the study does not quantitatively assess the impact of these uncertainties on its main findings.	2 - The model handles a regional scale (30 MPAs) efficiently, but scalability to larger or more complex systems is not discussed. The computational demands of the hydrodynamic model (ROMS) are noted, but parallelization or simplification strategies for broader applications are not addressed.	3 -The results directly inform marine conservation by identifying critical MPAs and recommending the inclusion of coastal MPAs in the MCCETP network. The study translates complex connectivity metrics into actionable insights, such as prioritising “stepping-stone” MPAs, which are highly relevant for policymakers.	3 -The results of GT can easily be linked to ecological concepts such as larval dispersal and resilience. Visualisations and clear explanations increase transparency and make the results accessible to those involved.	16

Table 25: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[81]	3 -The paper excels in ecological relevance as it directly addresses a complex problem in marine conservation which is the design of effective MPA networks.	3 -The paper is built on a strong theoretical foundations, integrating several established frameworks. It uses GT to model marine networks, where sites or subpopulations are represented as nodes and connections as directed edges.	2 -The paper acknowledges uncertainty in connectivity data (e.g., stochastic larval dispersal) but does not clearly quantify it.	3 -The method is designed for scalability, moving from brute-force searches (for small networks) to MCMC-based heuristics for larger networks (up to 100 nodes). The paper shows the efficiency by analysing only 0.0% of the possible solutions for a network with 20 nodes.	3 -The paper is highly focused on providing actionable insights for marine conservation and resource managers. It translates results into actionable insights, such as prioritising highly connected clusters or low-cost sites.	3 -The paper achieves a high level of interpretability. The paper provides clear diagrams and discusses structural shifts in optimal solutions.	17
[82]	3 -The paper excels in ecological relevance by linking landscape ecology principles to tropical marine ecosystems. It justifies the graph structure with empirical data and ecological theory.	3 -The paper robustly grounds its methodology in landscape ecology theory, it clearly explains key terms and justifying analytical tools. It discusses scale selection, organism-centered approaches, and the limitations of structural vs. functional connectivity, showing a well-organised and consistent approach.	2 -The paper thoroughly discuss various sources of uncertainty and limitations inherent in studying marine connectivity. However, it lacks explicit techniques to quantify uncertainty in connectivity metrics.	2 -The methods are scalable to larger seascapes, but the paper does not address computational efficiency or dynamic networks. While it suggests broad applicability, scalability challenges are not rigorously tested.	3 -The paper directly links findings to resource management and provides actionable insights. It translates complex connectivity concepts into practical recommendations, enhancing utility for policymakers.	3 -The paper uses simple visualisations and clear ecological analogies. GT and spatial metrics are well explained so that the results are accessible to non-experts.	16
[83]	3 -The paper excels in ecological relevance as it addresses a highly relevant and critical ecological problem in marine systems which is understanding the spatial dynamics of herbivorous fish and their implications for the stability and resilience of herbivory on coral reefs.	3 -The paper shows strong theoretical foundations by clearly defining its graph type and justifying the use of small-world network metrics. The application of Watts & Strogatz's framework is well-reasoned, and the analysis aligns with ecological questions.	2 -The paper acknowledges uncertainty and addresses it through sensitivity analyses. However, it lacks formal quantification for network metrics.	2 -The study scales well to the examined reef system but does not clearly test scalability to larger or more complex ecosystems. The linear reef array simplifies analysis, but the paper does not discuss computational challenges or solutions for broader applications.	3 -The results are highly actionable for stakeholders, offering clear conservation recommendations. The link between network properties and ecosystem collapse risks is clearly highlighted for decision-makers.	3 -The paper achieves high interpretability by using transparent network metrics with clear ecological analogs. Graph visualisations and power-law distributions are well-explained, and the discussion ties results to established concepts.	16

Table 26: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[84]	3 -This paper excels in ecological relevance as it addresses an important area in freshwater ecology, the application of patch-based spatial graphs to understand and quantify habitat connectivity.	3 -The paper shows strong theoretical foundations by rigourously defining graph types and justifying their applicability to freshwater systems. It discusses the suitability of patch-based graphs for dendritic river networks and contrasts them with terrestrial lattice networks.	1 -The paper minimally addresses uncertainty. While it acknowledges temporal variability and fragmentation effects, it lacks explicit techniques to quantify uncertainty such as sensitivity analyses.	3 -The paper fulfills scalability criteria as It shows the method’s applicability across hierarchical scales (microhabitat to watershed levels) and discusses computational feasibility for dendritic networks.	3 -The paper strongly supports stakeholder utility by translating graph-based results into actionable conservation insights. For example, it identifies critical segments for connectivity using betweenness centrality and prioritises habitats for conservation based on the integral index of connectivity.	3 -The paper is very easy to interpret through the use of easy to understand graphical representations and metrics. The hierarchical framework and examples further increase transparency and ensure that stakeholders understand the logic of the model and can trust it.	16
[85]	3 -The paper excels in ecological relevance as it addresses an important ecological problem, which is understanding coral larval dispersal and inter-reef connectivity patterns in the GBR.	3 -The paper shows strong theoretical foundations. It uses an unstructured-grid ocean model (SLIM) to resolve multi-scale hydrodynamics, validated with observational data. The graph-theoretic approach is well-justified for detecting reef communities, and the choice of metrics aligns with ecological questions.	2 -The paper acknowledges several uncertainties and limitations associated with the modelling approach. The paper does not include a comprehensive quantitative uncertainty analysis or sensitivity testing for all potential sources of error.	2 -The method scales well to the GBR’s complexity, simulating more than 1000 reefs with reef-scale resolution. This shows thoughtful scaling to a real-world system, but scalability beyond the studied region isn’t showed.	3 -The results are analysed from a management perspective, the community-specific dispersal distances are compared with the existing MPA distances and it is shown that the connectivity between MPAs is highly species-dependent. These results are directly relevant for conservation planning, indicating that the research has a high utility for stakeholders.	3 -The paper is highly interpretable the results and technical ideas are explained in a way that is easy to understand, making it useful for both scientists and people working in conservation. The visualisations effectively communicate results.	16
[86]	3 -The paper excels in ecological relevance as it addresses real-world ecology problem in metapopulation analysis, which is the mega-patch problem where traditional methods of defining habitat patches often ignore the actual dynamics of interacting subpopulations within seemingly contiguous areas.	3 -The paper shows strong theoretical foundations by using GT (modularity optimisation) and spatial autocorrelation to address the "mega-patch problem." The choice of network metrics is justified for identifying subpopulations, and the method is grounded in ecological theory.	2 -The paper acknowledges variability in population dynamics and tests sensitivity to parameters like the modularity threshold and linkage distance. However, it lacks explicit quantification of uncertainty.	3 -The study shows high scalability by applying its method to a large-scale dataset from the Southern California Bight, using 30-m Landsat pixels as nodes. However, the paper does not discuss computational limits or scalability to larger/more complex systems.	1 -The results could inform marine conservation, but the paper does not clearly translate findings into management recommendations. The modularity approach is conceptually useful, but lacks direct stakeholder engagement or policy-ready output.	3 -The graph-based approach is very easy to interpret. The nodes represent the habitats of the kelp species, and the edges reflect the synchrony of the population dynamics. Metrics such as modularity are explained in simple ecological terms. The visualisations effectively communicate the method and results.	15

Table 27: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[87]	0 -The paper focuses on marine oil spill detection using SAR images, which is an application with significant practical importance for marine ecological protection but does not incorporate any ecological processes or dynamics.	3 -The paper shows a comprehensive understanding and application of established theoretical foundations in DL and GNN.	1 -The paper acknowledges variability in SAR data but does not clearly address uncertainty quantification in its model predictions or in the construction of its graph structure.	2 -The study shows scalability by testing models on large SAR datasets (e.g., 10240×10240 pixels) and reporting computational efficiency. However, it does not clearly discuss challenges or solutions for scaling to even larger datasets or real-time monitoring systems.	3 -The paper strongly emphasises practical applications, such as real-time oil spill monitoring and cross-border collaboration. The results are presented in a way that is actionable for stakeholders. The discussion highlights how the model reduces economic losses and aids ecological protection, directly addressing conservation and management needs.	2 -The GNN-based models are less interpretable, but the paper mitigates this by visualising segmentation results and comparing performance across categories. However, it does not provide tools like feature attribution or decision rules to explain model behaviour, which could improve transparency for non-technical stakeholders.	11
[88]	0 -The paper focuses on SST prediction. While SST is acknowledged as an important factor in the marine environment with major impact on the global ecological environments and climate, the computational method itself does not directly represent biological or ecological dynamics.	3 -The paper provides a comprehensive theoretical foundation for its NGGCN model. It clearly articulates the problem of uneven SST distribution and justifies the need for a non-uniform grid, which is a main novel contribution.	1 -The paper evaluates model performance using standard metrics (RMSE, MAE, R^2) but does not clearly quantify uncertainty in predictions or address data noise. No sensitivity analysis is provided for key parameters.	2 -The method handles non-uniform grids and graph structures, which are computationally more efficient than dense grids for large areas. However there is no discussion of computational limits.	1 -While SST predictions are useful for climate modelling and fisheries, the paper does not translate results into actionable conservation or management insights. No stakeholder engagement or policy recommendations are provided. The focus is on technical performance rather than practical applications.	2 -The paper is moderately interpretable. The non-uniform grid and graph structure are visually interpretable and the GCN/GRU architecture is well described. The DL components lack transparency in how they encode ecological processes.	9
[89]	0 -The study addresses marine oil spill monitoring, an important environmental problem. However, the model itself does not directly represent or simulate biological or ecological dynamics in the sense of living organisms, populations, or their interactions.	3 -The paper provides a comprehensive theoretical foundation for its proposed GCAT network. It clearly justifies the use of GCN for spectral feature extraction and GAT for spatial feature extraction, highlighting their suitability for non-Euclidean data	1 -The study acknowledges potential sources of error related to the limited number of ground object types in hyperspectral images and challenges in the data preprocessing phase. However, the paper offers minimal formal quantification of the uncertainty associated with the model's predictions.	2 -The proposed GCAT model is tested on specific datasets, and its computational efficiency is implied by the use of GPU acceleration. However, the paper does not thoroughly evaluate scalability to larger or more complex marine environments.	2 -The study provides actionable outputs that are useful for marine conservation stakeholders, such as spill responders. However, the translation of results into explicit management recommendations is limited.	1 -The paper describes the model architecture clearly, it does not provide tools to explain how specific features contribute to classifications. Interpretability is further limited by the lack of visualisations or simplified summaries for non-technical stakeholders.	9

Table 28: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[90]	3 -The paper excels in ecological relevance. It focuses on the analysis of fish shoal behaviours, which is a worrying issue for scientists studying fish welfare and stress. It highlights that behaviour is a direct expression of fish welfare and contains a lot of information about living conditions.	2 -The paper provides a clear formulation of the GCN and justifies its use for fish behaviour analysis. However, while the GCN architecture is well-described, the paper lacks deeper theoretical justification for why GCN is superior to other graph-based methods for this specific problem.	2 -The paper acknowledges tracking inaccuracies as a source of uncertainty and tests the model under different tracking scenarios (manual vs. machine tracking). However, it does not clearly quantify uncertainty in the graph structure or model predictions.	2 -The study shows scalability by handling 29505 behavioural graphs with up to 20 nodes (fish) per graph. The GCN architecture is computationally efficient, but the paper does not explore larger-scale applications (e.g., high-density aquaculture systems) or dynamic graph updates over time.	2 -The outputs of the model are directly applicable to aquaculture management (e.g., monitoring the welfare of fish). However, the paper does not translate results into explicit conservation or management recommendations.	3 -The paper shows strong interpretability by using GCN explainability to highlight key features (e.g., dispersion for normal behaviour and posture for abnormal behaviour). Visualising node and edge weights links model decisions to ecological concepts, increasing transparency and usability for aquaculture practitioners.	14
[91]	3 -The paper clearly and directly addresses a significant marine ecological problem: stimulating seaweed growth to enhance ocean carbon sequestration, a form of geoengineering aimed at mitigating atmospheric CO ₂ accumulation.	3 -The paper employs a rigorous MDP formulation and integrates advanced DRL techniques (QR-DQN with deep dueling networks). The choice of algorithms is justified with references to their convergence properties and prior success in energy management.	2 -The paper acknowledges environmental stochasticity (e.g., tides, light intensity) and uses distributional RL (QR-DQN) to model value distributions, which implicitly accounts for uncertainty. However, it lacks explicit quantification for key parameters.	2 -The DRL approach is tested on a specific AUS configuration in Aoshan Bay, and the design of the neural network is adapted to this case. While the method could theoretically be scaled to larger systems, the scalability to more complex or larger marine environments is not shown. Computational efficiency is mentioned but not analysed in detail.	3 -The results offer practical insights for optimising AUS operations, with applications in marine conservation and aquaculture management. Although the DRL methodology is technical, the resulting policies are interpretable and easy to apply, allowing stakeholders to support real-world decision-making without needing DRL expertise.	3 -The paper achieves a high level of interpretability by thoroughly explaining its methodology and clearly linking results to ecological and operational concepts.	16
[92]	3 -The paper demonstrates strong ecological relevance by modelling fish schooling with hydrodynamic interactions. Its RL framework targets ecological objectives such as maintaining formations while minimising effort, directly addressing conservation related questions on collective behaviour.	3 -The paper shows strong theoretical rigour. The RL approach uses Q-learning, a well-established algorithm with proven convergence properties, and the study addresses the Markovian nature of the system.	2 -The study provides minimal explicit quantification of uncertainty, it lacks formal uncertainty quantification (e.g., confidence intervals) for the learned policies or dipole interactions.	2 -The dipole model and RL framework handle moderate-sized schools (up to 100 swimmers) effectively. The authors note computational limits of resolving finer-scale hydrodynamics but do not test scalability to very large groups or heterogeneous environments.	1 -The paper does not provide any direct or actionable insights for marine conservation or resource management stakeholders. Its primary focus is on theoretical insights rather than stakeholder engagement or policy recommendations.	2 -The paper excels in interpretability by clearly explaining its methods and the result, but they are not linked to broader ecological principles.	13

Table 29: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[93]	3 -High ecological relevance for underwater tracking of marine species and phenomena using AV fleets, with agents (vehicles) and environments modelled from real acoustic data and dynamics.	3 -Rigorously applies MARL with Transformer-based MAPPO (TransfMAPPO), defining states, actions, and rewards based on tracking objectives; justifies curriculum learning and GPU acceleration.	2 -Evaluates robustness in sim-to-real transfer but lacks explicit quantification of uncertainties like sensor noise or environmental variability; some sensitivity implied in multi-target scenarios.	3 -Demonstrates scalability to larger fleets (e.g., 5 agents) via GPU environment (30,000x speedup) and invariant policies, tested in high-fidelity sim.	3 -Provides practical pipeline for real-world deployment in marine research, reducing tracking errors to 5m for fast targets.	3 -Interprets results through trajectories and error metrics, visualized in sim; connects to ecological missions like species tracking.	17
[94]	3 -Strong ecological relevance for planktonic navigation in complex flows, modeling microswimmers with partial observations to mimic real marine dispersal.	3 -Critically assesses RL algorithms (Q-Learning, A2C, PPO), defining POMDPs with theoretical quasi-optimal policies for benchmarking.	3 -Quantifies uncertainties via sensitivity to hyperparameters, robustness tests across flows (e.g., turbulence), and comparisons to optimal policies.	3 -Scales to complex flows (e.g., 2D turbulence) with vectorized environments and hyperparameter optimization for efficient training.	3 -Highlights PPO’s superiority for discovering smart strategies, applicable to autonomous robots in oceans.	3 -Interprets policies in terms of navigation efficiency and flow exploitation, using visualizations and performance metrics.	18
[95]	3 -Ecologically relevant for sustainable tuna fishing, modeling purse seine dynamics to predict trajectories for capturing free-swimming schools.	3 -Applies DDQN with defined states (vessel/fish positions), actions (shooting points), and rewards based on catchability and gear behaviour.	2 -Acknowledges variability in fish movements but lacks systematic quantification; evaluates through simulations without explicit sensitivity.	2 -Applied to single vessel scenarios; scalable in principle but not demonstrated for multi-vessel or larger environments.	3 -Supports inexperienced fishers and automation in purse seine operations, improving success rates for FSCs.	3 -Interprets trajectories via gear analysis and catchability, visualized for practical understanding.	16
[96]	3 -High ecological relevance for jellyfish monitoring in marine environments, detecting 12 species to address blooms and surveillance.	3 -Proposes TBFRNet (Transformer-based) with RL module for feature reinforcement, justified by jellyfish morphology and optical challenges.	2 -Discusses robustness in diverse environments but quantification limited to ablation experiments; no extensive uncertainty analysis.	3 -Handles complex datasets with lightweight submodules; scalable for real-time underwater detection.	3 -Advances marine biology and surveillance, providing accurate detection for disaster mitigation.	3 -Interprets via accuracy metrics and visualizations, linking to human-like perception.	17

Table 30: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[97]	3 -High ecological relevance for predicting red tide outbreaks in coastal environments using hyperspectral data, addressing impacts on marine ecosystems and human life with spatial-temporal fusion.	3 -Integrates transformer encoder with GCN, defining graphs from multi-band spectral indices and extracting temporal or topological features for prediction.	2 -Evaluates accuracy with ablation but lacks explicit quantification of uncertainties like data noise or model sensitivity; implies robustness through input periods.	3 -Handles time-series data over interconnected points, scalable to large spatial scales with 92% accuracy on 5-day inputs.	3 -Provides early alarms for maritime agencies, reducing negative impacts on environments and economies.	3 -Interprets features via GCN and transformer, with visualizations of predictions and ablation results.	17
[98]	3 -Strong ecological relevance for forecasting SST in mesoscale-dynamic regions, supporting weather, transport, and fisheries applications from satellite images.	3 -Adapts GNN for oceanographic forecasting, defining graphs from satellite grids and using masked loss for spatial discontinuities.	3 -Quantifies uncertainties via comparisons with baselines, error metrics (e.g., RMSE), and handling data gaps/sparsity.	3 -Scales to regional grids (e.g., Canary Islands), outperforming ConvLSTM in efficiency for large datasets.	3 -Overcomes limitations of numerical models, offering flexible forecasting for ocean management.	3 -Interprets mesoscale capture through visualizations and metrics, linking to oceanographic processes.	18
[99]	3 -High ecological relevance for mapping seagrass in Italian Seas, addressing climate/human impacts for NbS and conservation.	3 -Applies GNNs (GCN, GAT) with spatial graphs, benchmarked against ML methods for distribution modeling.	3 -Includes sensitivity analysis, SHAP values for variable importance, and uncertainty via F1-scores/minority class handling.	3 -Handles 2244 units at 4km resolution, scalable for national seas with consistent predictions.	3 -Guides NbS implementation, policymaking for biodiversity and climate mitigation.	3 -Interprets via maps, SHAP, and variable rankings, explaining drivers like climate/human factors.	18
[100]	3 -Ecologically relevant for fish swimming behaviour recognition in aquaculture, assessing health via skeleton-based features for disease prevention.	3 -Proposes SGCN with ensemble (EMGCM), defining graphs from joints or bones or motions for spatiotemporal extraction.	2 -Validates on dataset with metrics but limited quantification of uncertainties like pose variations; implies robustness via ensemble.	3 -Scales to multi-behaviour dataset, efficient for real-time monitoring with 90.31% accuracy.	3 -Supports practical aquaculture management, health assessment, and optimization.	3 -Interprets via spatiotemporal features and ensemble predictions, with visualizations of skeletons or behaviours.	17

Table 31: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[101]	3 -The paper shows high ecological relevance by addressing a crucial challenge in movement ecology, which is accurately filling gaps in animal tracking data. It applies IRL to GPS trajectories of streaked shearwaters, a marine seabird, to predict realistic movement paths.	3 -The study is rigorous in its theoretical foundations, clearly basing its approach on the well-established Maximum Entropy and IRL framework. It defines the mathematical components of the RL problem, including the MDP formulation.	2 -Uncertainty is moderately addressed. The paper compares IRL to linear interpolation using the MHD but does not quantify uncertainty in the learned reward function or policy	2 -The method scales moderately. The discretization of space and time ensures computational feasibility, but the paper does not test scalability to larger datasets or more complex environments.	3 -The method provides high utility for stakeholders by offering a superior way to fill gaps in animal movement data, which is essential for understanding habitat use, migration patterns, and foraging strategies in animals.	3 -The paper shows high interpretability, the reward maps and feature weights clearly link environmental features to movement preferences. The policy’s avoidance of land aligns with ecological intuition. Visualisations effectively communicate how gaps are filled.	16
[102]	3 -The paper thoroughly addresses ecological relevance by aligning AI and automation technologies with marine conservation goals. It discusses how automated monitoring captures real-world ecological dynamics, such as fish abundance and coral reef health.	2 -The paper provides a good overview of AI techniques and their potential for marine conservation. However, it lacks detailed mathematical formulations, although the paper justifies their choice of methods the theoretical foundations of these techniques are not explored in depth.	1 -The paper acknowledges uncertainties in AI models (e.g., biases in the training data, variability in the marine environment), but does not quantify or address them systematically.	3 -The paper excels in scalability by discussing how AI and automation can handle large-scale marine monitoring. It highlights technologies like edge computing and IoT for real-time data transfer, as well as the cost-effectiveness of scaling DL for processing large amounts of image data.	3 -The paper emphasises the potential of AI to support conservation decision-making through user-friendly interfaces such as dashboards for managers and data-driven insights. It outlines benefits including faster assessment of management outcomes and enhanced support for adaptive management	1 -Interpretability is minimally addressed in this paper. The paper notes the "black box" nature of some AI models (e.g., DL), but does not explore techniques to improve transparency, such as attribution of features or simplified rule sets.	13
[103]	3 -The paper excels in ecological relevance by addressing the need for automated monitoring of marine ecosystems, particularly seabirds (e.g., Northern Gannets), which are indicators of ecological health.	2 -The paper employs a hybrid approach combining supervised ML and RL, with clear justifications for these choices. However, while the ML techniques are well-established, the integration of RL is less detailed.	1 -The paper reports high accuracy metrics (sensitivity, specificity) but does not quantify uncertainty in its results.	2 -The method shows scalability by processing large-scale aerial surveys and handling diverse marine environments. However, the computational demands of training detectors are noted without proposing optimisations for larger datasets.	3 -The study directly supports stakeholders by automating labor-intensive censuses and providing actionable insights (e.g., trends in seabird populations). The user-friendly Matlab interface and high accuracy rates make the tool practical for conservationists.	2 -The ML techniques are interpretable, and the paper contains visual examples of detected objects. However, the decision-making process of the RL component is less transparent and the internal logic of the ensemble approach is not fully explained.	13

Table 32: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[94]	3 - The paper shows high ecological relevance by directly addressing the problem of navigating in fluid flows for autonomous robots deployed in oceans for environmental monitoring.	3 -The paper lays a strong theoretical foundation by clearly formulating the navigation problem as a model-free POMDP, explaining why optimal control theory is insufficient given partial observations.	3 -The paper addresses uncertainty through repeated training trials (five random seeds) and reports confidence intervals for performance metrics.	2 - The paper shows a moderate consideration for scalability by testing flows of increasing complexity. However, it does not discuss computational costs, parallelization, or generalisability to larger-scale oceanic flows.	2 -The paper's results could inform the design of autonomous marine robots. However, it lacks direct translation to conservation or management actions (e.g., habitat monitoring, pollution tracking).	3 -The paper excels in interpretability by effectively using Visualisations (policy plots/heat maps) to show the learned behaviour of the RL agents.	16
[104]	3 -The paper shows strong ecological relevance by addressing a problem faced by planktonic organisms in turbulent marine environments, which is avoiding high-strain regions to minimise the risk of predation.	3 -The paper is mathematically rigorous and uses RL and analytical short-time expansions to derive optimal navigation strategies. The dynamics of the float are based on fluid mechanics, and the RL framework is well justified with clear convergence properties.	2 -The paper does not quantify uncertainty in its RL policies or flow models. While it tests robustness by varying parameters there is no formal confidence intervals.	2 -The methods can be easily transferred to different flow regimes (low/high Reynolds numbers) and the behaviour of swimmers. However, the study does not address the computational costs or convergence times for RL in larger, more complex ecosystems.	2 -The paper is primarily theoretical and cannot be directly applied to conservation or management. While the strategies could inform bioinspired robotics or plankton ecology, the paper does not provide actionable insights for stakeholders (e.g., for MPA design or fishery management).	3 -The paper excels in interpretability by making strong efforts to explain the internal logic and results of the models in ecologically meaningful terms. It presents learned strategies through clear policy Visualisations.	15
[105]	3 -The paper shows strong ecological relevance by directly addressing the problem of diel vertical migration (DVM) in zooplankton within turbulent marine environments.	2 -The study acknowledges stochasticity in training and tests robustness to parameter perturbations. However, it lacks formal uncertainty quantification (e.g., confidence intervals) for the learned policies.	3 -The paper is built on strong theoretical foundations, rigorously framing microswimmer navigation as a MDP. It clearly defines the agent (swimmer), environment (fluid flow), states (local hydrodynamic signals), actions (active reorientation), and reward function (vertical displacement efficiency), ensuring alignment with both RL principles and biological realism.	2 - The model handles 2D and 3D flows efficiently, but scalability to larger or more complex ecosystems is not tested. Tabular Q-learning may face limitations in high-dimensional state spaces.	2 -Although the findings are valuable for understanding plankton behaviour, the results of the study cannot be translated into direct conservation or management recommendations (e.g., habitat protection, climate protection).	3 -The paper excels in interpretability by making significant efforts to explain the internal logic and outcomes of the models in clear, ecologically and physically meaningful terms.	15

Table 33: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[106]	3 -This paper shows high ecological relevance by systematically reviewing graph-based indicators in ecological network analysis, addressing key questions on connectivity, species interactions, and ecosystem dynamics.	3 -The paper shows strong theoretical foundations by rigorously applying graph theory to ecological networks, justifying the use of indicators like Probability of Connectivity and Integral Index of Connectivity.	1 -The paper shows a low level of uncertainty quantification. As a review, it does not discuss or quantify uncertainties, limitations, or sensitivities in the application of the reviewed indicators or methods.	2 -Shows scalability in synthesizing data from diverse ecosystems, but it does not explicitly test or discuss computational limits for applying these indicators to larger, more complex networks or real-time scenarios.	3 -Directly supports ecological management by identifying credible graph-based indexes for conservation planning, with results actionable for prioritizing habitat connectivity in policy and restoration efforts.	3 -The paper excels in interpretability, clearly defining key concepts like "ecological network indicators" and "graph-based indexes," and linking findings to practical ecological applications in an accessible manner.	15
[107]	3 -Shows high ecological relevance by modeling connectivity hotspots for herbivorous reef fishes in a tropical atoll, directly addressing herbivory's role in coral reef resilience and restoration amid climate change and habitat degradation.	3 -Shows strong theoretical foundations by integrating habitat suitability modeling (MaxEnt) with GT for connectivity analysis, justifying species-specific choices based on ecological traits.	1 -The paper shows a low level of uncertainty quantification. It does not explicitly address or quantify uncertainties, limitations, or sensitivities in the modeling approach or results.	3 -The two-step modeling approach (habitat suitability + graph theory) demonstrates high scalability, efficiently handling atoll-scale data and suggesting applicability to larger seascapes or multi-species analyses.	3 -The study offers high stakeholder utility by mapping connectivity hotspots to guide restoration and conservation actions, with results directly informing MPA design and habitat management in island systems.	3 -The paper excels in interpretability, clearly explaining methods like graph-theoretic metrics and linking them to ecological processes (e.g., foraging movements), with visualizations for practical use.	16
[108]	3 -Shows high ecological relevance by addressing landscape connectivity in fragmented habitats, directly informing conservation strategies for organism movement and biodiversity persistence in agricultural and natural systems, with implications for marine analogs like coral reefs.	3 -Strong theoretical foundations by rigorously applying graph theory and network centrality measures, with clear justification for their use in modeling habitat patches and organism flows.	2 -Shows a moderate level of uncertainty quantification. It discusses assumptions in network representations and suggests further empirical validation but does not rigorously quantify uncertainties.	2 -The methods scale well for the studied landscape, but scalability to larger, dynamic marine networks is not tested, and computational demands for factor analysis on bigger graphs are implied but not detailed.	3 -Provides high stakeholder utility by identifying central patches for conservation prioritization, offering actionable insights for land managers in balancing ecological and societal needs.	3 -The paper excels in interpretability, providing clear explanations of complex centrality concepts and their ecological meanings, with visualizations that make results accessible.	16

Table 34: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[109]	3 -Shows high ecological relevance by assessing reinforcement learning for microswimmer navigation in complex flows, with direct implications for understanding planktonic organism movement and marine robotics in ocean monitoring.	3 -Shows strong theoretical foundations by critically evaluating RL algorithms in fluid dynamics contexts, justifying choices with benchmarks and POMDP formulations tailored to marine-like flows.	3 -Shows a high level of uncertainty quantification, rigorously testing robustness across flow types (e.g., turbulence) and incorporating techniques like generalized advantage estimation to handle variability.	3 -The RL implementations (e.g., vectorized environments) demonstrate high scalability, performing well in complex, dynamic flows and suggesting applicability to larger marine systems or multi-agent scenarios.	2 -The study provides moderate stakeholder utility, offering insights for ocean robotics but lacking direct translation to conservation policies or explicit stakeholder engagement.	3 -The paper excels in interpretability, clearly defining RL concepts (e.g., POMDP, reward functions) and linking results to ecological navigation challenges in accessible terms.	17
[110]	3 -Shows high ecological relevance by mapping trophic and non-trophic interactions in a Chilean rocky shore community, addressing how diverse links influence ecosystem resilience and functioning.	3 -The paper shows strong theoretical foundations by integrating food web theory with multiplex networks for non-trophic interactions, justifying the approach with empirical data and trait-based predictions.	2 -The paper shows a moderate level of uncertainty quantification. It mentions exploring consequences of uncertainty in network locations but does not rigorously quantify it through statistical validation or sensitivity tests.	2 -The network analysis scales well for the studied community (100 species), but scalability to larger marine ecosystems or dynamic scenarios is not explicitly addressed or tested.	3 -The study offers high stakeholder utility by providing a framework for predicting community responses to disturbances, actionable for marine conservation and resilience planning.	3 -The paper excels in interpretability, clearly defining interaction types (e.g., trophic vs. non-trophic) and linking network patterns to ecological theory in an accessible way.	16

Table 35: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[111]	3 -High relevance by integrating threat, function, and evolutionary distinctiveness for shark/ray conservation priorities, linking to global MPA strategies.	3 -Strong foundations using CAPTAIN RL algorithm with POMDP for optimized spatial prioritization based on ecological traits.	3 -Rigorous via sensitivity to fishing pressures and random expectations, with conflict analysis for robust outcomes.	3 -Scales to global elasmobranch ranges, handling large datasets and multiple regions effectively.	3 -High utility for policymakers via actionable insights on MPAs and conservation conflicts.	3 -Clear explanations of metrics, priorities, and overlaps for accessible understanding.	18
[112]	3 -High relevance by revealing pearl oyster connectivity patterns via genomics and dispersal models for sustainable management.	3 -Solid foundations integrating graph theory with multi-generational connectivity and coalescent models.	2 -Moderate via graph clustering, but limited sensitivity tests on dispersal variability.	2 -Scales to archipelagos, but not tested for larger oceanic systems.	3 -High utility for defining management units in pearl farming and conservation.	3 -Clear links between networks, genetics, and ecological processes.	16
[113]	3 -High relevance identifying meagre fish migratory corridors and aggregations for spatial planning.	3 -Strong use of graph theory with centrality metrics justified by telemetry data.	2 -Moderate via co-occurrence analysis, but lacks full sensitivity to data noise.	3 -Scales well to coastal networks with efficient spatial analysis.	3 -Useful for fisheries management and MPA design.	3 -Intuitive network visuals and metrics explained ecologically.	17
[114]	3 -High relevance for ocean health monitoring under climate stress using biohybrid-AI sensors.	3 -Strong foundations in hybrid CNN-LSTM and RL for adaptive sensing.	3 -Rigorous with ensemble methods and variance in model performance.	3 -Scales via nanostructured materials and ML for large marine areas.	3 -High utility for real-time conservation and ethical deployment.	3 -Clear model explanations linking to ecological stressors.	18
[115]	3 -High relevance by identifying causal links in food webs for ecosystem functioning.	3 -Strong asymmetry analysis grounded in topology and biomass correlations.	2 -Moderate pairwise correlations, but limited statistical validation.	2 -Scales to 34 webs, but not dynamic/large systems.	3 -Useful for assessing ecosystem health and resilience.	3 -Accessible asymmetry graphs and causality metrics.	16
[116]	3 -High relevance assessing human structures' role in benthic larval connectivity.	3 -Solid particle tracking and network metrics justified ecologically.	3 -Rigorous via simulations over years and residual current analysis.	3 -Scales to North Sea with extensive particle releases.	3 -Utility for decommissioning and invasive species management.	3 -Clear network clustering and connectivity explanations.	18
[117]	3 -High relevance revealing shark communities and risks via hourly bipartite graphs.	3 -Strong Fast-Greedy algorithm with NMI for dynamic networks.	2 -Moderate via telemetry subsets, but limited noise handling.	3 -Scales to 24-hour data and large acoustic arrays.	3 -High utility for bite risk management and public safety.	3 -Intuitive community clusters and fission-fusion dynamics.	17

Table 36: Template for evaluating the theoretical rigour of papers

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[118]	3 -High relevance for coral monitoring with AUVs addressing threats like bleaching.	3 -Strong integration of sensing, ML, and navigation theories.	3 -Rigorous with ensemble GNNs and perturbation tests.	3 -Scales via hierarchical RL and efficient architectures.	3 -Utility for conservation via disease detection and mapping.	3 -Explanations via attribution and heat maps.	18
[119]	3 -High relevance for sailboat navigation in ocean monitoring and exploration.	3 -Strong RL and path planning foundations with hybrid approaches.	3 -Rigorous uncertainty via Monte Carlo and variance reports.	3 -Scales to dynamic environments with efficient algorithms.	3 -Utility for sustainable marine data collection.	3 -Clear policy visualizations and learning explanations.	18
[120]	3 -High relevance clarifying marine functional connectivity for policy and management.	3 -Strong theoretical review integrating ecology and spatial planning.	2 -Moderate via literature synthesis, but no formal uncertainty.	3 -Scales globally across disciplines and ecosystems.	3 -High utility bridging science and stakeholder needs.	3 -Accessible definitions and recommendations.	17
[121]	3 -High relevance quantifying predator movements for seascape connectivity.	3 -Strong graph theory and ML habitat models justified by tracking data.	3 -Rigorous via network metrics and sex-specific analyses.	3 -Scales to remote reefs with broad telemetry networks.	3 -Utility for MPA effectiveness and ecosystem stability.	3 -Clear core use areas and connectivity patterns.	18
[122]	3 -High relevance reviewing ML in fisheries ecology from genomics to ecosystems, aiding conservation.	3 -Strong synthesis justifying ML methods with ecological logic and assumptions.	2 -Moderate discussion of biases and validation, but limited specific quantification.	3 -Covers ML for large datasets in genomics, video, and spatial analysis.	3 -High utility for management decisions, ethics, and stakeholder trust.	2 -Addresses black-box issues moderately, suggests explainability needs.	16

Table 37: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[91]	2 -Seaweed growth and carbon sequestration are genuine ecological goals and environmental variables are integrated into the reward function, but the ecological model is used purely instrumentally with no field validation.	3 -MDP formulation, distributional RL (QR-DQN), and deep dueling networks are rigorously grounded in established RL theory with full mathematical derivation.	2 -The distributional RL framework captures return stochasticity and 95% confidence intervals are reported for training rewards, but uncertainty in ecological outcomes is not quantified.	2 -Demonstrated on a single AUS over four months of simulation; scalability to larger or multisystem deployments is acknowledged as future work but not demonstrated.	1 -Operational guidance is offered implicitly through the Nutrient Transport Index, but no recommendations are made for conservation stakeholders and no co-design with end users is reported.	1 -Q-value distributions give limited insight into action selection but the learned policy is otherwise a black box, with no feature importance or ecological explanation of strategies.	11
[96]	1 -Jellyfish bloom monitoring is ecologically motivated, but the paper delivers bounding boxes only, with no linkage to population dynamics, habitat use, or ecosystem state.	2 -The Transformer backbone and ICA based attention are technically sound, but the use of reinforcement learning is non-standard as the auxiliary network is trained via supervised gradient flow rather than policy optimisation.	0 -No uncertainty quantification is reported at any level: no confidence intervals on mAP metrics, no analysis of detection uncertainty under varying conditions, and no ecological uncertainty considered.	2 -The model is lightweight (11.5 GFLOPs) and efficient, but scalability to continuous underwater monitoring or environments beyond the test dataset is not evaluated.	1 -Surveillance applications are mentioned generically and the dataset is made public, but no connection to management decision making or stakeholder workflows is developed.	2 -Heatmap visualisations clearly show feature activation regions and the ICA decomposition offers a theoretically grounded explanation of Q, K, V channel roles, though no ecological interpretation is offered.	8
[33]	3 -Diet data from three large Pacific regions ground the food webs empirically, and perturbation scenarios are directly tied to projected climate change outcomes for commercially important fisheries species.	3 -Graph theoretic key player analysis, REGE socialnetwork aggregation, and Levins style qualitative modelling are all rigorously applied with explicit mathematical definition and appropriate attribution.	2 -Sign determinacy probabilities are calculated for all perturbation predictions, but parametric and structural uncertainty in diet data and gear type sampling bias are acknowledged rather than formally quantified.	2 -The qualitative modelling approach is parsimonious but depends on extensive diet study databases; scalability to data poor systems or finer temporal resolution is limited.	2 -Predictions are framed explicitly for six commercially important species under ocean warming, accessible to fisheries managers, though operationalised harvest rules or MPA recommendations are not provided.	3 -Qualitative models are inherently interpretable as every prediction traces through explicit signed pathways in the community matrix, and predation tier diagrams make system structure visually transparent.	15
[21]	2 -The review spans food webs, biodiversity networks, MPAs, and ecosystem services across diverse marine contexts, but as a descriptive review no original ecological analysis is performed.	2 -Foundational network concepts and modelling approaches are clearly surveyed and contextualised within ecological network analysis, but theoretical synthesis across frameworks is shallow.	0 -Uncertainty quantification is entirely absent. Data limitations and model accuracy are acknowledged as challenges but no framework for assessing or communicating uncertainty is provided.	2 -Scalability challenges are discussed and future directions include GIS, AI, and remote sensing, but coverage of cross scale issues remains conceptual rather than demonstrated.	2 -MPA design, fisheries management, and ecosystem service assessment are clearly articulated as management applications, though the paper itself does not produce actionable tools.	2 -Network topology metrics are explained with ecological analogies, but the paper notes that many network studies remain predominantly descriptive, a limitation that applies partly to the review itself.	10

Table 38: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[54]	3 -The study uses real habitat maps, empirical connectivity estimates, existing MPA zonations, and actual fishing and exploitation costs, with conservation features tied directly to EU Habitat Directive and IUCN targets.	3 -Graph theoretic Probability of Connection via ConeforSensinode, MARXAN simulated annealing, and permutation tests (PERMANOVA, PERMDISP, MDS) are all applied rigorously with appropriate justification.	2 -Three-fold MARXAN replication and permutation based tests provide uncertainty assessment for scenario comparison, but dispersal distance uncertainty and the expert determined 60% propagule source threshold are not formally quantified.	1 -Explicitly limited to a small, semi enclosed gulf where simplified graph theoretic connectivity is defensible; the authors acknowledge that open coast application would require hydrodynamic or genetic data.	3 -Priority planning units are mapped spatially, gap analyses identify under protected habitats, scenarios minimise costs to local stakeholders, and stakeholder consultation is explicitly recommended as a necessary follow up.	2 -The PC metric and MARXAN selection frequency maps are spatially interpretable, but the connectivity model rests on Euclidean distance and patch area as dispersal proxies whose ecological validity is not fully explored.	14
[6]	2 -Species distributions and migratory behaviour motivate the network design, but biological connectivity is treated separately from governance interactions with limited ecological depth.	2 -Graph theory is competently applied (density, centrality, modularity) with appropriate citations, but the theoretical framing of socio-ecological network science is underdeveloped.	1 -Limitations of self reported data and lack of GIS validation are acknowledged, but no formal uncertainty or sensitivity analysis is conducted.	2 -The trina-tional scope is a strength, but the method is not generalised beyond the study region and transboundary data gaps limit broader application.	2 -Findings are framed around governance gaps and management recommendations, though actionable protocols for cross border cooperation remain vague.	2 -Network visualisations and centrality tables are clear, but the separation of biological and management networks across three countries adds complexity without sufficient synthesis.	11
[71]	2 -Movement ecology of bonefish, barracuda and permit is contextualised against habitat types and bathymetry, but ecological mechanisms driving community structure are not deeply explored.	3 -Six community detection algorithms are systematically compared with modularity scores and Wilcoxon tests; bipartite and spatial graph frameworks are theoretically well grounded.	2 -Detection probability issues and VPS array artefacts are discussed explicitly, but formal error propagation or sensitivity analysis on network metrics is absent.	1 -The study is explicitly exploratory and limited to one small island array; generalisability to other species or regions is noted but not demonstrated.	1 -Conservation and management implications are mentioned only briefly in discussion; the paper is primarily methodological with limited operational guidance.	3 -Bipartite and spatial graphs are clearly illustrated with degree distributions, dendrograms and habitat overlays, making results accessible and visually interpretable.	12
[13]	3 -The food web is reconstructed from 13,934 stomachs across 80 predator species spanning multiple trophic levels, providing strong ecological grounding across the full trophic spectrum.	3 -Small world and scale free structural models are formally tested against 1,000 random networks using Student's t- tests and AIC-based distribution fitting, demonstrating rigorous theoretical engagement.	2 -Random network comparisons provide a statistical baseline, but uncertainty in diet data quality, taxonomic aggregation effects, and binary interaction assumptions are acknowledged but not formally quantified.	2 -The approach is well documented and replicable, but is applied to a single ecosystem with no explicit discussion of how results would transfer to other food webs or management contexts.	2 -Ecosystem based fisheries management implications are discussed concretely, including prioritisation of sharks over shrimp, but operational management recommendations lack specificity.	2 -Network visualisations and centrality tables are informative, but the multiplicity of indices and the distinction between topological uniqueness and degree centrality may challenge non-specialist readers.	14

Table 39: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[7]	3 -Biophysical dispersal modelling incorporates spore biology, kelp density, ENSO variability and seasonal oceanographic dynamics, producing ecologically grounded connectivity estimates across the full species range.	3 -Network analysis is formally integrated with a hydrodynamic IBM, using Infomap clustering, GAMs with AIC selection, and a comprehensive suite of node level centrality metrics with ecological interpretations.	2 -ENSO phase and seasonal variability are explicitly decomposed and GAM model selection is reported; however, propagule decay rate sensitivity and spatial data gaps south of 43°S are flagged but not formally analysed.	3 -The framework spans approximately 6,600 km of coastline across two countries over 11 years, with management units delineated at multiple spatial scales and explicit discussion of application elsewhere.	3 -Results are directly translated into TURF based no-take zone recommendations, adaptive harvest closure timing and conservation prioritisation per management unit, with co-management framing for local fisheries.	3 -Node level metric maps, out-degree visualisations across ENSO phases and seasons, and clearly labelled management unit maps collectively make complex outputs highly accessible to managers and policymakers.	17
[113]	3 -Migratory corridors, feeding aggregations, and spawning grounds of a commercially and ecologically important sciaenid are directly addressed, with explicit links to MPA management and stock conservation; seasonal co-occurrence patterns are interpreted in full ecological context.	2 -Spatial network theory is well grounded with appropriate node/edge definitions, centrality metrics, and node removal analysis, but no temporal threshold between consecutive detections is applied and the choice of array level nodes introduces simplifications that are acknowledged but not fully resolved.	2 -A weighted residency index reduces bias from unequal monitoring periods and permutation tests assess transition heterogeneity, but propagation of tag expiration uncertainty and variable spatial coverage into centrality estimates is not formalised.	2 -The framework scales to 49 individuals, 122 receivers, and 25 arrays over six years and is designed for multiproject integration via the ETN database, but only 9 individuals drove half the transitions, limiting representativeness at population scale.	3 -Results are translated directly into tuna trap quota recommendations, temporal closures in the PNSACV, and marine spatial planning priorities, with explicit cross jurisdictional governance framing and calls for species specific regulation.	2 -Network visualisations with eigenvector and betweenness centrality overlaid on geographic maps are clear, and the abacus co-occurrence plot is effective, but the movement matrix and supplementary figures are dense and would benefit from more integrated narrative explanation.	14
[99]	2 -Seagrass ecosystem services and Mediterranean decline are contextualised thoroughly and key stressors are ecologically motivated, but the static 2017 snapshot limits temporal inference and 4 km resolution coarsens ecologically fine scale processes.	3 -GCN and GAT architectures are formally specified with full mathematical notation and their spectral and spatial foundations are clearly situated in the literature; comparison with RF, SVM, and MLP baselines is methodologically rigorous.	1 -No confidence intervals, prediction uncertainty maps, or cross validation uncertainty estimates are reported for spatial predictions; the Morris sensitivity analysis characterises variable importance but does not propagate model uncertainty into output distributions.	2 -The graph based framework operates on 2,244 grid cells and is described as transferable to other coastal ecosystems, but computational scalability is not benchmarked and temporal extension would require substantially richer data.	2 -Results are linked to EU MSFD targets and the Nature Restoration Law with spatially explicit restoration guidance near urban agglomerations, but practical implementation pathways for regional authorities remain generic.	2 -Error maps comparing RF and GCN predictions spatially are effective at illustrating the added value of spatial context, but the GCN graph construction and attention mechanism are difficult for non-specialist stakeholders to interpret.	12
[117]	3 -Diel fission fusion dynamics, sex structured habitat use, and harbour beach connectivity are interpreted in direct relation to shark bite risk and conservation tradeoffs, with explicit linkage to human activity patterns in Noumea.	2 -Bipartite graph construction, FastGreedy community detection, and NMIbased temporal clustering are clearly motivated and formally defined, but aggregating across all years without accounting for seasonal variation is a theoretical limitation acknowledged but not resolved.	1 -NMI clustering thresholds and dendrogram cut decisions are not validated with sensitivity analyses and no confidence intervals are placed on community membership or risk period estimates; the small and sex imbalanced sample is flagged but its effect on uncertainty is not quantified.	2 -The hourly bipartite framework is designed to scale to other species and locations and the Python/NetworkX implementation supports reuse, but the study covers only 30 individuals and 44 receivers and performance across larger arrays is not assessed.	3 -Risk periods and high risk zones are explicitly mapped onto human activity areas with direct recommendations for real time monitoring, time specific safety protocols, and public awareness campaigns, positioning the framework to inform local authority decision making.	2 -The six step workflow figure and colour coded community maps overlaid on geographic charts are accessible, but bipartite graph visualisations become complex at higher community counts and NMI heatmaps require familiarity with information theoretic concepts.	13

Table 40: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Rel- evance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[111]	3 -Threat sta- tus, functional uniqueness (FUSE), and evolutionary distinctiveness (EDGE2) are integrated for 1,000 species, directly address- ing ecosystem functioning, irreplaceable evolutionary heritage, and overfishing im- pacts across real biogeographic ecoregions.	3 -CAPTAIN re- inforcement learning, FUSE, and EDGE2 are formally speci- fied with explicit reward functions and weighting schemes; statisti- cal validation using beta regres- sion, permutation tests, Jonckheere- Terpstra, and quantile regres- sion is comprehensive and assumption tested.	2 -Null models with 100 randomisa- tion iterations provide spatially explicit SES val- ues and beta regression re- ports standard errors, but uncer- tainty in IUCN range maps, fishing pressure estimates, and CAPTAIN pre- diction variability across epochs are only partially propagated.	3 -The analysis covers 1,000 species across 131,327 global grid cells using a fully repro- ducible R and Python pipeline with publicly archived code; the CAPTAIN framework is ex- plicitly designed for flexible bud- get constraints and transferable reward functions.	3 -Findings are translated into regionally spe- cific conservation priorities, con- flict hotspots, and opportunity areas with ex- plicit alignment to the Kun- mingMontreal 30x30 target, distinguishing proactive low conflict pro- tection from intensive man- agement zones.	2 -Bivariate maps overlaying con- servation priority and fishing effort are informative and pairwise difference maps clarify spatial tradeoffs, but the reinforce- ment learning machinery of CAPTAIN and the discretisation of continuous scores limit in- terpretability for non-quantitative practitioners.	16
[93]	1 -Underwater tracking of ma- rine species and MPA manage- ment are cited as motivations, but the study is eval- uated entirely in simulated envi- ronments with no ecological target behaviour, habi- tat, or species specific context.	3 -The Dec- POMDP framework, MAPPO with clipped surro- gate objective, Transformer self attention ar- chitecture, and curriculum learn- ing progression are all formally specified with full mathemati- cal notation; the simplified dy- namics model is validated against Gazebo with ex- plicit R^2 and MAE reporting.	2 -Training results are averaged across five seeds with standard de- viations reported and Gaussian noise is injected during training to test robustness, but sim-to-real transfer uncer- tainty is assessed only qualitatively and no confi- dence intervals are reported for Gazebo evalua- tion episodes.	3 -JaxLrauv achieves up to 30,000x speedup over Gazebo and is bench- marked across five agent target configurations; the Transformer architecture is ex- plicitly designed to be invariant to fleet size, en- abling curriculum scaling from sin- gle agent to five agent five target scenarios.	1 -The paper is primarily a meth- ods contribution for robotics and MARL researchers; ap- plications to marine science management are mentioned in passing but no tools or decision support products for conservation practitioners are provided.	2 -Training curves, trajectory plots, and emergent co- ordination figures illustrate learned behaviours ac- cessibly, but the Trans- former attention mechanism and curriculum learn- ing dynamics are not visu- alised, limiting mechanistic in- terpretability of the policy.	12
[5]	3 -Models 16 eco- logically and economically im- portant coastal fish species using nursery habi- tat maps and species specific dispersal dis- tances grounded in mark recap- ture literature, directly linking connectivity to recruitment and population per- sistence.	3 -Applies graph theoretic degree centrality with a negative expo- nential dispersal kernel over three generations, incorporating least cost paths through water and a second model layer for anthropogenic disturbance resis- tance.	1 -Disturbance pas- sage probabilities are explicitly stated as as- sumed relative values with no empirical basis, and sensitivity of results to the dispersal kernel parameter or habitat quality thresholds is not tested.	3 -Framework op- erates at 250 m resolution over 49,469 km ² with 791,511 planning units, imple- mented in R and integrated with priori- tizr/Gurobi, making it appli- cable to other regions and species assem- blages.	3 -Directly assesses the existing Swedish MPA network, iden- tifies priority expansion areas, and provides an interactive web application and open GitHub code, making outputs immedi- ately actionable for managers.	3 -Results are presented as spa- tially explicit priority maps and species level dot plots of protec- tion levels against random expecta- tion benchmarks, with clear expla- nation of habitat connectivity ver- sus vulnerable connectivity dis- tinctions.	16

Table 41: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[82]	3 -Comprehensively addresses multi-habitat seascape connectivity for tropical marine fish across ontogenetic stages, linking spatial configuration of coral reef, seagrass, and mangrove patches to movement ecology and community structure.	2 -Reviews structural, potential, and actual connectivity frameworks from landscape ecology and graph theory, but the chapter itself does not implement new quantitative models, remaining primarily conceptual and synthetic.	0 -No uncertainty quantification is performed; the chapter is a review and framework discussion with no empirical analysis, model outputs, or error bounds.	2 -Discusses scale selection and multiscale analysis principles thoroughly, but practical guidance on computational scalability for large marine areas is limited and no demonstration at broad spatial extent is provided.	2 -Explicitly frames findings around MPA design and resource management decisions and identifies key questions for managers, but outputs are conceptual rather than spatially explicit or directly deployable.	2 -Concepts are clearly explained with illustrative figures and tables, but the absence of worked quantitative examples limits the interpretability of the framework for practitioners attempting implementation.	11
[108]	2 -Grounded in a real fragmented landscape with a biologically meaningful target species, but the ecological interpretation of centrality measures remains largely theoretical with limited species specific validation.	3 -Rigorously examines six network centrality measures across three graph representations with factor analysis, Spearman rank correlations, and a novel clumpiness coefficient formally derived.	1 -Patch removal simulations provide some sensitivity analysis, but the exponential dispersal kernel parameters and threshold distances are not varied and no confidence intervals are provided on centrality scores.	2 -Applied to hundreds of patches in Madagascar, but the authors acknowledge computational limitations of some centrality measures and do not demonstrate applicability to larger or more heterogeneous multispecies landscapes.	2 -Results identify which patches to conserve or exploit and frame outputs directly for land managers, but the case study is not linked to any actual conservation planning process or decision outcome.	3 -Two distinct families of centrality are clearly identified, ecologically interpreted, and validated through patch removal experiments, with the distinction well communicated for management use.	13
[12]	3 -Directly addresses domestication driven genetic erosion, trans Pacific dispersal history, and aquaculture extinction risk for <i>Gracilaria chilensis</i> , integrating archaeological, phylogeographic, and population genetic data in an ecologically meaningful framework	3 -Employs multiple complementary methods including Bayesian clustering (STRUCTURE), graph theoretic population graphs, median joining networks, mismatch distributions, Fu's Fs, and bottleneck tests, with appropriate molecular clock calibration.	2 -Confidence intervals are reported for demographic expansion dates and bootstrap resampling is used throughout, but ITS2 mutation rate uncertainty is acknowledged only partially and bottleneck timing cannot be precisely resolved.	1 -Limited to 19 populations across two ocean basins with modest sample sizes; methods are standard and transferable but no discussion of scalability to broader marine domestication or conservation genomics contexts is provided.	2 -Conservation implications for Chilean aquaculture including extinction vortex risk and clonal propagation danger are clearly stated, but no direct management recommendations or policy pathways are offered to industry or regulators.	3 -Results are clearly visualised through pie charts, network graphs, and allele frequency histograms, and the narrative linking archaeological evidence to genetic bottleneck to extinction risk is coherent and accessible.	14
[81]	2 -Motivated by cold water coral and North Sea benthic communities with ecologically justified use of larval connectivity and average shortest path as a resilience proxy, but habitat quality and species interactions are not incorporated.	3 -Formally develops a multi-objective Pareto optimisation framework using MCMC meta-heuristic search, with rigorous mathematical specification of the sample space, objective functions, edge weight to distance conversion, and convergence criteria.	1 -Convergence of the MCMC search is monitored pragmatically but no formal uncertainty bounds on Pareto front estimates are provided, and sensitivity to larval connectivity model assumptions or cost estimate errors is not explored.	3 -Tested on model random geometric graphs from 20 to 50 nodes and applied to a 104 node real world network, with explicit discussion of computational limits and parallel walk architecture enabling extension to larger networks.	3 -Pareto fronts and preferred solutions are directly interpretable for decision makers balancing cost and connectivity and real financial cost estimates are used, making tradeoff analysis immediately actionable.	3 -Step changes in optimal subnetwork structure and convex/concave Pareto front regions are clearly identified and mapped spatially, making the tradeoff surface highly interpretable for conservation planners without specialist optimisation knowledge.	15

Table 42: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Rel- evance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[106]	2 -Covers a wide range of species and ecosystems through 58 ecological network indices, but is a descriptive review with no applied ecological analysis of its own.	2 -Thorough cataloguing of indices via grounded theory with appropriate citations, but limited critical synthesis of how indices relate to one another or to ecological processes.	1 -Only Pearson correlation coefficients and p values are reported to compare indices; no formal uncertainty treatment of index behaviour across ecological contexts is conducted.	0 -Scalability of the reviewed methods across study systems, data types, or spatial extents is not addressed at any point in the review.	1 -A brief future directions discussion is provided but no direct stakeholder engagement or management application is demonstrated.	2 -Index tables and abbreviations are clearly presented, but limited interpretive synthesis of the ecological meaning of differences between indices reduces practical utility.	8
[68]	3 -Two contrasting real species with occurrence data and ecologically meaningful predictors are used, providing strong ecological grounding for the seascape connectivity and SDM integration.	3 -Well grounded combination of network centrality theory and boosted regression tree machine learning, with appropriate model selection and evaluation procedures.	2 -Cross validated AUC scores are reported throughout, but uncertainty in the centrality estimates themselves is not quantified and confidence bounds on partial dependence plots are absent.	1 -Applied to a single regional case study with no testing or discussion of whether the approach generalises to other regions, species, or data resolutions.	2 -Management implications are noted and fishing pressure limitations are discussed, but operationalised recommendations for spatial planning or MPA design are not provided.	3 -Relative influence scores and partial dependence plots give strong interpretability of predictor contributions, making results accessible to ecologists and managers alike.	14
[23]	3 -Four ecological groups differentiated by propagule duration are applied to a real Mediterranean reserve network, with results directly linked to larval supply, settlement, and population connectivity.	3 -Robust integration of biophysical modelling with betweenness and out strength graph theory, grounded in established larval dispersal and network ecology literature.	3 -Ten year interannual simulation explicitly captures temporal variability, and sequential node deletion sensitivity analysis tests the robustness of hub identification to removal of individual reserves.	2 -Applied across 70 reserves in the Mediterranean, though scalability of the biophysical model to larger or data poorer networks is not explicitly tested.	3 -Direct and explicit management recommendations are provided, including identification of priority reserves and reserves requiring improved larval retention, making outputs immediately actionable.	3 -Hub reserves are clearly identified with strong supporting figures and tabulated key nodes, making the connectivity hierarchy transparent and interpretable for marine spatial planners.	17
[14]	2 -Synthesizes many studies on human influence on connectivity across taxa but performs no independent ecological analysis, serving primarily as a supplementary study inventory.	1 -Presented only as a study inventory not theoretical at any level; no derivations, or critical syntheses of the graph theoretic methods viewed literature.	0 -Uncertainty is not addressed at any level; no error bounds, sensitivity analyses, or discussion of methodological consistency across reviewed studies is provided.	1 -Spans many systems and taxa but does not explicitly discuss scalability of methods or how approaches transfer across different data environments.	1 -Could inform management indirectly through the breadth of cases assembled, but no explicit stakeholder engagement, management recommendations, or decision support framing is shown.	1 -Tabular format aids browsing of the literature, but no interpretive framework, synthesis figure, or conceptual model is offered to help readers draw conclusions.	6

Table 43: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[72]	3 -Centrality measures are given explicit ecological interpretations and tested against a real MPA network with results directly linked to larval supply, retention, and population connectivity.	3 -Well grounded in graph theory and larval connectivity literature with a clear methodological roadmap linking model assumptions, threshold choices, or each centrality measure to an ecological concept.	1 -Little formal treatment of uncertainty; no sensitivity analysis of connectivity model assumptions, threshold choices, or confidence intervals is reported.	2 -Demonstrated on a 12 node system with scenario testing of converting OAAs to no-take zones, but not extended to other regions or larger reserve networks.	2 -Includes an explicit management scenario comparing current and proposed MPA configurations, though no direct stakeholder involvement or co-design process is reported.	3 -A detailed rationale table linking each centrality meaning makes the framework highly interpretable and provides a reusable reference for other practitioners.	14
[10]	3 -Covers 11 species with cultural and community input shaping habitat and spawning parameters, providing strong ecological and socio-cultural grounding for the connectivity analysis.	3 -The IBM is thoroughly parameterised with species specific biological data and explicitly grounded in graph theory, applied to identify functional population units.	2 -Limitations around fecundity assumptions, congener PLD estimates, and exclusion of ENSO years are discussed but not formally quantified through sensitivity analysis or confidence bounds.	2 -Analysis is confined to one island over a 2.5-year period, with explicit acknowledgment that broader spatial scale inference requires additional modelling effort.	3 -Direct collaboration with Native Hawaiian practitioners and the local fishing community resulted in clear management support tied to community conservation priorities.	3 -Rich region level visualisations, summary statistics, and community detection outputs support clear interpretation by both technical and non-technical audiences.	16
[88]	1 -Ecological relevance such as fisheries and maritime activities is mentioned in the introduction but not directly assessed; the work focuses on SST prediction accuracy rather than ecological outcomes.	2 -GCN and GRU modules and the gradient grid are clearly described, though the paper is grounded in machine learning methodology rather than ecological theory.	1 -Only point estimate error metrics (RMSE, MAE, R^2) are reported with no uncertainty intervals, ensemble spread, or probabilistic output provided.	2 -Tested across daily, weekly, and monthly timescales but only in one study region, with generalisation to other ocean basins left to future work.	1 -General references to maritime uses are made but no direct stakeholder engagement, usability evaluation, or management application is demonstrated.	2 -Visual comparisons of predictions versus ground truth and time series are provided, though the core GCN/GRU remains largely a black box without mechanistic explanation.	9
[84]	2 -A conceptual review grounded in real riverscape examples, but no primary data collection or empirical validation is performed and ecological depth is limited to illustrative cases.	3 -Thorough grounding in graph theory, hierarchical patch dynamics, and dendritic network model outputs, with appropriate synthesis of the theoretical landscape ecology literature.	0 -No uncertainty treatment of any kind; the paper is purely conceptual with no error bounds or sensitivity analysis.	2 -The framework is explicitly designed to operate across spatial scales from microhabitat to watershed, though practical computational scalability for large systems is not provided.	1 -Conservation applications are mentioned briefly but no direct engagement with managers or operationalisation of recommendations are provided.	2 -Clear conceptual figures and worked illustrative examples aid interpretation, but the absence of a quantitative model limits what practitioners can directly implement.	10
[69]	2 -Uses real larval connectivity data from 321 reefs of the Great Barrier Reef, but the ecological interpretation of the small world property remains somewhat abstract and is not linked to management relevant population outcomes.	3 -Strong basis in graph theory, small world and scale free network theory with formal mathematical derivations and comparison to random graph models.	1 -Sensitivity is tested via edge removal strategies but no statistical multiple parameter variations, error confidence intervals or error bars reported on any network metric.	2 -Analysis spans 32 years of simulated connectivity and scales across a large section of the GBR, though temporal generalisability is not discussed.	1 -Conservation implications for MPA design are noted but remain brief; no direct stakeholder engagement or specific management recommendations are provided.	2 -Centrality and clustering metrics are well explained with supporting figures, though technical graph terminology may limit accessibility for non-specialist readers.	11

Table 44: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Rel- evance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[121]	3 -Directly tied to real predator movement, habitat use, and MPA effectiveness across specific reef systems, with results interpreted in terms of conservation and management relevance.	2 -Draws on established network analysis and connectivity theory with appropriate application, but limited novel theoretical development beyond standard centrality and correlation methods.	1 -Some model fit statistics (R^2 , Spearman correlation) are given but no uncertainty intervals on network metrics themselves are reported and sensitivity of results to tagging effort is not assessed.	2 -Covers two reef systems, multiple species, sexes, and time periods, providing moderate spatial and biological coverage, though extension to broader regional scales is not discussed.	3 -Explicit and detailed management and conservation recommendations are provided, tied directly to MPA design, seasonal closures, and cross habitat protection priorities.	2 -Clear figures and centrality metrics support interpretation, but complex multi-panel network diagrams across species and sexes require considerable effort to synthesise.	13
[116]	3 -Directly addresses offshore structure decommissioning policy and real structure locations with clear management relevance to biodiversity and larval supply in the North Sea.	2 -Builds on established particle tracking and network connectivity theory, applying rather than extending it, with appropriate use of betweenness centrality and connectivity matrices.	2 -Compares two contrasting years and tests particle number and threshold sensitivity, though full statistical uncertainty on network metrics is not reported.	3 -The dataset is explicitly designed to be re-interrogated for any decommissioning structure without rerunning explicit simulations, making the framework highly scalable for policy scenarios.	3 -Directly framed around the OSPAR policy debate with explicit stakeholder relevant scenarios comparing full removal, rigs to reefs, and partial decommissioning outcomes.	2 -Multiple connectivity metrics and spatial maps are well presented, but dense technical content across many figures requires sustained effort to synthesise into actionable conclusions.	15
[11]	2 -Broad ecological relevance is discussed across many taxa and marine and freshwater systems, but the chapter is a review rather than a focused empirical case study with original analysis.	3 -Extensive and rigorous synthesis of landscape and seascape genetic theory, dispersal theory, and graph theoretic approaches, with thorough coverage of the theoretical literature.	1 -Uncertainty issues such as sampling power and IBD failures are discussed conceptually but no quantitative uncertainty treatment is provided within the chapter itself.	2 -Spans marine and freshwater systems at multiple scales, though the contribution is a synthesis chapter rather than a scalable analytical tool.	1 -Implications for conservation management are noted throughout, but the chapter is not a direct stakeholder facing product and no operationalised recommendations are provided.	2 -Well organised with figures and boxed resources, though the very dense and technical content may limit accessibility for non-specialist readers.	11
[90]	2 -Directly targets fish welfare behaviours in aquaculture including normal, resting, obstruction, and circular swimming, but the ecological framing is limited to stress indicators rather than broader population relevance.	3 -Well grounded in GCN theory with explicit graph construction rationale, normal and edge definitions, and systematic comparison of distance graph construction methods.	2 -Precision, recall, and accuracy metrics are reported across train, validation, and test splits, but no formal uncertainty or confidence intervals are provided.	2 -Tested across different tanks, growing periods, and different trackers (FairMOT and CenterTrack) showing some robustness, but confined to a single species and controlled lab setting.	2 -Framed explicitly for aquaculture practitioners and welfare monitoring, with a practicable low weight scheme proposed, though no direct engagement with farmers or production managers is reported.	3 -Includes a dedicated GN-NEExplainer based explanation section with detailed tracking figures showing which node and edge features drive each behaviour classification.	14

Table 45: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[1]	3 -The paper shows high ecological relevance by mapping sediment transport pathways in coastal systems using GT. The nodes (geomorphic cells) and links (sediment fluxes) are well-defined and justified with empirical data.	3 -The paper systematically applies GT, clearly defining directed, weighted networks and justifying choices (e.g., adjacency matrices, modularity, centrality).	2 -The paper acknowledges uncertainties (e.g., grain size variability), but does not systematically quantify them. Sensitivity analyses are implied (e.g., different grain size classes), but not clearly stated.	2 -Method scales well for the case study (25 nodes) but lacks demonstration for larger systems. The paper notes challenges in defining geomorphic cells at broader scales but does not test computational limits or propose optimisations.	2 -The authors link their framework to coastal management questions, indicating stakeholder relevance, but provide no applied examples or direct stakeholder use.	3 -The paper uses clear visualisations (e.g., adjacency matrices, pathway diagrams) and connects metrics to ecological concepts. The discussion contextualizes results in terms of system-wide behaviour and management implications.	15
[2]	3 - The paper excels in ecological relevance. It maps graph structures to real-world marine ecosystems, where nodes represent habitat patches or subpopulations, and edges represent dispersal or connectivity pathways. The authors justify their graph construction with empirical data and ecological theory.	3 -The paper is rigorous in its theoretical foundations. It clearly defines the types of graphs used and justifies these choices based on ecological logic.	2 - The paper acknowledges uncertainty (e.g., variability in dispersal data, model assumptions) but does not extensively quantify it.	2 -The paper shows scalability to some extent, as graph-theoretic methods are inherently efficient for large networks. However, challenges with dynamic or multi-layered networks (e.g., temporal changes in connectivity) are noted but not deeply addressed.	3 -The paper strongly supports stakeholder utility. It translates network metrics (e.g., betweenness centrality) into actionable conservation recommendations (e.g., prioritising key stepping-stone habitats). The visualisations and results are intuitive and directly applicable to MPA design and management.	3 -The paper provides a highly interpretable framework and results. Network metrics are explained in ecological terms, such as using betweenness to identify key dispersal pathways.	16
[3]	3 -The study thoroughly applies GT to real marine ecological systems like the GBR, using empirically-justified nodes and edges to represent actual ecological units and processes.	3 -The methods build on established ecological and GT with clearly justified metrics and derivations.	2 -The paper includes some sensitivity analyses but generally does not rigorously quantify uncertainty.	2 -Shows spatial scalability by analysing the 321-reef GBR and showing logarithmic MLT increase with network size, but temporal scalability is limited as dynamic or multilayer networks for future work.	2 -It targets conservation applications, indicating potential stakeholder relevance. For instance, it "provides a theoretical framework to integrate complex patterns of dispersal connectivity marine conservation planning".	3 -The results are interpretable through ecological reasoning, though the methods are technical meaning they are not easily accessible to non-experts.	15

Table 46: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[4]	3 -Shows high ecological relevance by analysing co-occurrence patterns of white sharks using acoustic telemetry data, mapping social interactions and spatial associations in a coastal ecosystem.	3 -Applies social network analysis rigorously, defining nodes, edges, and metrics like SRI, assortativity, and modularity based on established ethological and graph theory principles.	2 -Acknowledges uncertainties such as tag performance variability and ecological drivers, but lacks systematic quantification or explicit sensitivity analyses.	2 -Scales to 185 individuals and 365880 detections in a case study, but does not demonstrate applicability to larger systems or address computational challenges.	3 -Links findings to conservation implications, such as avoidance behaviours and habitat preferences, providing actionable insights for management of aggregation sites.	3 -Uses clear visualizations (e.g., network graphs) and interprets metrics in ecological terms, discussing implications for behaviour and population dynamics.	16

Table 47: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[8]	3 -The paper excels in ecological relevance by linking GT to marine connectivity studies. It justifies the use of nodes and edges with empirical data from Lagrangian simulations and discusses their ecological implications (e.g., larval dispersal, nutrient transfer).	3 -The paper provides a strong theoretical basis for its methodology. It rigorously defines graph-theoretic concepts and the betweenness measure. It critiques the misuse of transfer probabilities as edge weights in prior studies and proposes a mathematically sound alternative to ensure consistency with shortest-path algorithms.	2 -The paper acknowledges variability in connectivity matrices due to ocean circulation patterns but does not quantify uncertainty (e.g., confidence intervals or sensitivity analyses for edge weights).	2 -The method is scalable to moderately large networks (e.g., 32 nodes in the Gulf of Lion case study) and uses efficient algorithms (Dijkstra, Brandes). However, the paper does not discuss computational limits for larger or dynamic network.	2 -The paper's results could inform marine conservation decisions (e.g., MPA design). However, it does not translate findings into actionable recommendations for stakeholders.	3 -The paper achieves high interpretability by clearly defining nodes, edges, and metrics (e.g., betweenness) in ecological terms. The results are presented with intuitive visualisations and are explained in detail.	15
[9]	3 -Applies GT to relevant ecological problems (HABs, food-webs). - it excels by bridging network theory and marine ecosystem dynamics, making it valuable for both theoretical and applied ecology.	3 -The paper clearly defines GT concepts and applies appropriate network metrics with consistent definitions for ecological questions.	0 -The paper does not quantify uncertainty in data or parameters.	0 -The analysis uses a small hypothetical network (34 species) and does not show or discuss its scalability to larger, complex real-world ecosystems.	0 -The paper does not address practical use by decision-makers or provide management relevant outputs, offering only a general suggestion that network analysis could be “fruitful” for future applications.	3 -The model's logic is transparent, with nodes and edges having direct ecological meaning. Results and indices are explained with clear visuals and descriptions, they are clearly linked to ecological concepts.	9

Table 48: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[15]	3 -The paper addresses a critical ecological challenge, enhancing GBR resilience and recovery from disturbances like cyclones, bleaching, and COTS outbreaks, by directly modeling larval dispersal and connectivity, key processes for ecosystem function.	3 -The paper has strong theoretical foundations. It uses graph-theoretical measures (e.g., outer degree, strength, outer component) to quantify the connectivity of reefs and clearly justifies these choices.	2 -The paper acknowledges significant uncertainties inherent in marine systems, such as the variability and transient nature of demographic connections, and empirical uncertainty surrounding dispersal parameters. They do not fully quantify uncertainty ranges.	3 -The study scales well to the 3800 reefs of the GBR using efficient graph algorithms and high-resolution oceanographic modelling. It handles spatial and temporal complexity (e.g., seasonal spawning events) and shows computational feasibility for large ecosystems.	3 -The paper provides actionable insights for marine management by identifying 112 "robust source reefs" that are critical to GBR resilience. The results directly inform conservation prioritisation, such as protecting these reefs to promote recovery of the entire system.	3 -The paper effectively communicates its complex findings through clear explanations and comprehensive visualisations.	17
[16]	3 -The paper tackles the ecological challenge of designing marine reserve networks under global warming, addressing larval connectivity to support ecosystem services, resilience, and recovery after disturbances.	3 -The paper justifies its use of GT, using centrality metrics (eigenvector centrality, distance-weighted fragmentation) to identify hubs and stepping stones for connectivity. The choice of directed, weighted graphs aligns with asymmetric larval dispersal patterns.	2 -The paper acknowledges uncertainties like limited empirical data for dispersal patterns and the high uncertainty in future ocean warming impacts on larval duration. While addressing some by exploring extreme scenarios, it lacks comprehensive quantification for all identified uncertainties.	2 -The method scales well spatially and handles multi-species connectivity, but dynamic or multi-layer networks are not addressed. Computational constraints are noted (e.g., downscaling coarse connectivity matrices), but are not discussed.	3 -The paper directly supports MPA planning by translating graphical metrics into actionable recommendations (e.g., prioritising upstream nodes). It considers socio-economic costs (opportunity costs for fisheries) and compares trade-offs between different protected area configurations.	3 -The paper is highly interpretable, effectively communicating complex concepts and results.	16

Table 49: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[17]	3 -The paper is highly ecologically relevant, capturing COTS outbreak dynamics and linking nutrient enrichment and hydrodynamic conditions to larval survival and connectivity.	3 -The paper shows strong theoretical foundations. It uses GT to analyse larval connectivity, with clear justifications for the choice of metrics (e.g., cluster coefficient, CI) and their relevance to COTS outbreaks.	2 -The paper acknowledges some uncertainties (e.g., in identifying which flood events trigger plankton blooms, and it references confidence levels in model calibration), but it stops short of a full uncertainty analysis.	1 -The methods (GT and hydrodynamic modelling) are scalable to some extent, as showed by their application across a 30-year period and multiple reefs. However, the paper does not address computational challenges or scalability to larger spatial or temporal scales.	3 -The paper provides clear, actionable insights for marine management. It identifies the reduction of terrestrial nutrient loads (~20–40%) as a feasible measure to mitigate COTS outbreaks, consistent with existing water quality objectives.	3 -The results are very easy to interpret. GT metrics are linked to ecological concepts such as larval retention and local reef connectivity. The paper uses visualisations to illustrate relationships between environmental triggers and outbreaks.	15
[18]	3 -The paper addresses a highly relevant ecological problem on how spatial connectivity influences vital ecosystem functions, specifically herbivory, in the context of disturbances.	3 -The paper has a solid theoretical basis. It uses network theory (closeness centrality) to quantify connectivity, with clear justifications for the construction of graphs (static, undirected patches).	2 -The study addresses uncertainties moderately, considering variability in grazing rates and behavioural interactions, but does not provide confidence intervals or conduct further sensitivity analyses.	2 -The microcosm approach is in principle scalable, but limited by its small, controlled system. The methods could theoretically be scaled to larger systems, but the study does not show this empirically, or discuss the computational challenges for complex, dynamic ecosystems.	2 -The paper contains clear and actionable recommendations for marine managers. It shows that "greater connectivity significantly improves herbivory", which is a crucial ecological function for maintaining ecosystem resilience in disturbed environments.	3 -The paper is very interpretable. Network metrics are clearly linked to ecological concepts (patch accessibility), and the results are presented intuitively (e.g., herbivory rates vs. connectivity). The discussion contextualises the results for ecosystem management.	15
[19]	3 -The paper emphasises the ecological importance of Posidonia oceanica, highlights its role in important coastal services and the urgency of taking action to address its decline. It models realistic ecological processes such as seasonal dispersal using a high-resolution oceanographic framework.	3 -The paper shows strong theoretical foundations. It justifies the use of GT metrics (e.g., In-strength, Betweenness, Eigenvector centrality) to analyse connectivity, which are well-suited for ecological questions about dispersal and metapopulation dynamics.	2 -The paper addresses uncertainty, but could be more comprehensive. Sensitivity analyses are conducted to test the impact of key parameters (e.g., drift duration, coastal interaction types) and interannual variability in dispersal probability is recognised. However, uncertainty is not fully quantified.	2 -The methods are scalable, but have their limits. The graph-theoretic approach efficiently handles the spatial complexity of the Balearic archipelago, and the IBM simulates a large number of particles. However, the study does not address the computational challenges or scalability to larger regions or more complex ecosystems.	3 -The paper provides clear utility for stakeholders. It translates network analysis results into actionable conservation recommendations, such as prioritising Alcudia Bay for restoration due to its high In-strength centrality.	3 -The paper is easy to interpret, the GT metrics (e.g., centrality measures) are ecologically meaningful and clearly linked to concepts such as "stepping stone" patches or settlement hotspots. The methods are transparent and the results are presented in a way that combines technical rigour with practical insights.	16

Table 50: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[20]	3 -The paper shows a high ecological relevance by linking graph-theoretical methods to real-world marine conservation scenarios.	3 -The paper provides a rigorous theoretical foundation for using GT in marine reserve design. It clearly justifies the choice of graph metrics (e.g., Google PageRank, eigenvector centrality) and their ecological interpretations.	2 -The authors address the most important sources of uncertainty and variability in their analysis, but do not analyse the computational scaling in detail.	2 -The methods were applied to moderately large networks and multiple species, suggesting some scalability, but the paper does not analyse the computational limits in depth.	1 -The study makes only minimal connection to stakeholder or management perspectives.	2 -The methods and results are explained with moderate clarity, supported by descriptive summaries and examples, but are still technically complex.	
[22]	3 -The paper effectively connects ecological theory to real-world dynamics in MPAs by mapping nodes and edges to specific species and interactions.	2 -The paper uses GT with clear formulations and reasoned decisions based on ecological logic. However, it lacks mathematical depth as it lacks discussions on limitations and sensitivity analyses.	1 -The study acknowledges data limitations (e.g., reliance on expert validation) but does not systematically quantify uncertainty in edge weights or network metrics.	2 -The paper effectively show scalability by applying its methodology to four distinct MPAs and then integrating them into a "global network", but computational efficiency is not discussed.	3 -The paper excels in translating network results into actionable insights. visualisations and centrality metrics directly inform management priorities.	3 -The results are presented in a clear, visual, and qualitative way.	14

Table 51: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[24]	3 -The paper excels in ecological relevance. It models the dispersal of Sabellaria alveolata larvae, a reef-building polychaete, using a biophysical transport model.	3 -The paper shows strong theoretical foundations. It uses GT and hydrodynamic modelling (MARS3D) with clear justifications for methodological choices.	1 -The study acknowledges several sources of uncertainty and data limitations. However, it does not provide explicit quantitative uncertainty measures for the network metrics derived.	1 -There is no discussion of scaling methodology beyond this case, suggesting that scalability was only considered to a limited extent.	3 -The results directly inform conservation strategies, such as prioritising high-centrality reefs and stepping-stone habitats.	3 - Model outputs are presented and interpreted in clear ecological terms. The results are straightforward to interpret in ecological and conservation contexts.	14
[25]	3 -The paper shows strong ecological relevance by directly addressing a challenge in marine conservation, which is the need to incorporate ecological connectivity, particularly through adult movement, into the design of MPAs between climate change and data limitations.	3 -The paper shows strong theoretical foundations for GT, clearly justifying the use of multiplex networks to represent multiple habitat types and centrality metrics (e.g., betweenness, eigenvector centrality) to assess connectivity.	2 -The paper shows a moderate level of uncertainty quantification. It acknowledges various sources of uncertainty and data limitations inherent in marine ecological modelling but does not carefully quantify uncertainty.	2 -The methods scale moderately well to the regional level, with 100 km planning units and multiplex networks handling 16 habitat categories. The paper does not explore dynamic networks or parallelization, which would be needed for larger/more complex systems.	3 -The paper directly informs the ongoing MPA planning process by translating results into actionable insights (e.g., identifying connectivity hotspots for candidate MPAs).	3 -The paper excels in interpretability, making considerable efforts to clearly explain the internal logic and outcomes of the models in accessible, ecologically meaningful terms.	16
[26]	3 -The paper demonstrates high ecological relevance by directly addressing the critical need for understanding ecosystem connectivity in the design and adaptive management of MPA networks.	3 -The paper shows strong theoretical foundations by carefully applying GT (EDENetworks) and population genetics (FST metrics) with clear justifications for methodological choices. It clearly defines the graph components, with sampling localities as nodes and pairwise genetic differentiation (FST) quantifying connections.	2 -The paper demonstrates a moderate level of uncertainty quantification. It acknowledges limitations (e.g., species-specific idiosyncrasies, autocorrelation between variables) but does not carefully quantify uncertainty.	2 -The paper shows moderate consideration for scalability. The methods are scalable to larger datasets, as evidenced by the integration of multiple species and localities. However, the study focuses on 16 localities and 5 species, leaving open questions about performance with finer-scale or more taxa.	3 -The study provides actionable insights for marine park management, such as identifying key hubs and recommending monitoring priorities. Results are translated into clear conservation implications (e.g., protecting connectivity pathways for climate adaptation).	3 -The paper excels in interpretability, clearly explaining its methodology and findings in terms that are accessible and meaningful within an ecological context.	16

Table 52: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[27]	3 -This paper shows high ecological relevance by directly addressing the critical need for understanding ecosystem connectivity in the context of marine conservation and management, particularly for coral reefs in the GBR.	3 -The paper shows strong theoretical foundations by rigourously applying a biophysical modelling approach coupled with GT. The choice of GT is well-justified for identifying spatially isolated reef clusters.	2 -The paper shows a moderate level of uncertainty quantification. It acknowledges various sources of uncertainty and limitations inherent in its modelling approach but does not rigourously quantify them.	2 -The unstructured-grid model efficiently handles the GBR's spatial complexity, but scalability to larger or more dynamic systems is not fully tested. The Lagrangian model's computational demands are noted, but limitations for very large-scale applications remain.	3 -The study directly supports marine conservation by identifying reef communities and dispersal barriers, which can inform MPA design. The results are presented as spatially explicit maps and connectivity metrics, making them actionable for managers.	3 -The paper excels in interpretability, clearly explaining its complex methodology and findings in accessible and ecologically meaningful terms. It provides clear definitions of key ecological and modelling concepts such as "connectivity network", "larval dispersal", "genetic connectivity", and "demographic connectivity".	16
[28]	3 -The paper is ecologically relevant, addressing how temperature and dispersal shape population connectivity and species range boundaries, key for predicting distributions and assessing ecosystem resilience.	2 -The paper shows strong theoretical foundations by carefully applying biophysical dispersal modelling and GT, but the paper lacks formal justification for choosing specific metrics (e.g., betweenness centrality) over alternatives.	2 -The paper shows a moderate level of uncertainty consideration. It acknowledges inherent variability and influencing factors but does not rigourously quantify uncertainty .	2 -The study shows scalability within the tested system but does not discuss computational limits for larger networks or higher-resolution models.	3 -The paper clearly shows its potential for stakeholder utility by informing conservation and management of species living in variable environments. Results directly inform conservation by identifying critical sites for prioritising monitoring or protection.	3 -The paper excels in interpretability through its clear explanation of methods and direct linkage of results to ecological concepts.	15
[29]	3 -This paper shows high ecological relevance by directly addressing how climate-modulated ocean circulation influences larval transport and population connectivity, which are fundamental processes impacting species distribution and survival.	2 -The paper shows strong theoretical foundations by using Regional Ocean modelling System. It clearly defines the types of graphs used and justifies these choices based on ecological logic.	2 -The study shows a moderate level of uncertainty consideration. It acknowledges that connectivity is influenced by a multitude of factors, but it does not detail the use of formal uncertainty quantification techniques.	2 -The methods show scalability for regional systems. However, the study doesn't discuss computational limits or generalisability to larger/sparser networks.	2 -The paper directly addresses conservation applications and provides actionable insights. Results are translated into management recommendations though stakeholder engagement processes are not detailed.	3 -The paper excels in interpretability through its clear explanation of methods and direct linkage of results to ecological concepts. It defines its core concepts in easily understandable terms.	14

Table 53: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[30]	3 -The paper excels in ecological relevance by linking graph-theoretical concepts with ecological dynamics.	3 -The paper has a solid mathematical foundation and uses GT and linear algebra to develop its ideas. It builds on solid, established theories and concepts.	1 -The paper acknowledges variability in food-web structure but does not quantify uncertainty in its network metrics.	2 -The methods scale well to the 17 food webs analysed, but the paper does not address computational limits for larger or dynamic networks.	1 -The paper identifies functionally integrated species subgroups and discusses conservation implications but does not provide actionable management recommendations.	3 -The graph-theoretic framework is highly interpretable, with clear links between network metrics and ecological concepts, supported by clear figures and tables.	13
[31]	3 -The paper excels in ecological relevance. The paper justifies its graph construction with empirical data (carbon flow networks) and ecological theory.	3 -The paper is rigorous in its theoretical foundations. It clearly defines the use of directed graphs for energy flow networks and justifies the choice of SCCs to identify cycling behaviour.	2 -The paper moderately addresses uncertainty. It tests robustness by pruning weak edges and observing SCC fragmentation, but this is limited to a single ecosystem.	2 -Scalability is moderately addressed. The linear-time algorithms are efficient for the studied networks, but the paper does not test scalability to larger or more dynamic systems.	1 -The study offers moderate utility for stakeholders, but the paper does not translate results into explicit management recommendations.	3 -The paper is highly interpretable. and GT concepts are clearly linked to ecological processes. visualisations and table summaries enhance transparency, making the results accessible to ecologists.	14
[32]	3 -The paper shows high ecological relevance by applying the dominator tree method to the Barents Sea food web, which was constructed from empirical data on species and their interactions.	3 -The paper rigorously justifies its use of GT (dominator trees) with clear references to foundational algorithms. The choice of a rooted network and immediate dominators is well explained.	1 -The paper minimally addresses uncertainty. Although prediction observation discrepancies are noted, uncertainty in the food web structure and dominator tree results is not quantified.	2 -The method is applied to a moderately complex food web, showing scalability to real-world systems. However, the paper does not discuss computational efficiency for larger/dynamic networks.	2 -The paper provides usable findings, but lacks explicit recommendations for conservation. The results can be interpreted by ecologists, but the implementation of management actions is implicit.	3 -The paper excels in interpretability. It clearly defines the abstract concept of a dominator into simple understandable ecological terms.	14
[34]	3 -The paper excels in ecological relevance by linking GT to metapopulation dynamics in marine ecosystems, particularly coral reefs.	3 -The paper shows strong theoretical foundations by adapting Drechsler's (2009) metapopulation model to a graph-theoretical framework.	1 -The paper acknowledges variability in edge weights and patch areas but does not fully quantify uncertainty.	3 -The study shows scalability by testing networks of varying sizes (10–160 vertices) and discusses computational trade-offs	2 -The method has promise for practitioners but its practical utility is only moderately showed.	3 -The results on paper can be interpreted and the visualisation of the GBR network increases clarity.	15

Table 54: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[35]	3 -The paper is highly ecologically relevant. The paper deals with a real ecological system and establishes a direct link between connectivity, population resilience and fisheries.	3 -The paper has a strong theoretical foundation, it uses biophysical modelling and connectivity matrices with clear justifications for methodological choices.	2 -The paper acknowledges uncertainties and conducts sensitivity analyses for PLD, spawning timing, and dispersion coefficients but does not quantify it.	2 -The paper shows strong scalability. However, the study does not test or discuss scaling up (e.g., multi-species or larger domains) and is computed for a single 7-year period of one fjord.	3 -The results are directly framed in terms of fisheries management and stock protection.	3 -The paper explains complex graph-theory results in understandable ecological terms. GT results are clearly mapped to ecological concepts.	16
[36]	3 -The work shows high ecological relevance by linking its biophysical and bioenergetic models to the real ecological dynamics of <i>Pinctada margaritifera</i> in French Polynesia.	3 -The work is methodologically rigorous and combines Lagrangian particle tracking with dynamic energy budget (DEB) theory. The choice of graph metrics is well justified, and the DEB model is based on established principles of ecology.	1 -The paper analyses the variability between scenarios (e.g., seasonal, interannual effects), but does not quantify uncertainty (no confidence intervals or error bounds are given).	2 -The model efficiently handles a large spatial scale and 10-year simulations by using ocean parcels for Lagrangian tracking. However, the scalability to more complex systems has not yet been tested.	3 -The study provides direct information for pearl oyster management by identifying connectivity clusters and important atolls for spat collection. The results are translated into actionable insights and the discussion emphasises policy relevance.	3 -The graph-theoretical approach and the metrics (e.g., local retention, self-recruitment) are ecologically interpretable. The results of the DEB model are clearly linked to environmental factors.	
[37]	3 -The paper excels in ecological relevance. The structure of the graph is well reasoned, with the nodes representing coral reef habitats and the edges representing larval dispersal links across ocean currents.	3 -The paper shows strong theoretical foundations, it clearly defining graph concepts and connectivity metrics.	2 -The paper acknowledges uncertainty by testing different PLDs and inter-annual variability, but it lacks explicit quantification of uncertainty	2 -The model handles a large spatial domain and 457 reef sites, demonstrating moderate scalability. The graph-theoretic approach is computationally efficient, but the paper does not discuss limitations for larger datasets.	3 - The results are directly applicable to marine conservation, identifying key stepping-stone reefs and dispersal barriers, and offering actionable insights for prioritising reef protection based on connectivity metrics.	3 -The results are interpretable and graph models (nodes=reefs, edges=larval flows) are explained in plain terms and network metrics (degree, betweenness) are mapped to ecological meanings.	
[38]	3 -The paper excels in ecological relevance. The graph structure is well-justified, with nodes representing hydrothermal vent sites and edges based on species assemblage similarity.	2 -The paper shows strong theoretical foundations by using established graph-theoretic methods and justifying their application to metacommunity dynamics, but could benefit from deeper theoretical justification	1 -The study recognises data limitations (e.g., presence-only species records), but does not consistently quantify uncertainty.	2 -The methods handle the studied region (36 vent sites, 117 species), but the scalability to larger systems is not tested. The paper does not address computational efficiency or potential challenges.	3 -The paper directly supports conservation decisions by identifying key vent sites critical for regional connectivity. Results are translated into actionable insights and align with management needs.	3 -The network representations are highly interpretable, with clear mappings of nodes and edges . Metrics like centrality and modularity are well-explained in ecological terms.	

Table 55: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[39]	3 -The paper excels in ecological relevance by linking GT to metapopulation dynamics in marine systems.	3 -The work builds on established metapopulation and network theory and uses Leslie matrices and spectral graph analyses to derive clear persistence criteria that indicate a strong theoretical foundation.	1 -The paper focuses primarily on deterministic models and does not address parameter uncertainty (e.g., dispersal rates, survival probabilities) or network structure.	2 -The method scales well for theoretical analysis of metapopulation networks using efficient graph-theoretic metrics but does not address computational challenges for very large or dynamic networks.	3 -The results directly inform conservation priorities, such as identifying "critical links" in cyclic components for MPA design.	3 -The results are very easy to interpret and the work avoids black box methods. The relationship between the topology of the network and the viability of the population is transparent.	
[40]	3 -The study excels in ecological relevance. The graph structure (nodes as estuaries, edges as larval dispersal pathways) directly maps to the blue crab metapopulation dynamics.	3 -It justifies the use of GT as a mathematical field that "analyses the encoding and properties of the connections among a set of discrete objects" and offers "many usable metrics for analysis of Lagrangian model output".	2 -The authors acknowledge and partially address sources of uncertainty, but do not fully quantify it.	2 -The model handles a spatially explicit network of 14 estuaries and scales for Lagrangian tracking of millions of particles, but does not address dynamic graphs or multi-layer networks.	3 -The results are clearly linked to stakeholder needs and decision-making. The graph metrics (e.g., betweenness centrality) identify critical estuaries for maintaining connectivity, which can guide MPA design.	2 -The authors present many of their results in simple terms (percentage of retained, settled or oiled larvae), which makes interpretation easier, but full understanding requires some technical background knowledge.	15
[41]	1 -The paper focuses on mathematics, particularly network science and linear algebra. While it notes that large networks appear in biology, it does not explore real ecological systems or issues in detail.	3 - The work is built on solid graph-theoretic and linear-algebraic results. Definitions and measures are connected to classic centrality concepts and spectral GT.	0 -As a purely theoretical and mathematical paper, it does not address empirical data limitations or quantify uncertainty in the way conservation ecology models typically would.	2 - The authors acknowledge large, sparse networks, but the paper lacks complexity analysis or large-scale algorithms, leaving practical scalability to bigger ecological networks uncertain.	0 -The paper's explicit aim is to unify and extend mathematical concepts for network analysis, not to provide direct, actionable recommendations for conservation or management.	2 -The metrics make sense when looking at graphs, but they still feel a bit abstract or hard to relate to for most users.	8
[42]	3 -The paper thoroughly addresses ecological relevance by defining connectivity in terms of biological and ecological dynamics. It discusses direct and indirect methods for quantifying connectivity.	3 -The paper provides a strong theoretical foundation for connectivity research, distinguishing structural, functional, and genetic connectivity, and justifies various modelling approaches based on ecological questions.	2 -The paper acknowledges uncertainties in connectivity research, however while it mentions challenges like spatial autocorrelation and imperfect detection, it does not extensively quantify uncertainty.	2 -The paper discusses scalability in terms of handling large spatial and temporal extents. However, it notes computational challenges for high-dimensional landscapes and does not deeply address solutions.	3 -The paper strongly links connectivity research to conservation management. It contains clear recommendations for the integration of connectivity into operational strategies, such as the prioritisation of habitat patches or corridors.	3 -The paper excels in interpretability by clearly defining connectivity concepts and the use of intuitive metrics. It builds a bridge between theoretical models and ecological applications and makes the logic accessible.	16

Table 56: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[43]	3 -The paper excels in ecological relevance, it directly addresses a crucial aspect of marine conservation: integrating larval dispersal connectivity into MPA network design.	3 -The work is based on GT and metapopulation modelling with a clear mathematical formulation and justifying their use for marine connectivity.	2 -The study acknowledges stochasticity in larval dispersal and uses sensitivity analyses for parameters. However, the uncertainty in the topology of the graph is not fully quantified.	2 -The model handles a 50-site network efficiently and discusses scaling to larger systems. However, dynamic or multi-species networks (common in marine ecosystems) are not addressed.	2 -The paper is designed to provide practical guidance for marine conservation planning. But there is no indication of direct stakeholder engagement in the model's development or validation process.	3 -The study excels at providing clear, accessible interpretations of complex results, translating findings into practical guidance for protected area placement for conservation planners.	15
[44]	3 -The paper excels in ecological relevance, it directly addresses marine ecology by focusing on larval dispersal and MPA network connectivity, key processes for species persistence.	2 -The paper uses GT well and explains its choices clearly, but it does not fully test or validate its graph methods with detailed math.	1 -The study acknowledges uncertainty in dispersal distance estimates (e.g., PLD-to-distance conversion) but does not rigorously quantify it.	2 -The graph-theoretic approach scales well to the Gulf of California's 54 habitat patches, and the methods are computationally feasible for this case. However, the paper does not discuss scalability to larger networks.	3 -The paper excels in stakeholder utility by providing actionable insights for the design of MPAs. It evaluates existing spacing rules and proposes habitat-specific adaptations.	3 -The graph-based approach is highly interpretable as nodes and edges are mapped directly to habitats and connectivity. The figures visually illustrate the connectivity patterns, making it easier for stakeholders to understand.	
[45]	3 -This study is highly ecologically relevant as it addresses the critical issue of MPA network effectiveness, particularly for an overexploited, conservation dependent species.	3 -The paper uses GT rigorously, with clear definitions of nodes MPAs and edges (larval dispersal probabilities). Metrics such as neighbourhood size, cluster identification and betweenness centrality are well chosen for the ecological question.	2 -The paper acknowledges uncertainty through sensitivity analyses and reports variability in connectivity metrics. However, uncertainty is handled in a piecewise way and the study assumes an optimistic scenario.	3 -The study shows scalability by simulating larval dispersal across 115 MPAs over a decade, using high-performance computing. The paper discusses computational constraints but proves the approach is feasible for large-scale systems.	3 -The results directly inform MPA network design, identifying critical nodes and highlighting gaps in larval supply to fished areas.	3 -Model outputs and metrics are clearly defined and interpreted. The visualisations (connectivity matrices, larval export maps) enhance clarity.	17
[46]	3 -This paper is highly ecologically relevant as it directly addresses a key challenge in marine conservation.	3 -The paper shows strong theoretical foundations. It clearly defines the graph type and justifies its construction using established genetic distance metrics. GT concepts are applied appropriately to address ecological questions about connectivity.	1 - The paper acknowledges general challenges in marine connectivity research and limitations of hydrodynamic models, but it does not quantify uncertainty in its core findings or data.	2 -The study handles a moderate-sized network (47 nodes, 1081 edges) efficiently using GT tools. While the methods are scalable to similar or slightly larger systems, the paper does not address computational limits.	2 -The results provide actionable insights for conservation (e.g., identifying resilient clusters and long-distance dispersal pathways). However, the paper does not translate findings into explicit management recommendations.	3 -The paper excels in interpretability. Graph metrics are clearly linked to ecological concepts. The network visualisation and comparison to null models make the results accessible.	

Table 57: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[47]	3 -This paper is highly ecologically relevant as it directly addresses the complexity of ecosystems by focusing on food webs, which describes predator-prey relationships and flow of energy between species.	3 - The paper shows strong mathematical rigour and internal consistency by rigorously applying GT concepts to ecological networks. The authors justify their choices and reference prior work to validate their approach.	1 -The study acknowledges variability in food web construction (e.g., species aggregation biases) but does not quantify uncertainty in graph metrics or edge definitions.	2 -The methods scale moderately well to the 21 food webs analysed, but the study does not address computational challenges for larger or more dynamic networks.	2 -The study identifies structural patterns (e.g., common sub-graphs) that could inform conservation, but it does not translate findings into explicit management recommendations.	3 -The results and the paper are highly interpretable, metrics like clique number and diameter are clearly linked to ecological concepts	14
[48]	3 -The study is highly ecologically relevant as it addresses the critical problem of estimating animal abundance from passive acoustic data, which is useful for conservation in the presence of environmental stress and studying behaviour.	3 -The paper rigorously applies GT to solve the problem of bounding animal counts. The paper cites foundational works and justifies choices.	2 -The paper addresses uncertainty by using confidence intervals for call locations. However, it does not quantify uncertainty in other parameters.	2 - The method scales well for moderate call numbers (e.g., 12 calls in simulations) but does not show performance for very large datasets.	3 -The study has clear and direct utility for conservation and management. It translates graph-theoretic outputs into clear lower-bound estimates, which could inform policy.	3 -The results (minimum number of animals and possible call groupings) are intuitively clear and are illustrated with straightforward examples and figures	16
[49]	3 -The paper directly addresses the critical ecological problem of understanding the relationships between habitat heterogeneity and benthic species diversity, which is linked to the diversity of habitats.	3 -The paper shows strong theoretical foundations by clearly formulating the type of graph and justifying its use for studying species-habitat interactions.	1 -The paper provides only limited quantification of uncertainty. It acknowledges variability in ecological data and uses standardized centrality measures to account for uneven sampling.	2 -The method handles a moderately complex system (442 species, 6 habitats, 146 sites) using graph-theoretic algorithms. However, it does not address scalability challenges for larger or more dynamic systems.	3 -The paper connects its findings to conservation and management, showing clear stakeholder relevance.	3 -The graph-theoretic results are highly interpretable in ecological terms. The paper defines all network metrics in understandable ways.	15

Table 58: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[50]	3 -The paper excels in ecological relevance, this directly addresses a critical ecological and conservation challenge on how overfishing and localized depletion impact the population connectivity of queen conch.	3 -The graph-theoretic approach is well-formulated (directed, weighted edges for larval connectivity) and justified by oceanographic and biological data.	2 -The paper acknowledges uncertainty through bootstrapping and compares two hydrodynamic models. However, it does not provide formal confidence intervals for specific derived network metrics.	2 -The CMS model handles 1359 habitat polygons and multi-year simulations, demonstrating moderate scalability. However, dynamic changes are not tested, and computational limits are unclear.	3 - The results directly inform regional management. The paper translates network metrics into actionable recommendations.	3 -The authors provide detailed explanations of their methods and results, aiding interpretability. The paper clearly links results to genetic studies and management needs.	16
[51]	3 -The paper excels in ecological relevance as it directly addresses a significant real-world ecological problem.	3 -The study builds on well-established theory, using it to assess habitat connectivity. It applies established metrics, such as the proportion of patches in the largest component and the IIC.	1 - The study acknowledges data limitations (e.g., patch detection biases, salinity interpolation) but does not rigourously quantify uncertainty.	2 -The paper shows scalability within a single estuary but does not discuss broader applicability (e.g., to other coastal systems). Dynamic connectivity is not addressed computationally.	3 - The results directly inform conservation (e.g., prioritising areas for water quality improvements to enhance connectivity).	3 -The methods and findings are clearly explained. Key metrics are defined in simple terms and results are interpreted in ecological terms.	15
[52]	3 -The paper shows high ecological relevance by constructing a food web from empirical data. The study justifies its approach with ecological theory and aligns with real-world conservation concerns (e.g., overfishing impacts).	3 -The paper is rigorous in its theoretical foundations, it uses metrics like clustering coefficient, average path length, and topological redundancy.	1 -The paper acknowledges data limitations (e.g., using binary interactions due to inconsistent prey quantification) but does not deeply explore and quantify uncertainty.	2 -The method handles a moderately sized network (139 nodes, 837 edges) using standard GT tools. It is scalable to similar ecosystems but does not discuss computational limits or adaptations for larger, more complex networks.	2 -The results are relevant to fisheries management (e.g., highlighting sharks as low-redundancy species and squids as highly connected). However, does not translate findings into explicit conservation actions or policy recommendations.	3 -The results are well-interpreted in ecological terms, making them accessible. visualisations and straightforward simulations (species removal) enhance interpretability.	14
[53]	3 -The paper shows strong ecological relevance by mapping nodes (habitat mosaics) and edges (fish movement pathways) to real-world ecological units and processes.	3 -The paper has a solid mathematical foundation and uses GT (network analysis) with clear definitions of nodes (habitat mosaics) and edges (movement probabilities).	2 -The paper addresses uncertainty but could be more comprehensive, does not quantify uncertainty in edge weights or node attributes.	2 -The method shows moderate scalability, efficiently handling a regional seascape with static network analysis. Dynamic changes and multi-layer networks are not addressed, limiting scalability for larger or more dynamic systems.	3 -The study excels in stakeholder utility, it compares results directly with existing marine reserves, highlighting gaps while proposing actionable recommendations.	3 -The results and metrics are explained in a clear way, enhancing its interpretability. It thoroughly defines key concepts and terms from both ecology and GT. visualisations make results accessible to non-experts.	16

Table 59: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[55]	3 -The paper shows high ecological relevance by modelling sea-grass dispersal and connectivity in Torres Strait, a real-world marine ecosystem.	3 -The paper is methodologically rigorous, combining a hydrodynamic model (SLIM) with Lagrangian particle tracking and network analysis. IBMs and connectivity metrics are well justified and grounded in ecological theory.	2 -Key uncertainties are acknowledged and explored but also there are limitations that introduce unquantified error. Thus uncertainty is partially quantified but not exhaustively.	3 -The model handles spatial and temporal complexity, the study discusses extending the approach to other systems.	3 -The study provides actionable insights for marine conservation, such as the optimal number of particles for dispersion modelling and connectivity metrics for the design of protected areas.	3 -The paper excels in interpretability. Network metrics are clearly linked to ecological concepts, visualisations and regression analyses improve understanding.	
[56]	3 -The paper excels in ecological relevance as it directly addresses a critical aspect of marine conservation which is understanding connectivity between MPAs for effective regional management.	3 -The paper builds on established frameworks in biophysical modelling and network analysis. The GT approach is well formulated, with clear justification for using connectivity matrices, self-recruitment rates, and betweenness centrality.	1 -The study minimally addresses uncertainty. While it acknowledges stochasticity in larval dispersal, it does not quantify uncertainty in key parameters or model outputs.	2 -The method shows moderate scalability. The particle-tracking model efficiently simulates millions of larvae over large spatial scales and multi-year periods, using HYCOM's global output. However, the study does not clearly test scalability for more complex scenarios.	3 -The study provides high stakeholder utility. Results are directly applicable to MPA network design, highlighting key "source" and "sink" regions.	3 -The paper is highly interpretable, presenting its methods and results in a clear and logical manner.	15
[57]	3 -The paper excels in ecological relevance by using habitat types and depth ranges as proxies for species connectivity, which aligns well with real-world ecological dynamics. The study links GT to adult movement in marine ecosystems, justifying the choice of nodes and edges.	3 -The paper shows strong theoretical foundations by clearly defining the graph structure and justifying the use of centrality metrics for connectivity analysis. The methodology is internally consistent and aligns with established GT principles.	2 -The paper acknowledges uncertainty, particularly in the limitations section, where it discusses data constraints and assumptions. However, it lacks formal quantification of uncertainty, such as sensitivity analyses.	2 -The method scales moderately, handling 1597 planning units and 16 habitat categories. GT and multiplex networks are computationally feasible, though the paper does not address efficiency or challenges with larger datasets.	3 -The paper strongly supports stakeholder utility by providing actionable insights for MPA network design. The results are translated into clear recommendations (e.g., prioritising connectivity hotspots, upgrading protection for existing MPAs).	3 -The paper achieves high interpretability by using intuitive GT concepts and straightforward metrics. The ecological meaning of the results is transparent, making it accessible to both scientists and managers.	16

Table 60: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[58]	2 -The study shows high ecological relevance, however the method itself relies on social-media hashtags as proxies for human values rather than modelling ecological processes or conditions. Thus it connects to real ecosystems but does not directly incorporate ecological variables.	3 -The methodology is grounded in established theory as it clearly formulates its use of GT and justifies choices (e.g., hashtag networks for CES assessment). The paper provides a clear and justified theoretical basis for its application of GT to social media data for CES assessment.	1 -While data biases are acknowledged and a data volume sensitivity analysis is performed, the paper lacks explicit quantification of uncertainty in the graph-theoretic outputs.	3 -The paper highlights scalability as a key advantage and shows it through its chosen methodology and application across multiple large datasets.	3 -The study is strongly oriented towards practical application and stakeholder relevance in conservation and management. The paper directly links results to stakeholder needs.	3 -The methods and findings are clearly explained, comparisons with photo content analysis enhance transparency. The results are presented with clear network visualisations that facilitate understanding.	15
[59]	3 - The study shows high ecological relevance by accurately representing the biological and ecological dynamics of the Stony Coral Tissue Loss Disease (SCTLD) in the Florida Reef Tract (FRT).	3 -The paper shows strong mathematical rigour. It clearly formulates the modified SIR model equations, including inter and intra-reef transmission rates and a novel smooth step function for the intra-reef "infection" threshold (I0), which accounts for coral resistance.	1 -The paper acknowledges uncertainty (e.g., in coral resistance thresholds and current parameterizations) but does not rigorously quantify it.	2 -The model handles a large spatial domain (FRT) with high-resolution reef polygons, demonstrating scalability in terms of spatial complexity. However, the paper does not discuss computational efficiency or generalisability to other systems.	3 -The study provides actionable insights for marine conservation, such as identifying neutrally buoyant material as the likely disease vector and highlighting reefs at risk. The results are directly applicable to disease management strategies.	3 -The model is highly interpretable, with intuitive visualisations of connectivity networks and disease spread. The use of GT metrics aligns well with ecological concepts, and the SIR framework is transparent.	15

Table 61: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[60]	3 -The paper excels in ecological relevance as it directly addresses a significant real-world ecological problem by understanding the connectivity of seagrass populations and how multiple disturbances affect their resilience.	3 -The paper shows strong mathematical and conceptual rigour. It clearly formulates the biophysical model (SLIM) for hydrodynamics and particle tracking, which forms the basis for connectivity matrices.	2 -The paper acknowledges variability in propagule dispersal through 68 simulations capturing wind, tide, and current variability. However, it lacks explicit quantification of uncertainty.	2 -The method scales well to the spatial and temporal complexity of the GBR ecosystem. The study shows scalability by analysing large datasets and dynamic oceanographic processes, though it does not address computational efficiency.	3 -The results provide actionable insights for conservation, identifying critical meadows for protection. The study links findings to management strategies, such as prioritising environmental recovery over active restoration in well-connected areas.	3 -The GT approach is highly interpretable, with nodes and edges directly mapping to meadows and dispersal pathways. Metrics like "local retention" and "betweenness centrality" are clearly linked to ecological concepts.	16
[61]	3 -The paper excels in ecological relevance as it directly addresses a real-world problem, which is the decline of coral reef ecosystems in the Western Indian Ocean (WIO) and the role of connectivity in MPA design for resilience.	2 -The paper uses well-established graph-theoretic metrics and Lagrangian particle-tracking models, which are grounded in ecological and oceanographic theory. However, it lacks formal justification for certain assumptions.	1 -The paper note some modelling assumptions that could bias results, but they do not perform formal uncertainty or sensitivity analyses.	3 -The model handles a large spatial scale (Western Indian Ocean) and multiple PLD scenarios, showing moderate scalability. e authors mention that their approach allows for evaluation of connectivity at a "reasonable computational cost and time", indicating practical scalability.	3 -The study provides clear conservation recommendations, identifying priority reefs for MPAs. The results are directly actionable for policymakers, and the discussion emphasises the practical implications for MPA network design.	3 -The paper excels in interpretability by using clear graph visualisations to explain technical outputs into clear and understandable ecological insights.	15
[62]	3 -The paper excels in ecological relevance as it directly addresses a critical and real-world problem in conservation by accurately inferring connectivity across landscapes to understand population persistence, particularly when ecological data are limited.	3 -The paper justifies the use of social network models (sender-receiver and latent space models) for ecological connectivity based on stochastic network theory. It contrasts these with traditional methods and clearly explains the assumptions and limitations.	2 -he paper addresses uncertainty through cross-validation (randomly removing data to test robustness) and model selection criteria. However, it lacks explicit quantification of uncertainty in network metrics.	2 -The methods are tested on two spatially distinct systems (a small cactus patch network and a larger wetland network) and show that they are applicable across scales. However, the paper does not discuss computational efficiency or challenges of very large networks.	3 -The paper provides direct and actionable insights for conservation managers. It warns against the widespread use of maximum distance construction, which it shows consistently overpredicted connectivity and led to misleadingly high estimates of metapopulation lifetime.	3 -The results are presented and explained in a transparent, understandable way. Key network metrics are clearly defined and related to ecological concepts.	16

Table 62: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[63]	3 -The paper excels in ecological relevance as it directly addresses a significant conservation problem: the effectiveness of MPA networks in stemming biodiversity decline and promoting resilience in European marine ecosystems.	3 -The paper is built on robust theoretical foundations, combining a well-established biophysical modelling approach with GT. The choice of network metrics is appropriate for assessing connectivity, and the model's assumptions are clearly stated.	2 -The paper recognizes sources of uncertainty, noting that assuming passive propagule drift may oversimplify small-scale movements and active larval behaviour, potentially biasing estimates. However, the paper does not quantify the uncertainty of its derived network metrics.	3 -The method scales well to the European marine reserve network, covering large spatial and temporal scales. The use of detailed ocean data and efficient particle-tracking algorithms ensures computational feasibility.	3 -The paper provides highly relevant and actionable recommendations for marine resource managers and conservation planners. It directly identifies large connectivity gaps and highlights regions that require additional marine reserves to function as stepping-stone connectivity corridors.	3 -The paper excels in interpretability, they effectively communicate their complex scientific findings to a broader audience.	17
[64]	3 -The paper shows high ecological relevance as it addresses a crucial issue in conservation ecology by understanding metapopulation persistence in fragmented marine landscapes to inform conservation planning.	3 -The paper is firmly based on established theoretical frameworks. It applies metapopulation theory and uses the dominant eigenvalue of the migration matrix as a quantitative measure of the persistence and growth rate of metapopulations, a concept widely used in ecological modelling.	1 -The study acknowledges several sources of uncertainty and simplification, however the paper does not explicitly present a formal quantification of uncertainty.	2 -The method is scalable but with limitations, it is successfully applied to a large and complex geographical region demonstrating computational feasibility, but dynamic or multi-layer networks are not addressed.	3 -The paper provides highly practical and actionable recommendations for marine resource managers and conservation planners. It directly identifies two important hotspots of local connectivity in southeast Australia that contribute strongly to the persistence of multiple species.	3 -The paper effectively communicates complex scientific concepts and findings. It also uses visual network maps to help interpret results.	15
[65]	3 -The paper excels in ecological relevance as it directly addresses a critical and real-world conservation problem by understanding the spatial ecology of highly mobile marine species, specifically sea turtles, to guide conservation prioritisation and management.	3 -The paper uses GT metrics (e.g., closeness, betweenness) appropriately and justifies their selection for analysing connectivity. It clearly defines the basic elements of GT, such as nodes and links.	1 -The paper acknowledges sampling biases (e.g., uneven species representation) and data limitations but does not rigorously quantify uncertainty in network metrics.	3 -The study handles a large dataset (1235 individuals across six species) and global-scale networks efficiently. GT methods are inherently scalable, and the paper shows this by analysing regional and species-specific subnetworks.	3 -The paper provides highly practical and actionable recommendations for marine conservation and management, such as prioritising areas with high closeness or protecting high-betweenness links.	3 -The results are presented in an interpretable way. Metrics like closeness and betweenness are well-defined and linked to conservation concepts	16

Table 63: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[66]	3 -The paper is ecologically relevant as it addresses a long-standing debate in ecology and biogeography on whether marine ciliate taxa exhibit global dispersal or are structured by habitat and geography.	3 -The paper rigourously justifies its use of GT, detailing the construction of SSNs and the choice of metrics (e.g., assortativity, closeness).	2 -The authors provide comprehensive qualitative discussions of limitations and potential biases, but does not fully quantify uncertainty.	3 -The method handles large datasets (85482 sequences) efficiently, using graph partitioning and scalable tools.	2 -While the study identifies biogeographical patterns and suggests primer design strategies for future surveys, it lacks direct translation into conservation actions.	3 -Network visualisations and intuitive metrics enhance interpretability. The paper clearly links network properties to ecological concepts, making the results accessible.	16
[67]	3 -The paper excels in ecological relevance by linking GT to marine metapopulation dynamics. The study justifies its focus on node self-connections with ecological theory and empirical evidence.	3 -The paper rigourously modifies and introduces network metrics (e.g., Total Flux, Self-influx) to account for node self-connections, with clear mathematical formulations.	2 -The authors assess robustness of their results through systematic sensitivity analyses, but the paper does not include formal statistical uncertainty estimates.	2 - The methods are applied to a 33-node network and tested on 10-node subnetworks, demonstrating feasibility for moderate-sized systems. While the paper does not address computational limits or dynamic networks, the metrics are computationally tractable and could scale to larger systems with similar structures.	3 -The paper directly informs marine reserve design by identifying key nodes for persistence. Results are presented as actionable insights and are linked to conservation goals.	3 -Network metrics are clearly defined and visualised with ecological interpretations. Correlations among metrics are clearly discussed, enhancing understanding of their ecological implications.	16

Table 64: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[70]	3 -The paper excels in ecological relevance as it directly addresses key ecological issues of larval dispersal and connectivity in a vulnerable reef fish, emphasizing implications for marine reserves and fisheries conservation.	3 -The study is firmly grounded in GT and uses directional weighted networks to model larval connectivity. It applies ecological interpretations of metrics such as betweenness centrality and clustering and validates the model with a population graph from genetic data.	3 -The paper shows comprehensive uncertainty quantification by acknowledging assumptions like passive particles and the lack of explicit description of the habitat for larval recruitment.	2 - The method handles the spatial complexity of the Gulf of California well, but scalability to larger or more dynamic systems is not clearly tested. The approach is theoretically scalable, computational demands for larger networks or dynamic graphs are not discussed.	3 -The results are directly applicable to marine conservation, with clear recommendations for MPA design. The study highlights practical implications for fisheries management and provides actionable insights, such as identifying key larval source and sink areas.	3 -The graph-based approach is very easy to interpret, whereby nodes and edges can be clearly assigned to ecological units. Network metrics are intuitively linked to conservation concepts. Visualisations improve understanding.	17

Table 65: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[73]	3 -The paper shows high ecological relevance by defining its graph structures to represent real-world ecological units and processes.	3 -The paper shows strong theoretical foundations. It formulates the graph as a connectivity matrix derived from ocean circulation models and justifies the use of GT metrics to analyse dispersal barriers and recolonization pathways.	2 -The paper acknowledges variability in larval dispersal due to ocean currents and environmental conditions but does not fully quantify uncertainty. While it uses multiple years of data to capture temporal variability, it lacks explicit techniques to assess confidence in connectivity metrics.	2 -The paper shows scalability within the tested region but does not address how the method would perform in larger or more complex systems. The paper does not discuss computational limits or strategies for handling larger datasets.	2 -The paper provides actionable insights but does not fully translate results into explicit management recommendations. The findings could inform MPA design or recolonization strategies, but the link to stakeholder decision-making is implied rather than developed.	3 -The paper is highly interpretable. The GT metrics are easily linked to ecological concepts such as dispersal barriers and population connectivity. The results are presented with clear visualisations and straightforward explanations.	15
[74]	3 -The study excels in ecological relevance. It establishes an explicit link between oceanographic connectivity and genetic diversity in marine forests by using biophysical models to simulate dispersal and GT to visualise connectivity networks.	3 -The paper shows strong theoretical foundations. It employs a biophysical model coupled with GT, justified by prior work in marine connectivity.	2 -The study qualitatively acknowledges several sources of uncertainty and limitations. However, there is no discussion of confidence intervals for network metrics or propagation of uncertainty from input data to the final model results.	2 -The methodology is highly scalable, with a global biophysical model (16489 hexagons) and GT efficiently handling multi-generation connectivity. It is applied to 30 species across different marine regions, though computational costs for finer resolutions or dynamic networks are not discussed.	3 -The paper positions its findings as crucial for guiding conservation efforts and informing spatial conservation planning.	3 -The paper is very easy to interpret. The GT metrics are clearly linked to ecological concepts. Plots (e.g., connectivity networks, dispersal weighting) enhances understanding.	16

Table 66: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[75]	3 -The paper shows high ecological relevance by constructing a symbiotic graph model that accurately represents reef fish communities.	2 -The paper provides a clear formulation of the symbiotic graph model and justifies its choices. However, it lacks formal proofs for the graph kernel method (PMA) and does not discuss convergence properties for the Louvain algorithm.	1 -The study acknowledges data limitations but does not systematically quantify uncertainty in the graph model or PMA method. While it mentions repeated experiments (20 trials) for similarity calculations, it does not provide confidence intervals.	2 -The method handles a multi-year dataset with up to 1427 nodes and 122696 edges, demonstrating moderate scalability. However, the paper does not discuss computational efficiency or performance under larger datasets.	3 -The study directly supports conservation decision-making by identifying priority areas for protection. Results are linked to actionable insights in an easy way to understand.	3 -The graph-based approach is very easy to interpret. The results are visualised using multi-year symbiotic graphs and module subdivisions, which enhances interpretation.	14
[76]	3 -The excels in ecological relevance as it shows comprehensive ecological relevance by focusing on mesozooplankton, which are identified as critical components of marine ecosystems.	2 -The paper clearly formulates the GNN approach, including spectral-temporal GNNs (StemGNN) and transfer entropy for interaction analysis. However, although GNNs are justified for capturing multivariate dependencies, the paper lacks deeper theoretical discussion on why the chosen graph metrics are optimal for ecological forecasting.	2 -The study acknowledges variability in mesozooplankton dynamics but does not systematically quantify uncertainty in model predictions.	2 -The StemGNN model handles multivariate time-series data effectively, but scalability is only moderately addressed. The paper notes computational efficiency improvements over traditional models but does not test the method on significantly larger datasets.	2 -The results could inform marine management, but the translation to actionable conservation strategies is limited. While the GNN's interpretability is highlighted, there are no explicit recommendations for policymakers.	2 -The paper highlights the interpretative power of GNNs and visualises the learned graph to infer variable relationships. Interpretability is discussed qualitatively but not quantified or formally evaluated.	13
[77]	3 -The paper shows high ecological relevance by mapping hydrothermal vent systems and species distributions to real-world ecological dynamics. The study justifies its graph construction with empirical data and aligns with ecological theory.	3 - The paper builds upon strong theoretical foundations by clearly applying GT to examine connectivity among vent systems.	1 -The study acknowledges that sampling effort across vent systems is not consistent and may influence beta-diversity outcomes, but does not rigorously quantify uncertainty.	2 -The methods are scalable to the 11 vent systems studied, but the paper does not address computational efficiency or dynamic network extensions. The analysis is feasible for the dataset but lacks discussion of limitations for larger or more complex systems.	3 - The study excels in stakeholder utility by translating results into actionable conservation recommendations. The network "hub" and endemic-rich systems are clearly identified, providing direct guidance for seabed mining policies.	3 -The paper achieves high interpretability through intuitive network representations and clear ecological mappings. Metrics such as corrected weighted endemism and β -diversity components are well explained and linked to the conservation results.	15

Table 67: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[78]	3 -The paper shows high ecological relevance by linking its GT approach to real-world ecological dynamics. The study justifies its habitat grid and spawning locations using empirical data and aligns with ecological theory on metapopulation connectivity.	3 -The paper has a strong theoretical foundations. It clearly defines the structure of the graph and justifies its choice to answer ecological questions about connectivity and invasion dispersal.	2 -The paper acknowledges uncertainties through sensitivity analyses (e.g., testing scenarios for the duration of planktonic larvae, effects of particle retention) and discusses limitations. Uncertainty is addressed moderately, but not in detail.	2 -The study scales well spatially (180 grid polygons) and temporally (13 generations), but computational constraints are noted (e.g., 7 km hydrodynamic resolution). The paper does not clearly discuss parallelization or approximations for larger systems.	3 -The results directly inform management priorities by identifying highly connected nodes and stable dispersal pathways. The paper translates findings into actionable insights for invasive species control.	3 -The graph metrics are ecologically interpretable, and the paper clearly links them to concepts like source-sink dynamics and biogeographic barriers. The methods are transparent, with detailed descriptions of model inputs and outputs.	16
[79]	3 -The paper excels in ecological relevance as it addresses a relevant ecological problem such as the decline of marine and coastal ecosystems due to human pressures and the need for effective conservation strategies that transcend artificial maritime boundaries.	2 -The paper uses GT and spatial prioritisation with clear justifications for the methodological choices. While the graph-theoretic approach is well justified, the theoretical guarantees are not explored in depth.	1 The study acknowledges several limitations and uncertainties but does not quantitatively assess their impact on the main findings or include sensitivity analyses to evaluate how changes in assumptions might affect the identified connectivity patterns.	3 -The methods scale well to the Western Indian Ocean's regional scope. The study shows scalability by integrating multi-country EEZs and high seas, though dynamic or multi-layer network complexities.	3 -The paper excels in stakeholder utility by translating results into actionable conservation priorities. The Marxan output directly informs policy negotiations, and the emphasis on minimising fishing conflicts enhances feasibility.	3 -The paper achieves a high level of interpretability. The graph-based metrics are ecologically interpretable, and visualisations clearly link results to conservation targets.	15
[80]	3 -The study excels in ecological relevance by accurately representing coral larval dispersal dynamics using a biophysical model.	3 -The paper shows strong theoretical rigour, it uses a Lagrangian particle tracking algorithm in conjunction with an ocean circulation model (ROMS). GT metrics (e.g., betweenness centrality) are used appropriately to analyse connectivity and their use in a conservation context is clearly justified.	2 -The study acknowledges several limitations and simplifications. While these limitations are clearly identified, the study does not quantitatively assess the impact of these uncertainties on its main findings.	2 - The model handles a regional scale (30 MPAs) efficiently, but scalability to larger or more complex systems is not discussed. The computational demands of the hydrodynamic model (ROMS) are noted, but parallelization or simplification strategies for broader applications are not addressed.	3 -The results directly inform marine conservation by identifying critical MPAs and recommending the inclusion of coastal MPAs in the MCCETP network. The study translates complex connectivity metrics into actionable insights, such as prioritising "stepping-stone" MPAs, which are highly relevant for policymakers.	3 -The results of GT can easily be linked to ecological concepts such as larval dispersal and resilience. Visualisations and clear explanations increase transparency and make the results accessible to those involved.	16

Table 68: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[83]	3 -The paper excels in ecological relevance as it addresses a highly relevant and critical ecological problem in marine systems which is understanding the spatial dynamics of herbivorous fish and their implications for the stability and resilience of herbivory on coral reefs.	3 -The paper shows strong theoretical foundations by clearly defining its graph type and justifying the use of small-world network metrics. The application of Watts & Strogatz's framework is well-reasoned, and the analysis aligns with ecological questions.	2 -The paper acknowledges uncertainty and addresses it through sensitivity analyses. However, it lacks formal quantification for network metrics.	2 -The study scales well to the examined reef system but does not clearly test scalability to larger or more complex ecosystems. The linear reef array simplifies analysis, but the paper does not discuss computational challenges or solutions for broader applications.	3 -The results are highly actionable for stakeholders, offering clear conservation recommendations. The link between network properties and ecosystem collapse risks is clearly highlighted for decision-makers.	3 -The paper achieves high interpretability by using transparent network metrics with clear ecological analogs. Graph visualisations and power-law distributions are well-explained, and the discussion ties results to established concepts.	16

Table 69: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[85]	3 -The paper excels in ecological relevance as it addresses an important ecological problem, which is understanding coral larval dispersal and inter-reef connectivity patterns in the GBR.	3 -The paper shows strong theoretical foundations. It uses an unstructured-grid ocean model (SLIM) to resolve multi-scale hydrodynamics, validated with observational data. The graph-theoretic approach is well-justified for detecting reef communities, and the choice of metrics aligns with ecological questions.	2 -The paper acknowledges several uncertainties and limitations associated with the modelling approach. The paper does not include a comprehensive quantitative uncertainty analysis or sensitivity testing for all potential sources of error.	2 -The method scales well to the GBR's complexity, simulating more than 1000 reefs with reef-scale resolution. This shows thoughtful scaling to a real-world system, but scalability beyond the studied region isn't showed.	3 -The results are analysed from a management perspective, the community-specific dispersal distances are compared with the existing MPA distances and it is shown that the connectivity between MPAs is highly species-dependent. These results are directly relevant for conservation planning, indicating that the research has a high utility for stakeholders.	3 -The paper is highly interpretable the results and technical ideas are explained in a way that is easy to understand, making it useful for both scientists and people working in conservation. The visualisations effectively communicate results.	16
[86]	3 -The paper excels in ecological relevance as it addresses real-world ecology problem in metapopulation analysis, which is the mega-patch problem where traditional methods of defining habitat patches often ignore the actual dynamics of interacting subpopulations within seemingly contiguous areas.	3 -The paper shows strong theoretical foundations by using GT (modularity optimisation) and spatial autocorrelation to address the "mega-patch problem." The choice of network metrics is justified for identifying subpopulations, and the method is grounded in ecological theory.	2 -The paper acknowledges variability in population dynamics and tests sensitivity to parameters like the modularity threshold and linkage distance. However, it lacks explicit quantification of uncertainty.	3 -The study shows high scalability by applying its method to a large-scale dataset from the Southern California Bight, using 30-m Landsat pixels as nodes. However, the paper does not discuss computational limits or scalability to larger/more complex systems.	1 -The results could inform marine conservation, but the paper does not clearly translate findings into management recommendations. The modularity approach is conceptually useful, but lacks direct stakeholder engagement or policy-ready output.	3 -The graph-based approach is very easy to interpret. The nodes represent the habitats of the kelp species, and the edges reflect the synchrony of the population dynamics. Metrics such as modularity are explained in simple ecological terms. The visualisations effectively communicate the method and results.	15

Table 70: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[87]	0 -The paper focuses on marine oil spill detection using SAR images, which is an application with significant practical importance for marine ecological protection but does not incorporate any ecological processes or dynamics.	3 -The paper shows a comprehensive understanding and application of established theoretical foundations in DL and GNN.	1 -The paper acknowledges variability in SAR data but does not clearly address uncertainty quantification in its model predictions or in the construction of its graph structure.	2 -The study shows scalability by testing models on large SAR datasets (e.g., 10240×10240 pixels) and reporting computational efficiency. However, it does not clearly discuss challenges or solutions for scaling to even larger datasets or real-time monitoring systems.	3 -The paper strongly emphasises practical applications, such as real-time oil spill monitoring and cross-border collaboration. The results are presented in a way that is actionable for stakeholders. The discussion highlights how the model reduces economic losses and aids ecological protection, directly addressing conservation and management needs.	2 -The GNN-based models are less interpretable, but the paper mitigates this by visualising segmentation results and comparing performance across categories. However, it does not provide tools like feature attribution or decision rules to explain model behaviour, which could improve transparency for non-technical stakeholders.	11
[89]	0 -The study addresses marine oil spill monitoring, an important environmental problem. However, the model itself does not directly represent or simulate biological or ecological dynamics in the sense of living organisms, populations, or their interactions.	3 -The paper provides a comprehensive theoretical foundation for its proposed GCAT network. It clearly justifies the use of GCN for spectral feature extraction and GAT for spatial feature extraction, highlighting their suitability for non-Euclidean data	1 -The study acknowledges potential sources of error related to the limited number of ground object types in hyperspectral images and challenges in the data preprocessing phase. However, the paper offers minimal formal quantification of the uncertainty associated with the model's predictions.	2 -The proposed GCAT model is tested on specific datasets, and its computational efficiency is implied by the use of GPU acceleration. However, the paper does not thoroughly evaluate scalability to larger or more complex marine environments.	2 -The study provides actionable outputs that are useful for marine conservation stakeholders, such as spill responders. However, the translation of results into explicit management recommendations is limited.	1 -The paper describes the model architecture clearly, it does not provide tools to explain how specific features contribute to classifications. Interpretability is further limited by the lack of visualisations or simplified summaries for non-technical stakeholders.	9

Table 71: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[92]	3 -The paper demonstrates strong ecological relevance by modelling fish schooling with hydrodynamic interactions. Its RL framework targets ecological objectives such as maintaining formations while minimising effort, directly addressing conservation related questions on collective behaviour.	3 -The paper shows strong theoretical rigour. The RL approach uses Q-learning, a well-established algorithm with proven convergence properties, and the study addresses the Markovian nature of the system.	2 -The study provides minimal explicit quantification of uncertainty, it lacks formal uncertainty quantification (e.g., confidence intervals) for the learned policies or dipole interactions.	2 -The dipole model and RL framework handle moderate-sized schools (up to 100 swimmers) effectively. The authors note computational limits of resolving finer-scale hydrodynamics but do not test scalability to very large groups or heterogeneous environments	1 -The paper does not provide any direct or actionable insights for marine conservation or resource management stakeholders. Its primary focus is on theoretical insights rather than stakeholder engagement or policy recommendations.	2 -The paper excels in interpretability by clearly explaining its methods and the result, but they are not linked to broader ecological principles.	13

Table 72: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[94]	3 -Strong ecological relevance for planktonic navigation in complex flows, modeling microswimmers with partial observations to mimic real marine dispersal.	3 -Critically assesses RL algorithms (Q-Learning, A2C, PPO), defining POMDPs with theoretical quasi-optimal policies for benchmarking.	3 -Quantifies uncertainties via sensitivity to hyperparameters, robustness tests across flows (e.g., turbulence), and comparisons to optimal policies.	3 -Scales to complex flows (e.g., 2D turbulence) with vectorized environments and hyperparameter optimization for efficient training.	3 -Highlights PPO's superiority for discovering smart strategies, applicable to autonomous robots in oceans.	3 -Interprets policies in terms of navigation efficiency and flow exploitation, using visualizations and performance metrics.	18
[95]	3 -Ecologically relevant for sustainable tuna fishing, modeling purse seine dynamics to predict trajectories for capturing free-swimming schools.	3 -Applies DDQN with defined states (vessel/-fish positions), actions (shooting points), and rewards based on catchability and gear behaviour.	2 -Acknowledges variability in fish movements but lacks systematic quantification; evaluates through simulations without explicit sensitivity.	2 -Applied to single vessel scenarios; scalable in principle but not demonstrated for multi-vessel or larger environments.	3 -Supports inexperienced fishers and automation in purse seine operations, improving success rates for FSCs.	3 -Interprets trajectories via gear analysis and catchability, visualized for practical understanding.	16

Table 73: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[97]	3 -High ecological relevance for predicting red tide outbreaks in coastal environments using hyperspectral data, addressing impacts on marine ecosystems and human life with spatial-temporal fusion.	3 -Integrates transformer encoder with GCN, defining graphs from multi-band spectral indices and extracting temporal or topological features for prediction.	2 -Evaluates accuracy with ablation but lacks explicit quantification of uncertainties like data noise or model sensitivity; implies robustness through input periods.	3 -Handles time-series data over interconnected points, scalable to large spatial scales with 92% accuracy on 5-day inputs.	3 -Provides early alarms for maritime agencies, reducing negative impacts on environments and economies.	3 -Interprets features via GCN and transformer, with visualizations of predictions and ablation results.	17
[98]	3 -Strong ecological relevance for forecasting SST in mesoscale-dynamic regions, supporting weather, transport, and fisheries applications from satellite images.	3 -Adapts GNN for oceanographic forecasting, defining graphs from satellite grids and using masked loss for spatial discontinuities.	3 -Quantifies uncertainties via comparisons with baselines, error metrics (e.g., RMSE), and handling data gaps/sparsity.	3 -Scales to regional grids (e.g., Canary Islands), outperforming ConvLSTM in efficiency for large datasets.	3 -Overcomes limitations of numerical models, offering flexible forecasting for ocean management.	3 -Interprets mesoscale capture through visualizations and metrics, linking to oceanographic processes.	18
[100]	3 -Ecologically relevant for fish swimming behaviour recognition in aquaculture, assessing health via skeleton-based features for disease prevention.	3 -Proposes SGCN with ensemble (EMGCM), defining graphs from joints or bones or motions for spatiotemporal extraction.	2 -Validates on dataset with metrics but limited quantification of uncertainties like pose variations; implies robustness via ensemble.	3 -Scales to multi-behaviour dataset, efficient for real-time monitoring with 90.31% accuracy.	3 -Supports practical aquaculture management, health assessment, and optimization.	3 -Interprets via spatiotemporal features and ensemble predictions, with visualizations of skeletons or behaviours.	17

Table 74: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[101]	3 -The paper shows high ecological relevance by addressing a crucial challenge in movement ecology, which is accurately filling gaps in animal tracking data. It applies IRL to GPS trajectories of streaked shearwaters, a marine seabird, to predict realistic movement paths.	3 -The study is rigorous in its theoretical foundations, clearly basing its approach on the well-established Maximum Entropy and IRL framework. It defines the mathematical components of the RL problem, including the MDP formulation.	2 -Uncertainty is moderately addressed. The paper compares IRL to linear interpolation using the MHD but does not quantify uncertainty in the learned reward function or policy	2 -The method scales moderately. The discretization of space and time ensures computational feasibility, but the paper does not test scalability to larger datasets or more complex environments.	3 -The method provides high utility for stakeholders by offering a superior way to fill gaps in animal movement data, which is essential for understanding habitat use, migration patterns, and foraging strategies in animals.	3 -The paper shows high interpretability, the reward maps and feature weights clearly link environmental features to movement preferences. The policy's avoidance of land aligns with ecological intuition. Visualisations effectively communicate how gaps are filled.	16
[102]	3 -The paper thoroughly addresses ecological relevance by aligning AI and automation technologies with marine conservation goals. It discusses how automated monitoring captures real-world ecological dynamics, such as fish abundance and coral reef health.	2 -The paper provides a good overview of AI techniques and their potential for marine conservation. However, it lacks detailed mathematical formulations, although the paper justifies their choice of methods the theoretical foundations of these techniques are not explored in depth.	1 -The paper acknowledges uncertainties in AI models (e.g., biases in the training data, variability in the marine environment), but does not quantify or address them systematically.	3 -The paper excels in scalability by discussing how AI and automation can handle large-scale marine monitoring. It highlights technologies like edge computing and IoT for real-time data transfer, as well as the cost-effectiveness of scaling DL for processing large amounts of image data.	3 -The paper emphasises the potential of AI to support conservation decision-making through user-friendly interfaces such as dashboards for managers and data-driven insights. It outlines benefits including faster assessment of management outcomes and enhanced support for adaptive management	1 -Interpretability is minimally addressed in this paper. The paper notes the "black box" nature of some AI models (e.g., DL), but does not explore techniques to improve transparency, such as attribution of features or simplified rule sets.	13
[103]	3 -The paper excels in ecological relevance by addressing the need for automated monitoring of marine ecosystems, particularly seabirds (e.g., Northern Gannets), which are indicators of ecological health.	2 -The paper employs a hybrid approach combining supervised ML and RL, with clear justifications for these choices. However, while the ML techniques are well-established, the integration of RL is less detailed.	1 -The paper reports high accuracy metrics (sensitivity, specificity) but does not quantify uncertainty in its results.	2 -The method shows scalability by processing large-scale aerial surveys and handling diverse marine environments. However, the computational demands of training detectors are noted without proposing optimisations for larger datasets.	3 -The study directly supports stakeholders by automating labor-intensive censuses and providing actionable insights (e.g., trends in seabird populations). The user-friendly Matlab interface and high accuracy rates make the tool practical for conservationists.	2 -The ML techniques are interpretable, and the paper contains visual examples of detected objects. However, the decision-making process of the RL component is less transparent and the internal logic of the ensemble approach is not fully explained.	13

Table 75: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[94]	3 - The paper shows high ecological relevance by directly addressing the problem of navigating in fluid flows for autonomous robots deployed in oceans for environmental monitoring.	3 -The paper lays a strong theoretical foundation by clearly formulating the navigation problem as a model-free POMDP, explaining why optimal control theory is insufficient given partial observations.	3 -The paper addresses uncertainty through repeated training trials (five random seeds) and reports confidence intervals for performance metrics.	2 - The paper shows a moderate consideration for scalability by testing flows of increasing complexity. However, it does not discuss computational costs, parallelization, or generalisability to larger-scale oceanic flows.	2 -The paper's results could inform the design of autonomous marine robots. However, it lacks direct translation to conservation or management actions (e.g., habitat monitoring, pollution tracking).	3 -The paper excels in interpretability by effectively using Visualisations (policy plots/heat maps) to show the learned behaviour of the RL agents.	16
[104]	3 -The paper shows strong ecological relevance by addressing a problem faced by planktonic organisms in turbulent marine environments, which is avoiding high-strain regions to minimise the risk of predation.	3 -The paper is mathematically rigorous and uses RL and analytical short-time expansions to derive optimal navigation strategies. The dynamics of the float are based on fluid mechanics, and the RL framework is well justified with clear convergence properties.	2 -The paper does not quantify uncertainty in its RL policies or flow models. While it tests robustness by varying parameters there is no formal confidence intervals.	2 -The methods can be easily transferred to different flow regimes (low/high Reynolds numbers) and the behaviour of swimmers. However, the study does not address the computational costs or convergence times for RL in larger, more complex ecosystems.	2 -The paper is primarily theoretical and cannot be directly applied to conservation or management. While the strategies could inform bio-inspired robotics or plankton ecology, the paper does not provide actionable insights for stakeholders (e.g., for MPA design or fishery management).	3 -The paper excels in interpretability by making strong efforts to explain the internal logic and results of the models in ecologically meaningful terms. It presents learned strategies through clear policy Visualisations.	15
[105]	3 -The paper shows strong ecological relevance by directly addressing the problem of diel vertical migration (DVM) in zooplankton within turbulent marine environments.	2 -The study acknowledges stochasticity in training and tests robustness to parameter perturbations. However, it lacks formal uncertainty quantification (e.g., confidence intervals) for the learned policies.	3 -The paper is built on strong theoretical foundations, rigorously framing microswimmer navigation as a MDP. It clearly defines the agent (swimmer), environment (fluid flow), states (local hydrodynamic signals), actions (active reorientation), and reward function (vertical displacement efficiency), ensuring alignment with both RL principles and biological realism.	2 - The model handles 2D and 3D flows efficiently, but scalability to larger or more complex ecosystems is not tested. Tabular Q-learning may face limitations in high-dimensional state spaces.	2 -Although the findings are valuable for understanding plankton behaviour, the results of the study cannot be translated into direct conservation or management recommendations (e.g., habitat protection, climate protection).	3 -The paper excels in interpretability by making significant efforts to explain the internal logic and outcomes of the models in clear, ecologically and physically meaningful terms.	15

Table 76: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[107]	3 -Shows high ecological relevance by modeling connectivity hotspots for herbivorous reef fishes in a tropical atoll, directly addressing herbivory's role in coral reef resilience and restoration amid climate change and habitat degradation.	3 -Shows strong theoretical foundations by integrating habitat suitability modeling (Max-Ent) with GT for connectivity analysis, justifying species-specific choices based on ecological traits.	1 -The paper shows a low level of uncertainty quantification. It does not explicitly address or quantify uncertainties, limitations, or sensitivities in the modeling approach or results.	3 -The two-step modeling approach (habitat suitability + graph theory) demonstrates high scalability, efficiently handling atoll-scale data and suggesting applicability to larger seascapes or multi-species analyses.	3 -The study offers high stakeholder utility by mapping connectivity hotspots to guide restoration and conservation actions, with results directly informing MPA design and habitat management in island systems.	3 -The paper excels in interpretability, clearly explaining methods like graph-theoretic metrics and linking them to ecological processes (e.g., foraging movements), with visualizations for practical use.	16

Table 77: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[109]	3 -Shows high ecological relevance by assessing reinforcement learning for microswimmer navigation in complex flows, with direct implications for understanding planktonic organism movement and marine robotics in ocean monitoring.	3 -Shows strong theoretical foundations by critically evaluating RL algorithms in fluid dynamics contexts, justifying choices with benchmarks and POMDP formulations tailored to marine-like flows.	3 -Shows a high level of uncertainty quantification, rigorously testing robustness across flow types (e.g., turbulence) and incorporating techniques like generalized advantage estimation to handle variability.	3 -The RL implementations (e.g., vectorized environments) demonstrate high scalability, performing well in complex, dynamic flows and suggesting applicability to larger marine systems or multi-agent scenarios.	2 -The study provides moderate stakeholder utility, offering insights for ocean robotics but lacking direct translation to conservation policies or explicit stakeholder engagement.	3 -The paper excels in interpretability, clearly defining RL concepts (e.g., POMDP, reward functions) and linking results to ecological navigation challenges in accessible terms.	17
[110]	3 -Shows high ecological relevance by mapping trophic and non-trophic interactions in a Chilean rocky shore community, addressing how diverse links influence ecosystem resilience and functioning.	3 -The paper shows strong theoretical foundations by integrating food web theory with multiplex networks for non-trophic interactions, justifying the approach with empirical data and trait-based predictions.	2 -The paper shows a moderate level of uncertainty quantification. It mentions exploring consequences of uncertainty in network locations but does not rigorously quantify it through statistical validation or sensitivity tests.	2 -The network analysis scales well for the studied community (100 species), but scalability to larger marine ecosystems or dynamic scenarios is not explicitly addressed or tested.	3 -The study offers high stakeholder utility by providing a framework for predicting community responses to disturbances, actionable for marine conservation and resilience planning.	3 -The paper excels in interpretability, clearly defining interaction types (e.g., trophic vs. non-trophic) and linking network patterns to ecological theory in an accessible way.	16

Table 78: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Rel- evance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[112]	3 -High relevance by revealing pearl oyster connectivity patterns via genomics and dispersal models for sustainable management.	3 -Solid foundations integrating graph theory with multi-generational connectivity and coalescent models.	2 -Moderate via graph clustering, but limited sensitivity tests on dispersal variability.	2 -Scales to archipelagos, but not tested for larger oceanic systems.	3 -High utility for defining management units in pearl farming and conservation.	3 -Clear links between networks, genetics, and ecological processes.	16
[114]	3 -High relevance for ocean health monitoring under climate stress using biohybrid-AI sensors.	3 -Strong foundations in hybrid CNN-LSTM and RL for adaptive sensing.	3 -Rigorous with ensemble methods and variance in model performance.	3 -Scales via nanostructured materials and ML for large marine areas.	3 -High utility for real-time conservation and ethical deployment.	3 -Clear model explanations linking to ecological stressors.	18
[115]	3 -High relevance by identifying causal links in food webs for ecosystem functioning.	3 -Strong asymmetry analysis grounded in topology and biomass correlations.	2 -Moderate pairwise correlations, but limited statistical validation.	2 -Scales to 34 webs, but not dynamic/large systems.	3 -Useful for assessing ecosystem health and resilience.	3 -Accessible asymmetry graphs and causality metrics.	16

Table 79: Template for evaluating the theoretical rigour of papers(Nicole)

Study	Ecological Relevance	Theoretical Foundations	Uncertainty Quantification	Scalability	Stakeholder Utility	Interpretability	Score
[118]	3 -High relevance for coral monitoring with AUVs addressing threats like bleaching.	3 -Strong integration of sensing, ML, and navigation theories.	3 -Rigorous with ensemble GNNs and perturbation tests.	3 -Scales via hierarchical RL and efficient architectures.	3 -Utility for conservation via disease detection and mapping.	3 -Explanations via attribution and heat maps.	18
[119]	3 -High relevance for sailboat navigation in ocean monitoring and exploration.	3 -Strong RL and path planning foundations with hybrid approaches.	3 -Rigorous uncertainty via Monte Carlo and variance reports.	3 -Scales to dynamic environments with efficient algorithms.	3 -Utility for sustainable marine data collection.	3 -Clear policy visualizations and learning explanations.	18
[120]	3 -High relevance clarifying marine functional connectivity for policy and management.	3 -Strong theoretical review integrating ecology and spatial planning.	2 -Moderate via literature synthesis, but no formal uncertainty.	3 -Scales globally across disciplines and ecosystems.	3 -High utility bridging science and stakeholder needs.	3 -Accessible definitions and recommendations.	17
[122]	3 -High relevance reviewing ML in fisheries ecology from genomics to ecosystems, aiding conservation.	3 -Strong synthesis justifying ML methods with ecological logic and assumptions.	2 -Moderate discussion of biases and validation, but limited specific quantification.	3 -Covers ML for large datasets in genomics, video, and spatial analysis.	3 -High utility for management decisions, ethics, and stakeholder trust.	2 -Addresses black-box issues moderately, suggests explainability needs.	16

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